

Optimization Under Uncertainty for Space-Based Just-in-Time Collision Avoidance

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(ABSTRACT)

Orbital debris in low Earth orbit poses a growing system-level risk: rare but catastrophic debris-debris collisions can generate long-lived fragment populations and raise future collision rates. This dissertation studies a complementary mitigation concept, dust-enabled just-in-time collision avoidance, in which targeted dust releases are designed to impart small momentum changes before closest approach. The mission is posed as an optimization-under-uncertainty problem with explicit service-reliability requirements and mission-resource tradeoffs. Operational post-intervention conjunction-risk reduction is outside the current validated evidence scope.

The dissertation makes four concrete contributions. It defines a catalogue-grounded hazardous-event workflow in which conjunction evidence, screened from hash-pinned catalogue captures into reusable event banks, is treated as a design input rather than an afterthought. It formulates dust-enabled just-in-time collision avoidance as a deployer-architecture problem, coupling transfer feasibility, dust-mass logic, and mission reliability against physically screened conjunction opportunities. It develops a layered uncertainty-propagation and scoring framework, enabled by a custom multi-method Rust propagator using Battin’s numerically stable Encke formulation, with explicit epoch and body-role fidelity assignment: the search lane screens transfers under mean-element J_2 physics, the hybrid lane requires high-fidelity transfer correction, and blinded canonical replay carries its own strict authority. Finally, it compares

three constellation families across six optimizer methods under common-random-number evidence control: Beta-posterior and [Highest Posterior Density \(HPD\)](#) diagnostics control adaptive search, while canonical reporting is designed to certify 58 of 64 unsafe scenarios through a reconstructed nondominated maneuver–released-mass front. Optimizer evidence is common-random-number aligned and descriptive; it supplies one realization per cell rather than paired replicates, and the protocol permits no global or per-family optimizer winner. Current attainment, atmosphere-model, and probability-qualified released-mass claims remain held until a campaign sealed under the current compiled science authority and receipt chain is independently reconstructed. Dust-covariance transport uses one Gaussian propagated by the seven-point Julier minimal-skew simplex [Unscented Transform \(UT\)](#) rule, with fixed spherical-equivalent dust physics shared by all sigma points; the compiled mixture-split controls remain dormant and form no child components. That combination defines the transfer and coverage evidence required for defensible future results.

In short, the dissertation asks when dust-enabled nudging is feasible, what it costs in maneuver and released mass, and what safeguards and limitations bound its use as a possible complement to existing debris-risk management.

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(GENERAL AUDIENCE ABSTRACT)

Earth orbit is crowded with thousands of human-made objects. Most are inactive satellites or small fragments from past missions and breakups. Although each piece is tiny compared to a spacecraft, a high-speed collision can create thousands more fragments, making the problem worse for everyone who relies on space services. This study examines one way to lower the chance of the rare, dangerous encounters that cause the most harm.

The basic idea is simple: release a small cloud of dust at the right place and time so that a risky object receives a very small nudge. The method tests whether that nudge changes predicted encounter geometry; it does not yet establish operational post-intervention risk reduction. Turning that idea into a workable system requires practical answers. Where should the service satellites orbit to reach likely trouble spots quickly? How much dust should be released, and how should it spread out? How should decisions be made when knowledge of object positions is imperfect and still improving?

The study uses computer models to define comparisons among strategies and optimization methods, asking which designs can respond quickly without demanding too much dust or maneuvering. Current quantitative conclusions remain on hold until an independent group can rebuild one full campaign from the archived record and obtain the same numbers. The methodology identifies safeguards and limitations that would matter before any real-world use, including careful monitoring of possible secondary hazards.

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List of Abbreviations

AGE-MOEA2 Adaptive Geometry Estimation Multiobjective Evolutionary Algorithm 2.

[80](#)

API Application Programming Interface. [70](#)

CRN Common Random Number. [29](#), [75](#), [78](#), [104](#), [108](#), [112](#)

CVaR Conditional Value-at-Risk. [21](#)

DART Double Asteroid Redirection Test. [30](#)

EMOA Evolutionary Multiobjective Algorithm. [viii](#), [80](#)

GCRS Geocentric Celestial Reference System. [8](#), [49](#)

GMM Gaussian Mixture Model. [74](#)

HF High-Fidelity. [85](#)

HPD Highest Posterior Density. [iii](#), [xvi](#), [2](#), [8](#), [10](#), [78](#), [79](#), [87](#), [102](#), [119](#), [122](#)

JB2008 Jacchia-Bowman 2008 thermospheric density model. [35](#), [58](#), [59](#), [118](#)

JCA Just-in-Time Collision Avoidance. [1](#), [5–7](#), [9](#), [14](#), [18](#), [23](#), [24](#), [26](#), [32–34](#), [39](#), [122](#), [127](#)

KDE Kernel Density Estimation. [74](#)

LEO Low Earth Orbit. [1](#), [30](#), [70](#), [71](#), [73–75](#), [101](#), [146](#)

MF Mean-Field. [85](#)

MOPSO Multiobjective Particle Swarm Optimization. [vii](#), [80](#)

NSGA-II Nondominated Sorting Genetic Algorithm II. [vii](#), [79](#), [80](#)

OMM Orbit Mean-Elements Message. [8](#), [39](#), [70](#)

RAAN Right Ascension of the Ascending Node. [15](#), [58](#), [66](#)

RNSDE Robust Nondominated Differential Evolution. [77](#), [78](#)

SATCAT Satellite Catalog. [8](#), [70](#)

SGP4 Simplified General Perturbations 4. [39](#)

SMS-EMOA S-Metric Selection Evolutionary Multiobjective Algorithm. [81](#)

SRP Solar Radiation Pressure. [39](#), [58–60](#), [62](#), [99](#), [121](#)

SSO Sun-Synchronous Orbit. [73](#)

TCA Time of Closest Approach. [43](#), [72](#)

TLE Two-Line Element set. [39](#)

UT Unscented Transform. [iii](#)

List of Symbols

α_0	Alpha parameter of the beta prior used in the adaptive reliability model (unitless).
β_0	Beta parameter of the beta prior used in the adaptive reliability model (unitless).
c	Configured confidence level used for posterior reliability intervals and HPD threshold bands (unitless).
C_D	Free-molecular drag coefficient applied within the dust-force model (unitless).
d_{conj}	Conjunction-truth miss distance between the primary and secondary objects at closest approach; an event is admitted to the bank when it lies below the 1 km unsafe threshold (km).
ΔV	Executed maneuver coordinate in canonical event-coverage resource boxes, including phasing, replayed transfer, and release-control burn norms (km/s).

d_{miss}	Maximum allowed miss distance at the release/intercept point for a transfer to remain acceptable (m).
$f_1(x)$	Raw maneuver coordinate of one exact canonical event-coverage box before search regularization (units of km/s).
$f_2(x)$	Raw released-dust coordinate of one exact canonical event-coverage box before search regularization (units of kg).
$H(x)$	Serviced fraction metric giving the proportion of conjunction events that admit a feasible intercept and dust release under the operational guardrails (unitless).
I_k	Binary success indicator for event k , equal to 1 when transfer and deterministic-mass feasibility checks both pass (unitless).
F_1	Final penalized mission objective associated with deployer maneuver burden after the log-space transform and policy penalties (km/s semantics).

F_2	Final penalized mission objective associated with released dust burden after the log-space transform and policy penalties (kg semantics).
κ	Momentum-enhancement factor in the deterministic intercepted-mass gate, where the delivered fraction $f(m) = \kappa m / (m_p + m)$ rises monotonically to κ . Production seals $\kappa = 1$; a $\{1, 1.25, 1.5, 2\}$ sensitivity is proposed but not configured. In the shared-target contact bound that sets the reported released mass, κ is inert and its admissible domain is $(0, 1]$, so values above unity are refused there and cannot be read as a released-mass sensitivity coordinate (unitless).
M^{dust}	Pc-inverted released dust population mass coordinate in canonical event-coverage resource boxes (kg); deterministic intercepted-mass sufficiency is enforced separately as a physical gate.
$m_{\text{int},k}$	Event-level intercepted dust mass required to satisfy the deterministic momentum-transfer feasibility gate for event k (kg).

n_f	Count of failed events accumulated during adaptive mission scoring (unitless).
n_s	Count of successful events accumulated during adaptive mission scoring (unitless).
Ω_{unsafe}	Reusable bank of screened unsafe conjunction events supplied to the mission-scoring workflow.
P_c	Collision probability for a predicted conjunction. In the concept-of-operations framing it is a screening quantity; in the runtime it is the short-encounter encounter-plane disk integral that the released-mass solve inverts against packet count at fixed covariance. No tracking-update model refines it during a campaign (unitless).
P_c^{adv}	Notional advisory collision-probability threshold used in the concept-of-operations framing; production simulations admit unsafe events through catalogue screening plus geometric, timing, and feasibility gates rather than a live P_c trigger (unitless).

P_c^{com}	Notional commit collision-probability threshold used in the concept-of-operations framing; production simulations use it as timing context, not as an implemented event-admission criterion (unitless).
q	Configured exact finite-bank coverage fraction; the baseline $q = 0.9$ requires $\lceil qN \rceil = 58$ of $N = 64$ unsafe events (unitless).
R_E	WGS84 equatorial Earth radius, 6378.137 km, used as the J_2 reference radius of the mean-element design lane; the high-fidelity gravity field carries its own reference radius. Constellation parameterizations convert between altitude and semi-major axis against a separate compiled altitude datum of 6378 km; the two are not interchangeable (km).
s_{LCB}	Posterior lower credible-interval diagnostic used only for adaptive-search band classification; canonical service feasibility uses the exact 64-scenario success fraction (unitless).

$\mathcal{S}(x)$	Set of successful unsafe events serviced by design x , used to define successful-event resource summaries.
τ	Time-of-flight or propagation horizon variable used in transfer solving and short-horizon uncertainty propagation (s or min).
θ	Latent event-success probability in the beta-posterior reliability model (unitless).
t_ℓ	Lead time between dust intercept and conjunction epoch available for the induced trajectory change to separate the encounter geometry (s or h).
$t_\ell^\star$	Minimum required lead time enforced by the operational policy and transfer-feasibility guardrails (s or h).
$\mathbf{u}$	Three-parameter transfer-solver decision vector controlling phasing duration, semimajor-axis adjustment, and waiting time.
v_{rel}	Relative speed between the dust plume and resident space object at intercept, tracked as a diagnostic of energy transfer (km/s).

x Constellation design vector evaluated by the mission-scoring workflow; its interpretation depends on the architecture family under study.

Chapter 1

Introduction

[Low Earth Orbit \(LEO\)](#) is no longer a sparse operating region. It is a coupled socio-technical environment in which active spacecraft, inactive spacecraft, upper stages, and fragmentation debris share the same orbital bands, and in which rare but catastrophic collisions can inject long-lived fragment populations back into the system. Recent agency reporting and sustainability studies continue to show that congestion, collision risk, and long-horizon environmental stability can no longer be treated as separable concerns [19, 52, 75]. For active spacecraft, conjunction assessment and collision avoidance are now routine operational functions. For the debris population itself, however, many high-risk encounters remain unmanaged even though debris-debris collisions are among the strongest endogenous drivers of future environmental deterioration [33, 71].

This work studies a space-based dust-enabled just-in-time collision-avoidance architecture, a [Just-in-Time Collision Avoidance \(JCA\)](#) concept in which deployer spacecraft release dust clouds after a hazardous encounter has been identified but before closest approach. The goal is to impart targeted perturbations to threatening debris objects early enough to change the predicted encounter geometry. The central question is whether a bounded deployer service architecture can identify hazardous events, reach them in time, deliver physically meaningful nudges, and do so with acceptable mission-level reliability and resource expenditure.

The planned evidence is organized around two complementary questions. Part A asks how optimizer families compare when every method is judged under matched hazardous-event

evidence, fixed mission semantics, and actual-work reporting controls. Part B is designed to interpret what those shared evaluation rules reveal about the deployer-constellation design space itself. Part A remains an evidence gate for future architecture interpretation rather than a free-form optimizer choice; no current optimizer winner is asserted. Introducing that split at the outset separates algorithmic conclusions from constellation-design conclusions throughout the manuscript.

1.1 Motivation

The practical motivation is straightforward. A debris-debris collision can produce thousands of fragments, and those fragments can remain in operationally important altitude-inclination bands for years to decades [19, 25, 36]. Because the most consequential events are infrequent, uncertain, and geometry-sensitive, designing a remediation architecture around deterministic average-case assumptions is inadequate. The architecture must instead be judged against shared hazardous-event evidence, explicit uncertainty, and mission-level tradeoffs between service reliability and resource demand.

This study treats uncertainty as part of the design problem from the start. Candidate constellations are not judged on abstract coverage alone. They are judged on whether they can service screened unsafe conjunction opportunities under transfer, timing, and dust-delivery constraints, and on the resource boxes required to cover that hazardous-event cohort. Beta-posterior means and [Highest Posterior Density \(HPD\)](#) bands control adaptive search only. Canonical reporting requires exact service of 58 of the 64 fixed unsafe events and is designed to reconstruct a nondominated maneuver–released-mass front whose points carry the transfer, release, and dust evidence produced by the lane that generated them. Current attainment under the strict lane remains unverified.

The manuscript therefore uses stochastic optimization in its mission-design sense: uncertainty enters the objective and constraints through event sampling, state knowledge, transfer feasibility, and dust-cloud evolution, while randomized metaheuristics are used as computational tools for exploring the resulting mixed design landscape. Chapter 2 separates those two meanings more formally.

The resulting study is organized around the sequence summarized in Table 1.1. The logic begins with a catalogue-grounded event bank, proceeds through transfer selection and dust sizing, and ends in mission-level scoring, optimizer comparison, and constellation interpretation. The later chapters unpack each stage in turn.

Table 1.1: End-to-end workflow used throughout this study.

Stage	Role in the study workflow
Unsafe event evidence	Public catalogue data are screened into a reusable bank of hazardous conjunction opportunities.
Transfer selection	Candidate deployers are screened and refined into physically admissible intercept plans.
Dust mass logic	Event feasibility is gated by intercepted-mass physics before released mass is scored on the encounter plane.
Mission scoring	Exact event-count coverage and nondominated maneuver-released-mass boxes are reconstructed over the hazardous-event cohort.
Optimizer comparison	Architecture and optimizer families are compared under the same evidence and the same mission semantics.

1.2 Orbital Debris as a System-Level Design Problem

The debris problem is often described through collision probability, catalog growth, or post-mission disposal policy, but from a systems perspective it is better understood as a coupled feedback process. Launch activity, mission operations, fragmentation events, environmental persistence, and tracking capability all influence one another. High-value operational shells

attract more traffic; more traffic raises the rate of close approaches; fragmentation events inject additional objects into those same bands; and incomplete observability means the true risk environment is always only partially known [19, 47, 52]. For this reason, the relevant design question is not simply whether a given conjunction can be retired, but whether a reusable mitigation architecture can reduce endogenous risk without introducing disproportionate new burdens in fuel demand, released material, or operational complexity.

This system-level framing matters directly for the present study. A one-off intervention can be evaluated mainly on whether it changes the geometry of a single threatening encounter. A reusable service architecture has to be evaluated on where deployers are staged, which orbital bands they can access repeatedly, how long they need to prepare an intercept, how uncertainty affects both event screening and dust delivery, and how mission resources accumulate across many potential remediations. The design variable is therefore the service architecture operating over a population of hazardous events, not just a maneuver or a dust release.

The study examines three constellation families that span a range of architectural structure: Walker constellations parameterized by shell altitude, inclination, plane count, satellites-per-plane count, and phasing; Flower constellations adding eccentricity, argument of perigee, and a bounded combination index under a bounded J_2 -secular repeat-groundtrack specialization; and Dual-Shell Walker-Delta constellations carrying per-shell altitude, inclination, topology, and phasing plus second-shell RAAN and mean-anomaly offsets. All three are analytically structured, and every family shares the same 350 km to 2000 km realized radial envelope and the same 40-satellite cap. An unstructured plane-constrained encoding, with a fixed three-plane by five-satellite layout carrying per-plane altitude, inclination, RAAN, and phase freedom, was scoped as a fourth family but is not implemented in the reported evaluation stack and is not evaluated in this work.

The architecture must also respond to an asymmetry that distinguishes debris-debris encounters from many active-spacecraft conjunctions. An active spacecraft can often maneuver itself. Two non-maneuverable objects cannot. This asymmetry helps explain why the debris problem remains operationally under-served even though conjunction assessment itself is now a mature activity. A debris-remediation architecture must therefore supply mobility, timing control, and intervention capability from outside the conjunction pair. The deployer constellation is the mechanism that provides that missing control authority.

For this reason, the manuscript treats debris remediation as a design-under-uncertainty problem rather than as a purely astrodynamical calculation. The event set is uncertain, the transfer feasibility is uncertain, the delivered dust density is uncertain, and the mission-level meaning of “good performance” cannot be reduced to a single deterministic scalar. The technical challenge is to organize those uncertainties in a way that is both scientifically defensible and computationally manageable.

1.3 Dust-Enabled JCA as an Architecture Problem

Dust-enabled [JCA](#) is especially useful for studying this question because it couples a low-impulse intervention concept to a demanding systems problem. The basic actuation idea is simple: if a threatening debris object is perturbed early enough, even a small change in velocity can alter the conjunction geometry enough to retire the event. The operational realization is not simple. The deployer must be in a favorable orbital region, the transfer must be feasible within the available lead time, the dust cloud must reach the target with sufficient geometric coverage, and the resulting intervention must meet both physical and mission-level acceptance criteria.

These requirements make architecture design central. A deployer constellation that is excellent

for broad nominal coverage may still be poor for time-critical event access. Conversely, a constellation that is efficient for a few favored orbital bands may fail to provide sufficient reach across the actual hazardous-event population. The dissertation therefore treats the deployer constellation, the transfer workflow, and the dust-delivery logic as parts of the same system. The methodology begins with a reusable event bank and ends with mission-level scoring because the architecture is judged by how it behaves across the hazardous-event evidence it is meant to service.

The dust concept also sharpens the role of uncertainty propagation. A deterministic intercept point is not enough; what matters is the overlap between the uncertain target location and the evolving dust-cloud density at the encounter plane. Catalogue pair screening and search-lane transfer selection use MF/ J_2 physics with high-fidelity acceptance disabled, while each admitted conjunction anchor is refined under strict HF propagation; every cell searches on that same physics, and the hybrid lane enables high-fidelity propagation with required HF transfer correction only at the blinded 64-event (H64) canonical replay, which carries its own strict authority. [Section 5.4](#) records which of those lanes actually executed in the reported campaign. The compiled single-Gaussian Julier-7 transform then preserves the separation between intercepted-mass sufficiency and Pc-inverted released population mass.

Dust-enabled [JCA](#) is also narrow enough to support a focused manuscript. The study does not attempt to solve every aspect of debris remediation. It concentrates on the design of an uncertainty-aware response architecture for hazardous debris-debris conjunctions over short horizons. That narrower scope makes it possible to connect environmental evidence, transfer solving, uncertainty propagation, and multiobjective optimization within one consistent framework.

1.3.1 Research Questions & Objectives

The study is guided by seven research questions that move from representation and physical modeling to architecture and optimization:

1. How should the debris environment and hazardous-event evidence be represented so that the optimization problem is grounded in physically meaningful conjunctions rather than in abstract synthetic proxies?
2. What transfer logic, staging geometry, and operating rules are required for a dust-enabled [JCA](#) service to be physically defensible and operationally responsive?
3. What descriptive differences in physical-front quality and realized-work behavior appear among optimizer methods on the resulting mixed discrete-continuous, event-driven, and stochastic design landscape under a shared single-realization evidence protocol, and what campaign shape would be required to place uncertainty bounds on those differences?
4. What does seeded, architecture-aware initialization contribute to convergence stability and solution quality, and where does local refinement fail to provide a reliable advantage?
5. How should model fidelity be allocated so that bulk search remains efficient while acceptance-critical decisions remain physically credible?
6. How should uncertainty be encoded in the evaluation itself so that reliability and resource demand are measured on consistent semantics?
7. Are structured single-shell constellations sufficient for the mission, or do structured multi-shell and resonant repeating-groundtrack architectures provide meaningful advantages within the same event-level mission semantics?

These questions map directly onto the methods developed in [Chapter 3](#). [Item 1](#) is addressed through the catalogue-first event workflow, in which the dissertation’s canonical event banks are generated offline from hash-pinned Space-Track [Orbit Mean-Elements Message \(OMM\)](#) and [Satellite Catalog \(SATCAT\)](#) captures, screened by a deterministic mean- J_2 natural-pair scan, and refined to strict-HF unsafe conjunctions carrying paired osculating-[Geocentric Celestial Reference System \(GCRS\)](#) and mean screening anchors; the older surrogate debris generators are reported as historical exploratory work rather than as a current sensitivity lane ([Section 3.7](#)). [Item 2](#) and [Item 5](#) are addressed through the mission concept, transfer solver, event-level dust-mass logic, and mixed-fidelity propagation policy ([Sections 3.2 to 3.5](#)). [Item 3](#) is addressed through the six-method optimizer matrix under a single-realization descriptive protocol, [Item 4](#) through the PRNSDE local-refinement contrast under uniform-random initialization, and [Item 7](#) through the structured single-shell versus structured multi-shell and resonant constellation study ([Sections 3.6, 3.8 and 3.12](#)). [Item 6](#) is addressed through the mission objectives, constraints, and dust-propagation framework ([Sections 3.9 to 3.11](#)).

Operationally, the study adopts a specific baseline. Event evidence is unsafe-only and catalogue-grounded; Beta-posterior and [HPD](#) statistics control adaptive search; and canonical resource fronts require exact service of $\lceil 0.9 \times 64 \rceil = 58$ unsafe events. Every event retains multiple verified transfer–release alternatives before finite-bank maneuver–mass coverage boxes are formed. Dust-cloud uncertainty uses one Gaussian propagated by the seven-point Julier simplex, with one fixed spherical-equivalent dust tuple across all points.

1.4 Scope and Contributions

This work makes four primary contributions.

1. It defines a catalogue-grounded hazardous-event workflow that treats conjunction evidence as a reusable design input rather than as an ad hoc afterthought.
2. It formulates dust-enabled JCA as a deployer-architecture problem in which transfer feasibility, dust-mass logic, and mission reliability are evaluated together.
3. It develops a layered uncertainty-propagation and scoring framework, enabled by a custom multi-method Rust propagator using Battin’s numerically stable Encke formulation, that keeps short-horizon remediation studies tractable without forcing every stage to use the same fidelity.
4. It compares three constellation families (Walker, Flower, and Dual-Shell Walker-Delta) across six optimizer methods spanning differential evolution, dominance-based and ϵ -dominance evolutionary algorithms, geometry-aware survivor selection, and particle swarm, all under a shared mission evaluation protocol with common-random-number evidence control so that algorithmic and architectural conclusions rest on the same evidence.

Just as important are the study’s deliberate boundaries. The dissertation does not claim to present a flight-ready operational system. It does not attempt a full legal or governance treatment of remediation deployment, nor does it build an end-to-end post-intervention traffic-management architecture. Its contribution is methodological and architectural: it provides a defensible framework for evaluating when a dust-enabled JCA service might be effective, what physical and computational assumptions it requires, and how architecture choices interact with reliability and mission burden.

Those contributions are summarized in Table 1.2, which maps the scientific role of each contribution to the chapters in which it is developed most fully. The corresponding scope boundaries are stated in Table 1.3.

Table 1.2: Contribution map for the manuscript front matter and methods core.

Contribution area	Core idea	Primary chapter anchor
Hazardous-event evidence	Catalogue-grounded unsafe conjunction bank with reusable evaluation semantics	Chapter 3
Transfer and dust logic	Event-level transfer solving, intercepted-mass gating, and released-mass objective accounting	Chapter 3
Uncertainty-aware mission scoring	Beta/ HPD adaptive-search control and exact $q = 0.9$ finite-bank coverage-box Pareto fronts	Chapter 3
Architecture and optimizer comparison	Three structured constellation families under shared event evidence	Chapters 2 to 3

1.4.1 Evaluation Philosophy

Before turning to the literature review and methods chapter, one framing point matters. The study is intentionally conservative in how it judges architecture quality. Candidate constellations are evaluated on hazardous-event evidence rather than nominal coverage surrogates; Beta-posterior and [HPD](#) diagnostics govern adaptive search only; and canonical resource fronts require exact finite-bank event coverage with full transfer/release/Pc support. These are methodological commitments, not presentation choices.

Four method commitments carry that philosophy through the manuscript: catalogue-first hazardous-event evidence, conservative reliability semantics, separation between physical event gates and optimized mission objectives, and matched evidence across optimizer families. Together they prevent the architecture comparison from being driven by easy-event dilution, inconsistent stochastic samples, or hidden changes in mission value.

This conservative evaluation philosophy reflects the problem setting. A remediation architecture that appears attractive only because easy events dominate its score is not operationally persuasive. A design that looks efficient only because demanding feasible events are diluted inside an optimistic resource summary is likewise misleading. The dissertation therefore prefers event-bank realism, explicit failure bookkeeping, and mission-level summaries that keep hazardous-event burden visible rather than rewarding numerical optimism.

That philosophy also explains why the manuscript is written as an architecture study rather than as a maneuver study. The central object of interest is not a single dust intervention in isolation; it is a reusable service that must behave credibly across a structured hazardous-event population. The literature review and methods chapter that follow should therefore be read through that lens: each modeling choice is motivated by whether it helps evaluate a remediation architecture honestly, not merely by whether it simplifies one isolated event calculation.

1.5 Manuscript Organization

The remainder of the manuscript follows the structure of the problem itself. Chapter 2 reviews the relevant literature in constellation design, optimization under uncertainty, and debris remediation, and identifies the gap that motivates the present work. Chapter 3 then defines the methodology in detail, from event-bank construction and transfer solving through dust-mass accounting, uncertainty propagation, and mission-level scoring. Chapter 4 reports the evidence in two parts: Part A compares optimizer families under matched evidence and mission semantics, while Part B interprets what the supported fronts imply about deployer siting and access geometry. Chapter 5 explains the scope of those patterns, Chapter 6 states the dissertation-level conclusions warranted by the evidence, and Chapter 7 closes with a

Table 1.3: Primary scope boundaries for the manuscript front end and methods core.

Boundary area	In scope	Out of scope
Event evidence	Hazardous-event construction and reuse from public catalogue data	A universal model of all future debris traffic states
Service design	Deployer architecture, transfer logic, dust sizing, and mission scoring	Flight hardware, propulsion/propellant sizing, launch mass, fleet cost, or stakeholder-ready operations
Canister and dust scope	Fixed spherical-equivalent trajectory bodies and Pc-inverted dust population mass	Canister capacity, payload mesh, grain/deployment hardware design, or hardware-feasibility claims
Uncertainty treatment	Short-horizon uncertainty propagation relevant to hazardous-event servicing	Full long-horizon uncertainty closure for every post-intervention environmental effect
Architecture comparison	Three structured constellation families under shared evidence	Unstructured plane-constrained encodings, which are scoped but not evaluated here, and exhaustive search over every conceivable orbital architecture family

short synthesis of the contribution chain and future extensions.

That ordering is deliberate. Because the study is interdisciplinary, the reader needs both a systems picture and a precise methods picture before the results can be interpreted correctly. Chapters 1–3 therefore do more setup work than they would in a purely single-discipline project: they establish why the problem matters, what prior work leaves unresolved, and how the present study turns that gap into a defensible evaluation workflow before the back half interprets the resulting evidence.

Table 1.4: Reader roadmap for the manuscript.

Chapter	Primary reader question	Role in the manuscript argument
Chapter 1	Why does this problem matter, and what is being claimed?	Establishes the debris-remediation motivation, the architecture framing, and the study’s conservative evaluation philosophy.
Chapter 2	What does prior work solve, and what gap remains?	Synthesizes constellation, optimization, and remediation literatures to identify the precise unresolved problem.
Chapter 3	How is the proposed architecture actually evaluated?	Defines the methodology: event evidence, transfer logic, dust sizing, uncertainty propagation, and mission scoring.
Chapter 4	What evidence exists, and what is it allowed to support?	States the Part A evidence contract and the historical campaign record, and scopes the Part B architecture interpretation to a post-correction matrix.
Chapter 5	How far do the patterns travel?	Closes each research question against its evidence, and states the limits: no replicates, no strict-HF execution in the reported campaign, unprovenanced sizing threshold, no covariance realism.
Chapter 6	What is concluded, and what is held?	Names the four methodological contributions and the claims that remain held until a reconstructed campaign exists.
Chapter 7	What is the one-paragraph take-away?	Closes on the contribution chain and the next research layer.

Chapter 2

Literature Review

This chapter brings together constellation design, uncertainty-aware optimization, orbital-debris remediation, and the transfer theory needed to connect them. Each literature contributes a bounded piece of the problem. Walker’s structured-shell formulations [72, 73], Mortari’s Flower-constellation framework [49, 51, 59], search-based constellation studies such as Williams et al. [77] and Feringer and Spencer [21], and recent uncertainty-aware mission-design work [16, 23, 53] establish adjacent foundations. Remediation and conjunction-assessment studies establish intervention mechanisms and probabilistic risk logic [24, 30, 55, 60]. The narrow gap addressed here is their operational integration: a catalogue-conditioned natural conjunction bank, short-horizon dust-enabled JCA transfer and uncertainty evaluation, and deployer-architecture comparison under explicit reliability and resource semantics. This is an integration claim, not a claim that any individual component is new.

2.1 Constellation Optimization

Constellation design is usually approached from two complementary directions. One is geometric and analytic: structured families are studied because their symmetry yields interpretable coverage, revisit, and maintenance behavior. The other is algorithmic: the constellation is treated as a design vector and searched against one or more mission objectives. Both perspectives matter here because deployer constellations must remain physically interpretable

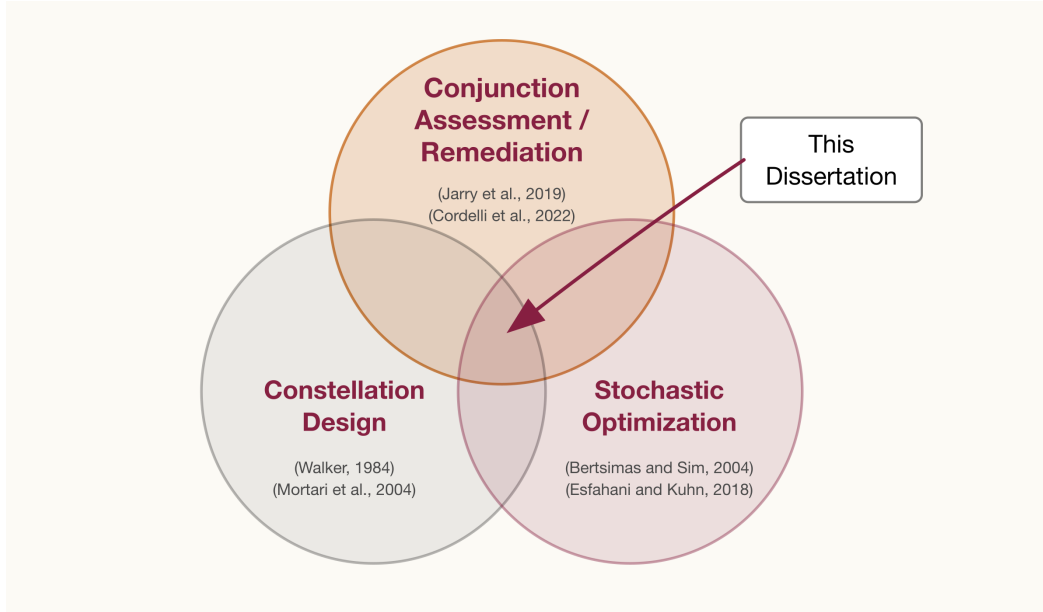


Figure 2.1: High-level synthesis of the study’s literature position. Constellation design, remediation or collision-avoidance interventions, and uncertainty-aware optimization are all established research areas; this work addresses the gap at their intersection by treating hazardous-event evidence, event-level physics, and architecture search as one integrated design problem.

while still being flexible enough to react to a highly anisotropic debris environment. For this dissertation, the useful question is what each family or optimizer adds to the design freedom needed for a hazardous-event response mission.

2.1.1 Analytic Families and Geometric Intuition

Walker [72] and its later formalization by Walker [73] remain the canonical starting point for structured circular-shell design. Those papers establish the logic of shared altitude, shared inclination, regular [Right Ascension of the Ascending Node \(RAAN\)](#) spacing, and controlled phasing in a form that is both interpretable and tunable. Later treatments preserve that insight [40]: a structured shell is useful because it is easy to specify and because its perturbation behavior and deployment logic remain readable. Real systems such as Galileo

and Starlink still show why that matters operationally [20, 63].

Analytic shell families remain attractive because they make the design variables readable, support fast first-order reasoning about access and maintenance, and often simplify deployment and station-keeping. Those same properties matter here, where the goal is not only to find a mathematically good constellation but also to explain why a given architecture performs well or poorly on a debris-remediation mission.

Mortari’s introduction of Flower constellations [51] and the later dynamical development by Mortari et al. and Ruggieri et al. [49, 59] generalize that structured viewpoint by replacing a circular-shell picture with a repeated closed path in a rotating frame. Their value in the present context is conceptual as much as practical: they show that symmetry and regular slotting need not be limited to circular Walker shells. Once the design problem is framed around access geometry rather than nominal global coverage alone, resonant or repeating-path families become natural comparison classes for debris-remediation staging.

Comparing Walker and Flower families is useful for the present study. Walker constellations provide the clearest structured-shell baseline, especially when the architecture question is largely about altitude and inclination selection. Flower constellations widen that structured family by allowing repeating-path behavior that can alter access geometry without abandoning regularity altogether. The literature establishes both as physically interpretable, but it does not answer which is better for hazardous-event response under uncertain transfer and dust-delivery constraints. That unresolved question motivates their side-by-side treatment here.

2.1.2 Search-Based Constellation Design

Search-based constellation design emerged in parallel with these analytic families. Ely’s early genetic-algorithm study [17] showed that even traditional constellation design problems

become awkward once discrete architecture choices and nonlinear performance metrics are optimized simultaneously. Williams et al. [77] then demonstrated that multiobjective evolutionary search could expose trade sets rather than one nominal optimum, while Ferringier and Spencer [21] applied related logic to realistic orbital-coverage trades in a way that connected better to engineering interpretation. Later work broadened this toolbox with diversity-preserving selection, parallel evaluation, semi-analytical coverage estimators, and mixed-integer formulations for more tightly constrained mission settings [12, 22, 42]. The common theme is that architecture design quickly becomes too nonlinear, too multimodal, or too combinatorial for closed-form optimization alone.

This is especially relevant when the constellation itself is not the only object being optimized. In many realistic mission settings, the usefulness of a constellation depends on an inner operational problem: sensor pointing, access timing, inter-satellite coordination, or trajectory planning. Once one of those inner problems becomes part of the evaluation loop, architecture design inherits the nonsmoothness and multimodality of that operational layer. The present study is a direct example. A constellation is not valuable simply because of where its satellites are; it is valuable because those staging locations change transfer feasibility, lead times, and dust-delivery geometry for hazardous events.

For the present problem, search-based methods are more than convenient. They accommodate mixed variable types, tolerate discontinuous feasibility boundaries, and expose tradeoffs among multiple objectives in a single run. They also create a comparison challenge: when several algorithm families are available and objective evaluation is itself uncertain, it becomes difficult to tell whether one optimizer is genuinely better or simply lucky under a particular evidence sample. That challenge leads directly to the next section of the review.

2.1.3 Constellation Design Under Uncertainty

Much of the constellation literature still shares a nominal-performance emphasis. Even when the objective is multiobjective, the problem is often posed in terms of deterministic revisit or coverage proxies rather than uncertain event feasibility. The more recent papers that depart from that pattern are especially important. Gondelach and Linares [23] show how uncertainty in mission demand can materially shift constellation choice. Eddy and Kochenderfer [16] treat task uncertainty explicitly in constellation planning rather than as a downstream afterthought. Norman and Rivest [53] likewise show that uncertainty-aware constellation design can change both the ranking of architectures and the meaning of “good” performance. They do not study dust-enabled JCA, but they do make the point this dissertation needs: a deployer constellation for an uncertain service mission has to be judged on uncertain service outcomes, not on nominal geometry alone.

The literature therefore motivates a design stance rather than a single preferred family. Structured families such as Walker and Flower provide interpretable baselines, useful seeds, and geometry that can be explained physically. Less structured or irregular encodings remain useful as stress tests because the debris environment is not itself organized as a single neat shell. The dissertation works within the first view: it studies three structured families that span single-shell, stacked-shell, and resonant repeating-groundtrack geometry, on the reasoning that they may provide most of the required access geometry. The complementary question of whether an unstructured encoding leaves the structured baselines meaningful performance on the table is scoped but not evaluated here, so no evidence in this work bounds it in either direction.

2.1.4 Architecture Realism and Design-Space Parameterization

Another recurring issue in the constellation literature is the tension between expressive design spaces and interpretable architectures. Highly expressive decision vectors can search a very large family of possible constellations, but they often become difficult to interpret physically and difficult to compare fairly across families. Highly structured parameterizations are easier to understand and often easier to deploy operationally, but they may hide real advantages available to more irregular architectures. This tradeoff matters for the present study because the goal is not simply to discover one mathematically attractive orbit set; it is to understand how architecture structure itself affects mission behavior.

The literature addresses this tradeoff in several ways. One strategy fixes the shell structure and optimizes only a small set of orbital parameters, which favors interpretability and computational tractability. Another allows multiple structured families and compares them under shared controls. A third introduces a bounded plane-constrained arbitrary family as a stress test for how much performance the structured baselines leave unrealized. This study follows the second strategy: three structured families are compared under shared controls, spanning single-shell, stacked-shell, and resonant repeating-groundtrack geometry. The third strategy's plane-constrained arbitrary family was scoped as a stress test but is not implemented in the reported evaluation stack, so the structured-versus-unstructured reading is identified as future work rather than answered here.

This parameterization issue is also tied to fairness in optimizer comparison. Optimizers do not all respond equally to changes in design-space dimension, variable discreteness, or geometry-induced multimodality. A fair comparison therefore requires the architecture encodings themselves to be controlled, family by family, rather than merged into one giant mixed search space. That is why the manuscript compares optimizers within each family and

then compares the families on the basis of their resulting trade spaces.

Table 2.1: Representative themes in the constellation-design literature and their relevance to this study.

Literature theme	Typical emphasis	Relevance here
Analytic shells	Structured geometry, phasing, coverage intuition	Supplies interpretable architecture baselines and low-dimensional encodings
Search-based architecture design	Multiobjective trade-offs, mixed-variable search, Pareto fronts	Supplies the computational paradigm for constellation-family comparison
Uncertainty-aware mission design	Stochastic task conditions, robust architecture choice, uncertain targets	Motivates evaluating constellations on hazardous-event evidence rather than nominal coverage alone

Table 2.2: Architecture-level contrast among the constellation families considered in this study. The first three families are evaluated; the plane-constrained Arbitrary family is scoped only.

Family	Primary strength	Primary limitation	Study role
Walker	Clear shell geometry, low-dimensional interpretation, natural structured baseline	Restricted to circular-shell staging behavior	Reference architecture and seed-friendly baseline
Flower	Structured noncircular repeating-path access geometry	More specialized parameterization and interpretation	Tests whether resonant structure improves response geometry
Dual-Shell Walker-Delta	Two independent structured shells for altitude diversity under satellite-count bounds matched to the Walker family	Larger search space than single Walker	Tests whether a second shell provides meaningful access improvement
Plane-constrained Arbitrary	Fixed 3×5 plane layout with per-plane altitude, inclination, RAAN, and phase freedom	Reduced interpretability, larger search burden	Scoped as a bounded stress test; not evaluated in this work

2.2 Stochastic Optimization

Once uncertainty is admitted as a central part of the design problem, the optimization literature offers several modeling choices. Expectation-based stochastic programming optimizes average performance. Chance-constrained methods enforce a specified probability of constraint satisfaction. Coherent risk measures, including [Conditional Value-at-Risk \(CVaR\)](#), emphasize tail behavior. Robust and distributionally robust formulations protect against uncertainty sets or ambiguity sets rather than one fixed nominal distribution [3, 6, 8, 18, 57, 62, 76]. For this dissertation, the relevant question is which of those frameworks map cleanly onto space-mission design and which still do not.

2.2.1 Classical Formulations for Uncertain Decisions

Each of these formulations emphasizes a different decision philosophy. Expectation-based models are natural when average performance is the main objective and when large numbers of scenarios can reasonably be regarded as samples from a stable process. Chance-constrained models are more natural when reliability itself is the core requirement, such as requiring a specified minimum probability of successful service. Risk-measure models are attractive when tail events matter disproportionately, which is common in safety-critical engineering. Robust and distributionally robust models become important when the analyst does not trust one precise probability law or wishes to protect against model misspecification.

The debris-remediation design problem has features of several of these formulations at once. It contains binary event-level success or failure, uncertain geometry, uncertain propagation, and mission-level tradeoffs that should be robust to sparse evidence. This makes it a poor fit for a single naive average-performance metric. A design that succeeds spectacularly on a few easy events but fails frequently on the rest is not useful operationally, even if its mean resource

expenditure appears low. Likewise, a design that achieves good expected performance by accepting a long tail of severe resource demand may still be unattractive as a reusable service architecture. The literature therefore suggests that reliability-aware and tail-aware semantics are more defensible than pure expectation-based scoring for this problem class.

2.2.2 Noisy Multiobjective Search

Space applications have adopted uncertainty-aware optimization in different ways. Hou, Vasile, and Hou [27] show that robust design choices in space-mission problems can change materially once epistemic and scenario uncertainty are modeled explicitly rather than buried inside deterministic point estimates. In conjunction assessment and collision avoidance, Laguel et al. [39], Gonzalo and Colombo [24], and Ryu et al. [60] reach a similar conclusion from the risk side: once the decision boundary itself is defined by uncertain geometry and state knowledge, deterministic optimization logic can become misleading.

At the same time, the optimizer families used in practice are often randomized or population-based rather than purely mathematical-programming solvers. Differential evolution, dominance-based evolutionary algorithms, epsilon-dominance variants, swarm methods, indicator-based EMOAs, geometry-aware EMOAs, and simulated annealing are attractive because the design landscape is mixed, event-driven, and only partially smooth [4, 10, 14, 35, 41, 54, 64]; this is a survey of what the field uses, and the six-method subset carried into the reported matrix is stated in [Section 3.8](#). Under noisy or scenario-based evaluation, naive comparisons can confuse algorithm behavior with sampling noise. That is why shared evidence banks, common random numbers, matched evaluation budgets, and explicit service-reliability semantics are methodological necessities rather than incidental details.

The present study inherits the same difficulty. Its objective values are not analytic expressions. They are the result of screening events, solving transfers, propagating uncertainty, and evaluating dust-delivery logic under finite evidence. A fair optimizer comparison is therefore impossible unless the evidence itself is controlled. The literature on noisy evolutionary optimization and paired experimental design does not remove that burden, but it makes clear what the comparison protocol must accomplish: each optimizer must be exposed to the same evidence basis, the same resource semantics, and the same reliability semantics, otherwise observed pairwise differences cannot be attributed to search behavior.

2.2.3 Methodological Takeaways

This study does not adopt a standard textbook formulation verbatim. Unsafe scenarios are screened into a catalogue-conditioned bank; Beta/HPD state is a search controller only; and canonical reporting builds nondominated maneuver–released-mass boxes that must cover $[0.9 \times 64] = 58$ events. CVaR remains useful background and a provisional scalar diagnostic, but exact event-front coverage is the publication authority. The result sits in the same conceptual family as robust and reliability-aware optimization, with semantics chosen for a reusable [JCA](#) architecture.

This literature also informs the optimizer-comparison design in the manuscript. The goal is to characterize descriptive differences among optimizer methods once event evidence, uncertainty semantics, mission constraints, and exact budgets are fixed under a single realization per cell. The comparison therefore spans multiple search families under matched evidence and scoring rules, but it does not rank, qualify, recommend, or promote an optimizer.

2.2.4 Uncertainty Propagation in Orbital Mechanics

The optimization literature alone is not enough to motivate the methodology adopted here. The design problem also depends on how uncertainty is propagated through orbital dynamics. Classical covariance analysis remains valuable because it is fast and interpretable, but it can under-represent the distortion that accumulates when uncertain states are propagated through nonlinear dynamics over multi-hour horizons [78]. This issue is particularly important in conjunction analysis and short-horizon remediation because the geometry of the encounter plane is sensitive to both mean-state drift and covariance deformation.

Recent orbital uncertainty-propagation studies emphasize that nonlinear or hybrid methods can outperform purely linearized approaches when the goal is to preserve conjunction-relevant geometry over short to moderate horizons [34]. Sigma-point approaches, semi-analytical methods, and Gaussian-mixture approximations each address different parts of this problem: sigma-point methods capture local nonlinear deformation efficiently, semi-analytical methods retain dynamical structure, and Gaussian-mixture methods allow one uncertain state to branch into several locally coherent components when a single Gaussian is no longer representative [32, 37].

For the present study, the implication is direct. If dust-enabled JCA is to be judged on whether the dust cloud overlaps the uncertain target state at the right time and geometry, then the uncertainty model has to be part of the architecture-evaluation logic, not just a plotting aid. On the propagation side, Battin’s $f(q)$ reformulation of the Encke gravity correction [2] eliminates the catastrophic cancellation that arises when the perturbation is small relative to the reference orbit, which is directly relevant to the short-horizon Encke integration used in the numerical dust-propagation model adopted here. The methodology chapter therefore treats dust-cloud uncertainty propagation as part of the main workflow

rather than as a secondary sensitivity analysis.

2.2.5 Gaussian-Mixture and Encounter-Plane Methods

The literature on nonlinear uncertainty propagation is especially relevant once the state uncertainty is no longer well described by one Gaussian ellipsoid. This is a familiar difficulty in orbital mechanics: state transition matrices propagate only the first-order covariance geometry, while nonlinear dynamics can stretch, skew, and even split the effective support of the probability density over time. Entropy-based and hybrid uncertainty-propagation studies therefore highlight the limitations of purely linearized reasoning when nonlinear flow structure matters [15, 78].

Gaussian-mixture models offer one principled response to that difficulty. Rather than representing uncertainty by one distribution

$$\mathcal{N}(\boldsymbol{\mu}, \mathbf{P}), \quad (2.1)$$

the state can be represented as a weighted mixture

$$p(\mathbf{x}) = \sum_{k=1}^K w_k \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \mathbf{P}_k), \quad (2.2)$$

where the weights satisfy $w_k > 0$ and $\sum_k w_k = 1$ [26, 70]. Each component can then be propagated locally, allowing the overall distribution to capture multimodality or strong anisotropic deformation more accurately than a single Gaussian. For orbital uncertainty problems, that is especially useful when the relevant question is how the support of the uncertainty distribution deforms relative to a future geometric event.

The next question is how to construct the mixture itself. Moment-matching approaches based

on Gauss-Hermite quadrature provide a systematic way to split one Gaussian into several components while preserving the original low-order moments [43]. More recent work by Kulik and LeGrand shows that the splitting direction itself can be selected using nonlinearity-aware criteria rather than simple maximum-variance rules, which improves accuracy when the underlying dynamics are strongly curved or scale-sensitive [37, 38]. These methods are particularly relevant here because the short-horizon dust-cloud problem is exactly a case in which the shape of the uncertainty distribution matters for the decision outcome.

For conjunction and interception problems, the encounter plane plays a special role. The geometric question is often not whether two uncertain objects occupy exactly the same state in six-dimensional phase space, but whether their relative projected density places mass inside a finite hard-body disk in the plane normal to the relative velocity. A generic finite-region form is

$$P_{\text{HB}} = \int_{\|\mathbf{r}^\perp\| \leq R_{\text{HB}}} p_{\text{rel}}(\mathbf{r}^\perp) d\mathbf{r}^\perp, \quad (2.3)$$

where R_{HB} is the hard-body radius and p_{rel} is the projected relative density. This perspective is directly relevant to a dust-based intervention because required release mass depends on finite overlap over likely target locations, not merely on nominal point coincidence.

This literature ties stochastic optimization to the event-level physics the dissertation actually uses. Nonlinear uncertainty propagation motivates moving beyond linear covariance transport; Gaussian-mixture methods motivate representing short-horizon deformation more flexibly; and encounter-plane reasoning motivates evaluating success through uncertainty overlap rather than deterministic point coincidence. Chapter 3 turns those ideas into the workflow used for dust-enabled JCA.

The dissertation implements that literature through a production (K=1) Gaussian propagated by the seven-point Julier minimal-skew simplex. The compiled authority retains `maxvar`, a

scaled-TOF α schedule, and rank 1 as dormant split mechanics, but those controls form no child components at (K=1). Historical (K=3), Van der Merwe-13, adaptive- α , Monte Carlo, and alternate-rule studies are counterfactual context rather than the reported production profile.

2.2.6 Transfer Theory as the Physics Bridge

Lambert targeting supplies the two-point boundary-value bridge between a deployer state and a selected intercept epoch; modern universal-variable formulations make multi-revolution branch enumeration practical [2, 29]. That two-body core is not, by itself, an accepted mission transfer. The operational chain must close the endpoint under the selected perturbation model and preserve the physical target. The runtime also computes a rendezvous-arrival diagnostic in inertial coordinates,

$$\Delta \mathbf{v}_{\text{arr}} = \mathbf{v}_{\text{target}} - \mathbf{v}_{\text{payload}},$$

but the current objective transfer burden remains phasing plus transfer and does not include this arrival correction. Final canonical acceptance is designed to require the HF validation and post-HF endpoint residual described in Chapter 3; Section 5.4 records which acceptance lane actually executed in the reported campaign.

Secular J_2 theory provides the fast closure and phasing model, while numerical propagation is reserved for acceptance-sensitive and post-release quantities [67]. The same separation appears in uncertainty transport: sigma-point or mixture methods preserve nonlinear short-horizon geometry at lower cost than dense Monte Carlo, then a 2×3 encounter-plane projection converts relative state uncertainty into the finite hard-body-disk calculation [32, 78]. Finally, constellation-architecture studies and multiobjective search provide the outer decision layer

[21, 72, 77]. The bridge is therefore a chain of established theories, not a new transfer or uncertainty-propagation theorem.

Table 2.3: Representative uncertainty-propagation themes and their relevance to the present problem.

Method class	Core idea	Main strength	Relevance here
Linear covariance propagation	State-transition transport of first and second moments	Fast and interpretable	Useful for screening intuition but limited under nonlinear deformation
Sigma-point methods	Deterministic nonlinear sampling around the mean	Captures local curvature efficiently	Supports short-horizon nonlinear transport without Monte Carlo cost
Gaussian-mixture methods	Represent one uncertain state as several coordinated Gaussian components	Captures multimodality and strong anisotropy better than one Gaussian	Relevant for dust-cloud evolution and encounter-plane coverage
Encounter-plane formulations	Judge success by projected overlap at the critical geometry	Ties uncertainty directly to collision or intercept geometry	Directly relevant to dust-delivery success semantics

Table 2.4: Representative optimization themes and their role in the methodology used here.

Theme	Typical question	Role in this study
Stochastic programming	What is the expected or probabilistically feasible decision?	Provides the conceptual basis for reliability-aware mission scoring
Robust / DRO methods	How should designs behave under uncertainty or model misspecification?	Motivates conservative semantics and sensitivity studies
Noisy multiobjective search	How should Pareto-search algorithms be compared under noisy evaluation?	Motivates shared evidence, CRN control, and actual-work comparison rather than raw final-row totals

2.3 Orbital Debris & JCA

Orbital debris is an environmental and statistical problem, not just a series of isolated conjunctions. Recent agency reporting continues to emphasize the growth of the resident object population, the increasing congestion of operational altitude bands, and the system-level consequences of fragmentation events [19, 47]. Environment-evolution studies likewise show that collision risk, fragmentation, and long-horizon sustainability are coupled phenomena rather than separable concerns [25, 52]. That matters here because this work is motivated by endogenous risk: the debris population itself can generate future debris if high-consequence debris-debris conjunctions are allowed to pass unmanaged.

2.3.1 Debris Environment, Persistence, and Sustainability

Several themes recur across current debris-environment studies. First, the most operationally important orbital bands are not uniformly populated. Certain altitude-inclination corridors, especially those associated with Earth observation, communications, and legacy mission infrastructure, contain disproportionately large numbers of catalogued objects. Second, fragmentation events do not simply add independent hazards; they reshape the future state of the environment by creating persistent fragment populations that then participate in later conjunctions. Third, sustainability and traffic growth cannot be separated from collision-risk analysis because the long-horizon viability of [LEO](#) depends on the same fragmentation and persistence mechanisms that matter for near-term conjunction risk [\[25, 52, 71\]](#).

These themes directly motivate the manuscript’s event-selection logic. If collision risk and sustainability are concentrated in particular regions of orbital-element space, then a reusable mitigation architecture should not be designed around generic or isotropic hypothetical events. It should be designed around hazardous events drawn from the structured, anisotropic environment that actually exists. This is one of the core methodological departures of this work from many more generic optimization studies.

2.3.2 Remediation and Collision-Avoidance Concepts

The remediation literature contains several relevant strands. Jarry et al. [\[30\]](#) study plume- or dust-mediated perturbation ideas aimed at collision prevention rather than traditional capture. Cordelli et al. [\[11\]](#) and Phipps et al. [\[55\]](#) treat laser-based momentum transfer as contactless deflection mechanisms. The [Double Asteroid Redirection Test \(DART\)](#) mission [\[9\]](#) demonstrated that momentum transfer from kinetic impact can exceed the direct impulse through ejecta enhancement, with reported κ values ranging roughly from 2.2 to 4.9 depending

on assumed bulk density and an equal-density nominal estimate near 3.6. That result does not validate dispersed dust-cloud delivery. This dissertation uses $\kappa = 1$ as the reportable baseline only within the current monotone direct-momentum model, pending evidence that real dust interaction cannot produce effective transfer below unity. Values above unity are parametric enhancement coordinates, not conservative or dust-validated values. Another strand studies collision avoidance more directly, including debris-debris avoidance logic and probabilistic assessment of hazardous conjunctions [7, 24, 44, 60]. Across those studies, the recurring result is that small perturbations, if delivered at the right time and geometry, can materially change the outcome of an otherwise hazardous encounter.

What this literature less often addresses is how the intervention should be staged as a reusable service. A one-time or one-target analysis can treat staging and access as secondary concerns. A service architecture cannot. It has to confront how deployers are placed, what geometries they can reach repeatedly, how long they need to prepare an intercept, how much resource margin they need to carry, and how event-level feasibility should be aggregated into system-level performance. The present study focuses on that shift in viewpoint.

2.3.3 Architecture-Level Remediation Studies

Architecture-level remediation studies are especially important for the present study because they bridge the gap between mechanism and service design. Wagner et al. [71] examine remediation effectiveness at the environmental level rather than at the level of one isolated maneuver. Muciaccia et al. [52] tie congestion and long-term capacity directly to intervention policy. These papers make clear that intervention value depends not only on whether one object can be influenced, but on how interventions scale across a population of hazardous opportunities. This is precisely the level at which constellation design becomes unavoidable.

Recent distributed or coordinated remediation concepts reinforce the same systems perspective. Rogers et al. [58] make the architecture issue explicit by studying a distributed laser concept in which timing, geometry, and platform coordination are inseparable from the actuation physics. The present work differs in mechanism and methodology, but the shared lesson is that a physically plausible intervention is not enough; the surrounding architecture must bring that mechanism to the right events often enough and cheaply enough to matter.

Table 2.5: Representative remediation and collision-avoidance concept classes relevant to this study.

Concept class	Typical actuation logic	Typical strength	Typical limitation
Dust or plume interventions	Momentum exchange or drag augmentation through particulate interaction	Potentially low-impulse and time-sensitive intervention geometry	Requires careful uncertainty and density modeling
Ground-based or distributed laser systems	Photon-pressure or ablation-driven perturbation	Contactless engagement and strong architecture-level scheduling problem	Sensitive to engagement windows, infrastructure, and operations
Long-horizon removal concepts	Deorbit or capture over longer timescales	Strong environmental-risk reduction potential	Less aligned with short-warning conjunction retirement

2.3.4 Just-in-Time Collision Avoidance as the Integrating Concept

Just-in-time collision avoidance is the operational concept that ties this chapter together. A JCA action is not intended to deorbit the threatening object immediately, nor is it simply an abstract risk metric. It is a time-sensitive intervention delivered after a hazardous encounter has been identified but before closest approach, with the aim of producing enough trajectory

separation to retire the event [24, 30]. For non-maneuverable debris objects, that makes JCA especially attractive as a complement to traditional spacecraft avoidance.

The dust-based version of JCA adopted here is especially useful as a research vehicle because it lays out the full chain of decisions. The event must first be deemed hazardous. A deployer must then be able to reach a useful staging geometry in time. The dust release must be sized against an uncertain encounter geometry. Finally, the event-level outcomes must be aggregated into system-level architecture judgments. Those layers turn a simple intervention idea into a full architecture-design problem, which is why this work uses it to connect debris evidence, uncertainty propagation, and optimization in a single study.

2.3.5 Literature Gap

These strands leave a specific opening for this dissertation. Debris-environment and capacity studies motivate catalogue-grounded evidence [19, 47, 52]; JCA, laser, plume, and dust concepts show that small perturbations can be meaningful when timing and geometry are favorable [7, 11, 30, 55]; constellation-design and stochastic-optimization literatures supply tools for architecture search under uncertainty. What is missing is their joint use as one reusable service-evaluation workflow with a catalogue-conditioned natural conjunction bank, explicit transfer closure, finite encounter-plane mass conversion, and fair architecture search.

The gap addressed here is narrow and operational: the cited near-neighbors do not combine catalogue-conditioned natural-pair conjunction screening, short-horizon dust-enabled JCA uncertainty transport, explicit Lambert/ J_2 transfer closure, finite encounter-plane mass physics, and optimizer-family comparison under shared reliability semantics. Chapter 3 then turns each element into an operational step without claiming novelty for the component theories themselves.

Table 2.6: Near-neighbor gap mapping for the dissertation workflow.

Near-neighbor literature	Evidence or uncertainty focus	Architecture / actuation focus	Remaining gap addressed here
Uncertainty-aware constellation design [16, 23, 53]	Stochastic demand or task uncertainty	Constellation design under uncertainty	Does not study dust-enabled debris-debris JCA with event-level transfer and mass physics
Dust, plume, and laser JCA concepts [7, 11, 30, 55]	Time-sensitive perturbation mechanisms	Single-event or mechanism-level intervention	Does not compare deployer-constellation families under shared unsafe-event evidence
Distributed remediation architectures [52, 58, 71]	Environmental or platform-level remediation value	Multi-platform or policy-scale intervention framing	Does not combine dust-cloud uncertainty transport with matched optimizer-family architecture search
Probabilistic conjunction and collision-avoidance studies [24, 39, 60]	Risk under uncertain encounter geometry	Avoidance or assessment logic	Does not turn catalogue-grounded unsafe-event cohorts into reusable deployer-service objectives
Nonlinear uncertainty propagation and mixtures [26, 37, 70]	Short-horizon distribution deformation	Propagation method rather than service architecture	Does not evaluate constellation siting or optimizer behavior under mission-level reliability semantics
Noisy multiobjective optimization [5, 13]	Algorithm behavior under uncertain objectives	Generic optimizer methodology	Does not supply the mission physics, natural conjunction bank, or deployer-architecture encoding

Chapter 3

Problem Formulation and Methodology (Dust JCA)

This chapter defines the methodology used throughout this work. It does not document every implementation detail. The goal is to state the scientific workflow clearly enough for the reader to see what is evaluated, on what evidence, under which physical assumptions, and with which mission-level semantics. The chapter moves from the overall architecture of the study to the detailed treatment of transfer solving, dust mass accounting, uncertainty propagation, constellation encoding, and optimizer comparison.

Candidate constellations are evaluated on the catalogue-conditioned natural conjunction bank; each scenario is screened through transfer feasibility, dust-delivery physics, and uncertainty-aware mass accounting; and the resulting outcomes are aggregated into mission-level reliability and resource metrics for optimizer comparison. The remaining sections unpack each stage of that workflow.

Held pending independent model authority. Current quantitative attainment is held, and the descriptive replication policy prohibits global or per-family optimizer-winner claims. Atmosphere decision D1 selects the [Jacchia-Bowman 2008 thermospheric density model \(JB2008\)](#) in its compiled fitted-kernel form; the versioned synthetic thermosphere proxy is not used, and drag-qualified

claims are conditioned on the sealed driver scenario described in [Section 3.5.1](#). The probability-qualified released-mass claim remains held pending immutable-receipt sensitivity and exact-commit/archive validation; D2 now defines finite physical-grain mass probability, while historical effective-packet arithmetic is non-authoritative. Momentum enhancement and operational risk-reduction interpretations remain parametric or out of scope until their respective authority gates close.

ECF concept traceability and limits. The *Fitzgerald ECF 23 Summary* is concept authority for coupling hazardous-event access, transfer opportunity, dust interception, and nondominated architecture tradeoffs; it is not numerical-solver or validation authority. This dissertation’s public-catalogue-conditioned natural conjunction bank differs from the literal proposal pseudocatalog and must be identified as such in any cross-document claim. The production baseline permits a minimum lead time of 6 h, whereas the concept summary describes roughly half-day timing; a matched 6 h/12 h sensitivity is required before any proposal-faithful timing claim. Two configured release fractions and a solved release-control vector provide partial cloud adaptability, but fixed grain and dispersion parameters mean the study does not jointly optimize all cloud designs described in the concept. Each modeled unsafe event is one debris pair; a breakup-cloud multi-fragment event is outside scope. Response time and plane count remain design metadata or constraints, not independent publication-front axes, whose physical coordinates are executed maneuver and released mass. Finally, trajectory feasibility and architecture analysis provide no flight-hardware or technology-readiness-level validation.

3.1 Dust JCA Problem Formulation

In operational terms, the task is to evaluate deployer-constellation designs intended to retire unsafe conjunction opportunities under strict physics and timing constraints while minimizing mission resource demand. Let x denote a candidate constellation design, and let Ω_{unsafe} denote the catalogue-conditioned natural conjunction bank.

Evaluating x over Ω_{unsafe} yields:

- event-level transfer/intercept feasibility outcomes under active solver limits;
- event-level released dust mass requirements from encounter-plane coverage conversion;
- event-level deterministic intercepted-mass validity under the active strict event-feasibility policy;
- mission-level reliability/resource summaries for optimization and reporting.

The final mission objective pair comprises F_1 for deployer maneuver burden and F_2 for released dust mass burden. Their raw inputs remain distinct and are reconstructed from event-level nondominated fronts. For a configured coverage fraction q , the canonical objective is the nondominated set of resource boxes (d, M) that can service at least $k = \lceil qN \rceil$ members of the fixed N -event bank with per-event executed maneuver no greater than d and released dust mass no greater than M . The production bank has $N = 64$ and $q = 0.9$, hence $k = 58$. Beta/HPD state is an operational search controller only; canonical coverage is an exact finite-bank count. Deterministic intercepted-mass sufficiency is not optimized as a soft preference; it remains a strict event-level acceptance gate.

3.2 Mission Overview

The end-to-end workflow begins with a reusable catalogue-conditioned natural conjunction bank. A candidate constellation is then decoded into deployer orbits, screened against shared scenario evidence, and passed through the transfer solver and dust-mass logic. Event-level outcomes are then aggregated into mission-level reliability and resource scores so that optimizer comparisons rest on the same evidence and the same operational semantics.

Under routine cataloging and screening operations, endogenous debris threats are often identifiable days before conjunction. Let $P_c(t)$ denote the current estimate of collision probability for a predicted conjunction at epoch t_0 . Following initial detection near $t_0 - 7$ days, an automated monitor ingests refined tracking updates and recomputes P_c as state uncertainties contract.

A notional two-threshold triage policy helps balance responsiveness against unnecessary actions: (i) an advisory level, for example $P_c \geq P_c^{\text{adv}}$, which triggers active tasking and reachability checks; and (ii) a commit level, for example $P_c \geq P_c^{\text{com}}$, above which the system prepares to execute remediation. In baseline simulations, execution is ultimately driven by minimum lead-time and feasibility constraints, so the 1% commit level is used here as timing context. If by $t_0 - 48$ h the updated probability remains above P_c^{com} , the system commits subject to required lead time $t_\ell \geq t_\ell^*$ between dust intercept and closest approach. This lead time keeps the required effective ΔV in a feasible range and leaves margin for post-intercept state confirmation.

For each conjunction pair, one object is designated as the primary target, that is, the object to be nudged. The decision balances expected risk reduction, reachable geometry, and the operational burden of delivering the intervention. A deployer is then selected through a staged transfer workflow: candidate pairs are screened with fast heuristics, phasing and

multi-revolution Lambert branches generate a transfer Pareto set, and every replay-verified transfer retained for canonical evaluation is crossed with every configured release fraction. The catalogue intercept epoch is fixed; final evaluation does not retime it or collapse the transfer set to one preselected candidate.

For the short horizons relevant to [JCA](#) (hours to two days), simulations allocate fidelity by body role and epoch. Catalogue scenario construction uses the tractable analytical/ J_2 -class lane. Lambert and J_2 machinery provide transfer seeds, while production transfer-front candidate generation/verification and canonical replay require strict HF authority; the $E_0 \rightarrow L$ transfer body uses a gravity-only HF context. The canister coast and released-dust flight are separate HF arcs with separate immutable ballistic tuples. The dust arc uses fixed 5×5 spherical-harmonic gravity, drag, [Solar Radiation Pressure \(SRP\)](#), Sun, and Moon terms with interpolated ephemerides; its configured physical window is at most two hours. [Simplified General Perturbations 4 \(SGP4\)](#) is used only to propagate the pinned [OMM](#) captures to the common source anchor; no [Two-Line Element set \(TLE\)](#) ingestion path exists.

Dust deployment is adaptive to the remediation task through the solved release-control vector, configured release fraction, and propagated uncertainty state. Across all modes, the mission planner enforces a minimum lead time between dust intercept and conjunction so that the induced orbital change meaningfully separates the target from the conjunction funnel before closest approach.

The modeled canister is a trajectory-reference body, not a hardware sizing model. It uses the fixed spherical-equivalent tuple $A/m = 0.01 \text{ m}^2/\text{kg}$, $C_D = 2.2$, and $C_R = 1.3$ on the HF coast from transfer burn epoch L to release epoch R . No mesh, dry mass, payload capacity, or dust-mass feedback is modeled. At R , one solved ECI release-control vector is applied exactly once; the released dust then starts from that post-control state. Direct release, dust-spacecraft recontact scoring, hardware feasibility, and operational re-tasking

logic are outside the implemented study workflow.

Maneuver ΔV is therefore an executed kinematic impulse burden under an external-impulse assumption. The model does not convert ΔV into propellant, wet mass, launch mass, or monetary cost. This deliberate “resources unconstrained except maneuver and dust” scope asks how much impulse and dust the concept requires before imposing a particular spacecraft or budget architecture.

After dust-primary intercept, a bounded verify-and-guard concept would update the primary state, rescreen the former conjunction, check nearby protected objects, and assess dust dilution or decay. The implemented study records these as safeguards and limitations; it does not simulate closed-loop tracking, a second release, deployer trim, or fallback escalation.

For clarity, the nominal response timeline is:

- $t_0 - 7$ d: Detection; begin intensive monitoring and advisory tasking.
- $t_0 - 72$ h: Candidate deployers scored on reachability and expected P_c reduction; phasing plans generated.
- $t_0 - 48$ h: Commit if $P_c \geq P_c^{\text{com}}$ under refined tracking. Lock fixed dust model, release-fraction grid, and control/coverage requirements.
- $t_0 - 24$ h: Refine the selected intercept geometry and verify guardrails under the latest tracking update.
- $t_0 - t_\ell$: Eject the modeled canister and execute dust release to achieve intercept. Begin verify-and-guard assessment.
- t_0 : Former conjunction epoch; screen for residual risk and for secondary hazardous geometries created by the intervention.

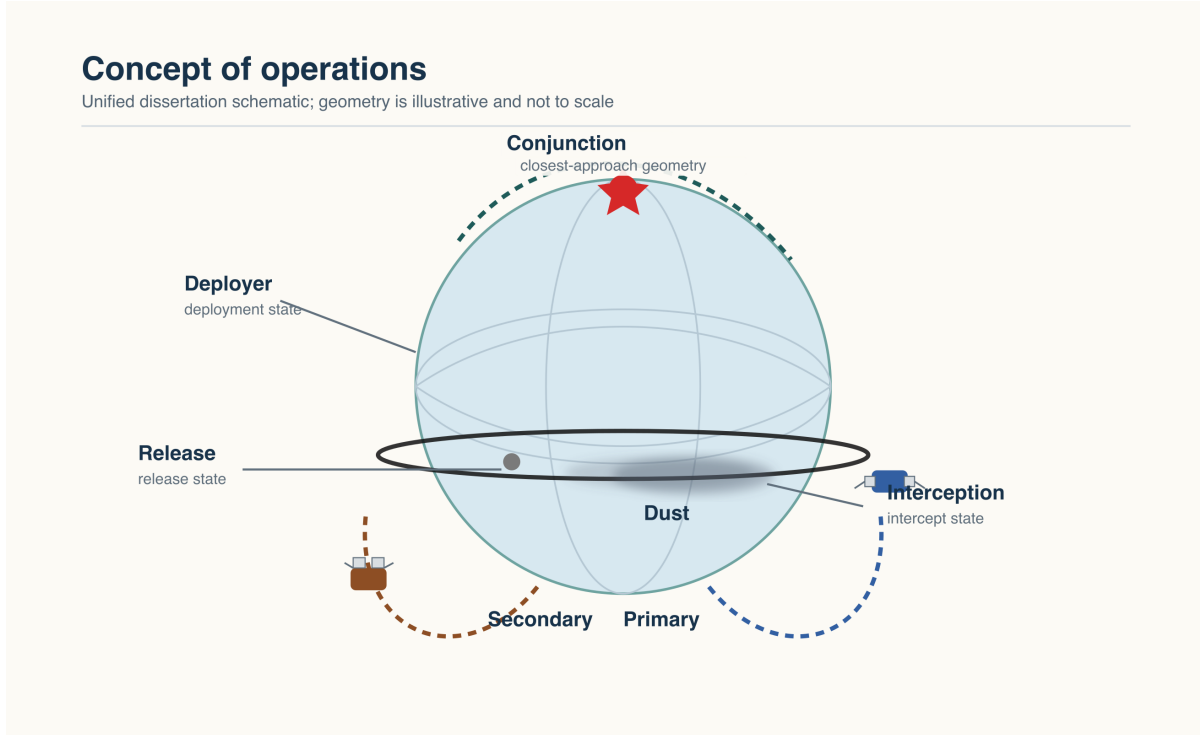


Figure 3.1: Concept of operations schematic for dust-based JCA. Representative deployer trajectories stage a dust release ahead of closest approach so the cloud intercepts the target before conjunction and imparts a small separating perturbation. Geometry is illustrative and not to scale.

Table 3.1: Operational triage thresholds and timing parameters (illustrative policy context; numerical defaults are specified in the run configuration as detailed in [Sections 3.10](#) and [3.12](#)).

Parameter	Meaning	Baseline value
P_c^{adv}	Advisory threshold to begin tasking	10^{-4} (representative; adjusted per scenario)
P_c^{com}	Commit threshold for remediation	0.01 (1%)
t_{BP}	Decision/intercept offset reference	selected relative to the conjunction epoch within the active scenario
t_ℓ^*	Minimum lead time to CA	baseline configuration enforces at least 6 h
d_{miss}	Max release/intercept miss distance	transfer tolerance is 25 m

The threshold values in [Table 3.1](#) provide policy context only; quantitative acceptance is

governed by the constraints in [Sections 3.3, 3.10 and 3.12](#).

3.2.1 Shared Hazardous-Event Evidence

A central design choice in this work is that candidate constellations are not evaluated on independently generated event sets. They are evaluated on a shared, catalogue-conditioned natural conjunction bank. In practice, each study run begins with public catalogue objects, constructs and screens candidate conjunctions, retains unsafe cases under the study policy, and exposes all optimizers and constellation families to that same bank. The bank is a controlled experimental object, not a sample from a claimed population probability law.

The catalogue-conditioned natural conjunction bank is not a side product of optimization; it is one of the primary scientific inputs. Each scenario carries encounter timing, geometry, miss distance, and object metadata sufficient for downstream transfer solving and dust-mass evaluation. After selection, each event has equal weight in the exact finite-bank coverage count, so neither bank selection nor finite-bank scoring is presented as a population probability estimate.

This shared-evidence design also clarifies the role of stochastic sampling in the study. The optimization loop does not regenerate new scenario banks for each algorithm. When adaptive prefixes are used, the scenario order is fixed before the run: each island view rotates the sealed bank order by that island’s compiled offset and places that island’s sealed balanced orthogonal-array selection at the head of the ladder. Generation, optimizer, and design identity do not alter scenario order; island identity does, by construction, and identically for every method compared within an island. The resulting stochasticity comes from controlled prefix exposure, not from drift in what counts as a scenario.

In operational terms, the catalogue-conditioned natural conjunction bank is the finite set of

unsafe service opportunities used by the architecture evaluation. A constellation is judged on whether it can service members of that set under the transfer, timing, and dust-delivery semantics defined later in the chapter. The bank is fixed before architecture scoring and is not interpreted as an observed event-rate record or population-probability sample.

Algorithm 1 Catalogue-conditioned natural conjunction-bank workflow used by the baseline evaluation.

- 1: **Input:** public debris catalogue, response horizon, study-band filters, unsafe-event admission policy, and event-selection policy
 - 2: Select the debris subset relevant to the study altitude band and time window
 - 3: Enumerate candidate object pairs using adaptive orbital proximity screening
 - 4: **for** each surviving pair **do**
 - 5: Refine the encounter to a local-minimum close-approach geometry
 - 6: Reject the pair if the miss-distance or perigee geometry gates fail
 - 7: Compute conjunction-time metadata, miss distance, object masses, and risk-score inputs
 - 8: **if** the event is unsafe under the run policy **then**
 - 9: Add the event and its object metadata to the candidate pool
 - 10: Assign each candidate to one of 32 strata by rank-half splits on [Time of Closest Approach \(TCA\)](#) offset, miss distance, mean geocentric radius, encounter latitude, and relative speed
 - 11: Draw a population-proportional seeded stratified cohort, then assign equal scoring weight to each selected event and cache Ω_{unsafe}
 - 12: **Return:** reusable catalogue-conditioned natural conjunction bank Ω_{unsafe}
-

3.3 Intercept and Transfer Solver

The transfer solver is deterministic for fixed scenario evidence and fixed solver settings. Its logic is staged rather than monolithic: candidate deployer–target pairs are first screened and ranked with fast heuristics, then each assessed pair enters the native OxyMOO NSGA-II transfer-front stage over the compact three-parameter decision vector [u](#). Deterministic geometry seeds are evaluated beside the OxyMOO population, retained transfer candidates are polished locally, and a preliminary fixed-target J_2 closure improves seeds. Canonical

evaluation then replays stored candidate controls from their original base epoch, holds the catalogue intercept epoch fixed, applies all hard checks, and preserves every feasible replay-verified trajectory for downstream transfer–release Pareto construction.

Solver input-output specification.

- *Inputs:* event geometry/time metadata, deployer state candidates, analytical/ J_2 -class transfer settings, and solver/timing limits from configuration.
- *Primary solver limits:* phase and transfer ΔV caps, Lambert multi-revolution limits, intercept tolerance, shortlist depth, and orbital-safety guards.
- *Primary timing limits:* minimum lead time and allowable intercept timing relative to conjunction.
- *Outputs (feasible):* a replay-verified transfer set carrying deployer, target, phasing/wait/TOF, Lambert branch, stamped states, burn vectors, and fixed-intercept authority.
- *Outputs (infeasible):* explicit failure status with reason code; event contributes a failure in downstream reliability accounting.

Algorithm flow. The solver operates on a three-parameter decision vector

$$\mathbf{u} = (r_{\text{phase}}, r_{\text{sma}}, r_{\text{wait}}) \in [0, 0.95] \times [0.5, 1.5] \times [0, 0.95], \quad (3.1)$$

which encodes the phasing duration $t_{\text{phase}} = r_{\text{phase}} \cdot t_{\text{max}}$, the phased semimajor axis $a_{\text{phase}} = r_{\text{sma}} \cdot a_{\text{dep}}$, and the wait time $t_{\text{wait}} = r_{\text{wait}} \cdot t_{\text{max}}$, subject to the feasibility condition $r_{\text{phase}} + r_{\text{wait}} < 1$. The residual timing headroom $t_{\text{max}} - t_{\text{phase}} - t_{\text{wait}}$ provides an upper bound for the transfer time of flight; the actual τ is selected by the bounded time-of-flight search

between the configured minimum and that upper bound, further limited by the revolution cap when applicable. The full staged sequence is:

1. *Deployer screening.* Build deployer-target candidates and screen with fast Hohmann plus plane-change heuristic cost estimates. Rank survivors, retain the configured verification superset, and in the production profile verify eight ranked candidates without coarse early termination.
2. *Transfer-front optimization.* For each assessed candidate pair, optimize $\mathbf{u}$ with the native OxyMOO NSGA-II stage at the sealed fast policy: population 20, 3 generations, best-one initialization, seeded by a 52-point deterministic TOF $\times$ phase $\times$ wait grid. This is a bounded seeded polish rather than a converged inner Pareto search. Its two transfer objectives are total ΔV and time-to-intercept per intercept relative velocity, and the deterministic geometry seeds remain in the evaluated set so obvious low-cost transfers are not missed by population initialization. Each transfer-front evaluation performs:
 - (a) propagate the deployer through the phasing duration t_{phase} and apply the phasing ΔV to reach a_{phase} ;
 - (b) propagate through the wait period t_{wait} ;
 - (c) search the configured time-of-flight sample budget, which is 256 in the dissertation production profile, and solve multi-revolution Lambert branches up to the configured revolution cap;
 - (d) polish retained transfer candidates locally with a bounded Nelder–Mead-style ΔV refinement before final verification.
3. *Preliminary endpoint closure.* Apply the fixed-target J_2 seed closure described below (up to 5 iterations, gain $\alpha = 1.0$, target 10 m); this is not canonical HF acceptance evidence,

but on the mixed-fidelity lane it is acceptance-bearing: once it has run, a candidate whose J_2 endpoint residual exceeds the 10 m target plus a 1 m numerical tolerance is rejected by verification.

4. *Exact fixed-epoch replay.* Replay each stored candidate from its original base state and epoch without re-optimizing or retiming. Propagate the selected target independently from catalogue epoch E_0 to intercept I and fail closed on any invalid replay status.
5. *Hard acceptance checks.* Apply timing positivity, lead-time requirement, ΔV caps, minimum Lambert time (300 s), perigee and apogee guards, a minimum deployer–target separation at intercept (120 m), a revolution cap, and intercept miss-distance tolerance (25 m).
6. *Set return.* Preserve all feasible replay-verified trajectories. If none survives, return typed infeasibility; never substitute a first-feasible or proxy-mass choice.

Two transfer sets remain deliberately distinct. The *transfer-domain Pareto set* retains nondominated opportunities in the transfer solver’s own coordinates, total ΔV and time-to-intercept per intercept relative velocity, so multiple transfer designs remain available for inspection. The *verified transfer superset* contains every trajectory that survives exact replay and hard acceptance, including trajectories not selected by transfer-domain dominance alone. Downstream deterministic intercepted-mass and dust-capture evaluation crosses this verified transfer superset with every configured release fraction. Transfer-domain dominance therefore never removes a replay-verified trajectory before its transfer–release resource consequences are known.

3.3.1 Operational Transfer Variant

The production profile uses a transfer variant designed to preserve physical interpretability while keeping architecture-scale search tractable. The Lambert solve remains a two-body boundary-value core and the J_2 closure remains a seed-improvement stage, but canonical acceptance belongs to exact fixed-epoch HF replay. Search may rank a bounded proxy subset; reporting sets `final_replay_picks_to_verify=0`, attempts every native input pair, and preserves every feasible verified trajectory.

The key idea is to preserve a fixed physical intercept target while iteratively adjusting the synthetic Lambert endpoint used to seed the next solve. Let $\mathbf{x}_{\text{nom}}$ denote the physical target position at the selected intercept epoch, and let $\mathbf{x}_{J_2}(\mathbf{v}_k, \tau)$ denote the payload position obtained by propagating the k th Lambert departure under the active J_2 -aware transfer dynamics over time of flight τ . The pre-HF closure residual is

$$\mathbf{r}_k = \mathbf{x}_{J_2}(\mathbf{v}_k, \tau) - \mathbf{x}_{\text{nom}}. \quad (3.2)$$

A corrected synthetic endpoint is then formed as

$$\mathbf{x}_{\text{syn},k+1} = \mathbf{x}_{\text{syn},k} - \alpha \mathbf{r}_k, \quad (3.3)$$

where α is the closure gain. The important feature is that the physical target $\mathbf{x}_{\text{nom}}$ is not moved. Only the synthetic endpoint used for the next Lambert resolve is updated. This keeps the closure numerically useful without redefining the mission target itself.

The physical intercept epoch is never moved by this closure or by canonical replay. Final acceptance requires valid HF replay status at the same catalogue target and an endpoint miss within the 25 m intercept tolerance. Independent target-corroboration bounds of 25 m

in position and 2×10^{-5} km/s in velocity are sealed in the compiled authority but are not evaluated at runtime; a separate, undigested stage-1 witness gate does enforce 25 m and 2.65×10^{-5} km/s, so the sealed velocity bound is a declared future acceptance term, not a check the reported runs passed. Runtime records rendezvous arrival $\Delta V_{\text{arr}} = \mathbf{v}_{\text{target}} - \mathbf{v}_{\text{payload}}$ in ECI coordinates as a diagnostic; it is not an executed burn. Event objective maneuver is the sum of nonnegative executed norms for phasing, replayed transfer, and release control.

The configured search contract supports multi-revolution verification up to four revolutions, eight proxy-ranked candidate pairs, a 256-sample time-of-flight budget, and no coarse early stop. Canonical replay removes the pair cap, validates all native input pairs, and requires the HF status, target corroboration, and provenance checks described above.

The solver’s admissible timing window is inherited from the scenario timeline and from the requirement that the intervention leave usable lead time before closest approach. This timing contract is evaluated together with screening, closure, exact replay, and final HF acceptance.

The authoritative epoch chain is

$$E_0 \rightarrow P \rightarrow L \rightarrow R \rightarrow I \rightarrow C_j, \quad t_{E_0} \leq t_P \leq t_L \leq t_R \leq t_I \leq t_{C_j}. \quad (3.4)$$

Here E_0 is the stored catalogue/base epoch, P is the end of phasing, L is the transfer-burn boundary, R is the sole release-control epoch, I is the fixed catalogue dust–target intercept, and C_j is the later conjunction diagnostic epoch. The transfer body is replayed from E_0 through L ; the fixed canister propagates from L to R ; and the fixed dust model propagates from the post-control state at R to I . The $I \rightarrow C_j$ calculation is explicitly nonblocking and diagnostic: it can report an available separation or a typed propagation failure, but it never mutates the I -state distribution or executed ΔV ledger. Every recorded duration is checked against its endpoint JDs, including $t_R + \tau_{R \rightarrow I}/86400 = t_I$ to floating-point JD tolerance.

Time-scale authority is explicit in canonical artifacts. Catalogue-derived source anchors are osculating Cartesian [GCRS](#) states at source-derived UTC astronomical Julian dates, with the mean-element screening seed derived from that anchor; propagation intervals use SI seconds; and each endpoint is formed by adding its elapsed SI seconds to the source UTC JD. The modeled arcs do not cross an inserted leap second. Earth-fixed rotation is evaluated as UT1 under the declared study approximation $\text{DUT1} = \text{UT1} - \text{UTC} = 0$, rather than leaving UTC and UT1 interchangeable by implication. Astropy converts the UTC grid epochs to its internal TDB dynamics scale. Dates beyond the frozen leap-second table use the last-known offset and are study-coordinate approximations, not operational UTC predictions.

Embedded Sun, Moon, Venus, and Jupiter catalogues use same-epoch body-minus-Earth barycentric vectors aligned to ICRS axes: they are geometric Earth-centered force inputs with no light-time or stellar-aberration correction. A deterministic Astropy built-in builder records Astropy/ERFA versions, complete grid headers, exact payload sizes, and binary hashes. JPL Horizons DE441 geometric ICRF vectors independently bound source error at campaign start, perihelion season, aphelion season, and campaign end. The preregistered two-hour third-body free-displacement proxy remains below 0.025 m for source differences and for 257 campaign-spanning interpolation midpoints per body. Raw source-position tolerances remain body-specific because distant-planet position error is not itself the propagated force error. Reinterpreting the grid as TT, or perturbing the assumed DUT1 within its civil range, can exceed the 25 m target-corroboration tolerance; these choices are documented model assumptions, not negligible formatting details.

The transfer specification is summarized in [Table 3.2](#). This summary is more useful than a one-off orbit sketch because it states exactly which stages and checks distinguish the active transfer path from a generic phasing-plus-Lambert description.

Table 3.2: Summary of the transfer path used in event evaluation.

Stage	Study-specific role
Pair screening	Fast reachability heuristics rank deployer-target pairs before exact solve.
Phasing grid	A deterministic phasing search produces candidate departure states.
Lambert targeting	Bounded transfers close from the phased state to candidate intercept epochs.
Preliminary J2 closure	A perturbation-aware seed step improves closure while leaving the catalogue target and epoch unchanged.
Exact HF replay	Stored controls are replayed from original epoch; all feasible trajectories survive for downstream Pareto evaluation.

Table 3.3: Transfer-solver failure taxonomy used in event-level bookkeeping.

Failure class	Trigger condition	Recorded outcome
Timing-window failure	Non-positive phasing, wait, or transfer timing, or lead-time violation	Event infeasible (timing)
Maneuver-budget failure	Phase or transfer ΔV exceeds the active caps	Event infeasible (DV)
Geometry failure	Intercept miss distance exceeds tolerance, or transfer closure fails	Event infeasible (geometry)
Orbit-safety failure	Lambert-time, perigee, apogee, or related safety guard fails	Event infeasible (safety)
Deterministic-mass failure	Downstream deterministic intercepted-mass gate fails status or finite checks	Event infeasible (deterministic)
Numerical/status failure	Numerical convergence, interpolation quality, or solver status fails	Event infeasible (numerical)

Given a feasible transfer, the next task is determining how much dust to release.

3.4 Dust-Cloud Mass Estimation

Dust-cloud mass estimation follows a four-step event-to-mission logic. A feasible transfer first supplies the intercept state and timing context. The released dust cloud is then propagated to the intercept epoch, and its reconstructed mean velocity feeds a deterministic momentum-transfer gate that establishes the minimum intercepted mass needed to perturb the target sufficiently. A probabilistic coverage conversion on the encounter plane, with the target uncertainty description, then determines the released mass required for the event. Mission-level dust burden is formed only after those event-level released-mass values are folded into the unsafe-event scoring and aggregation layer downstream.

The first step in the mass logic is a deterministic physics gate that determines the minimum dust mass that must physically strike the target to achieve the desired orbital perturbation. Under the study’s parametric momentum-balance model, the velocity change imparted to a primary of mass m_p by intercepting dust of mass m_{int} at relative speed v_{rel} is

$$\Delta V = \kappa \cdot \frac{m_{\text{int}}}{m_p + m_{\text{int}}} \cdot v_{\text{rel}}, \quad (3.5)$$

where κ is a parametric momentum-enhancement factor. Under D3, $\kappa = 1$ is the reportable baseline only within the current monotone direct-momentum model, pending evidence that real dust interaction cannot produce effective transfer below unity. A proposed but unconfigured $\{1, 1.25, 1.5, 2\}$ sensitivity would treat values above unity as parametric enhancement, never as conservative, realistic, validated, or material-specific without dust-specific evidence. The DART result [9] concerns a different impact regime and does not validate dust-specific

enhancement. The required target velocity change ΔV_{req} is whatever is needed to open the primary–secondary miss distance at conjunction to the configured minimum separation goal, sealed at 1 km. That goal is an assumption of this study and traces to no external requirement document; because the deterministic intercepted mass is approximately linear in it, an operational screening threshold of 5 km to 10 km would raise every reported mass by roughly the same factor, and all deterministic-mass and released-mass results are conditioned on it. For that ΔV_{req} , the deterministic-mass path obtains a J_2 seed and validates or repairs the required *intercepted* mass with the HF dust-to-intercept model. That accepted intercepted mass is a strict model gate, not the mission objective. Only the separate capture conversion below produces the effective released-mass surrogate. Non-finite, nonconverged, or below-practical-threshold deterministic outcomes remove that transfer–release alternative from the event front; the event is infeasible only when no alternative survives.

The distinction among intercepted mass, released mass, and mission-level dust burden is summarized later in the objective and workflow discussion. The relevant question is local overlap: how much dust density is required to cover the likely target locations on the encounter plane once both dust-cloud uncertainty and target-state uncertainty are represented there? That is why a deterministic intercept point alone is not enough for sizing the release.

If the event passes that gate, the required *released* dust mass is computed from encounter-plane coverage statistics. The released dust cloud is propagated under the post-release dust model. The target position covariance is fixed at the intercept in the configured local RIC frame, transformed into the common encounter-frame coordinates, and is not propagated by the dust model. The coverage solver then applies the probabilistic conversion model used in the production profile to determine the released dust mass needed to achieve the required coverage on that plane.

The active post-release model intentionally separates dynamics from payload sizing. The

compiled production profile carries one Gaussian component and propagates its seven off-centre Julier minimal-skew simplex points, each with weight $1/7$. Every point uses the same spherical-equivalent dust tuple, $A/m = 1.948 \text{ m}^2/\text{kg}$, $C_D = 2.2$, and $C_R = 1.3$; only state varies across points. Released mass scales the represented particle population and does not change these trajectories. Propagation begins at the native post-control state and JD returned for R , and the raw propagated mean and covariance feed dust finite-disk capture inversion without affine centroid recentering. The former 13-point Merwe construction and its $(\alpha_{\text{UT}}, \beta_{\text{UT}}, \kappa_{\text{UT}})$ controls are historical comparison machinery, not the compiled method.

3.4.1 Encounter-Plane Mass Conversion

The released dust mass objective answers a precise geometric question: given the evolved dust-cloud density on the encounter plane and the target's projected positional uncertainty, how much dust must be released so that the required intercept probability is achieved? The calculation proceeds in four steps.

First, the encounter plane is constructed from the relative velocity between dust and target at the projected intercept epoch. Let $\hat{\mathbf{v}}_{\text{rel}} = (\mathbf{v}_{\text{dust}} - \mathbf{v}_{\text{target}}) / \|\mathbf{v}_{\text{dust}} - \mathbf{v}_{\text{target}}\|$ be the relative-velocity unit vector. An orthonormal basis $\{\hat{\mathbf{b}}, \hat{\mathbf{h}}\}$ spanning the plane perpendicular to $\hat{\mathbf{v}}_{\text{rel}}$ is formed by Gram–Schmidt orthogonalization, and the 2×3 projection matrix $\mathbf{H} = [\hat{\mathbf{b}} \ \hat{\mathbf{h}}]^\top$ maps three-dimensional position offsets into the two-dimensional encounter plane.

Second, the single propagated Gaussian and the fixed intercept-local RIC target covariance are projected onto this plane. Let $\mathbf{R}_{\text{RIC} \rightarrow \text{ECI}}^{\text{int}}$ be the intercept-frame rotation and define $\Sigma_{\text{tgt, pos}}^{\text{int, ECI}} = \mathbf{R}_{\text{RIC} \rightarrow \text{ECI}}^{\text{int}} \Sigma_{\text{tgt, pos}}^{\text{int, RIC}} \mathbf{R}_{\text{RIC} \rightarrow \text{ECI}}^{\text{int}^\top}$. The component index below is retained because the packed calculation is component-shaped, but compiled production has only $k = 1$. This

target covariance is fixed at intercept and is not propagated by the dust model:

$$\begin{aligned}\boldsymbol{\mu}_{k,\text{rel}}^{2\text{D}} &= \mathbf{H}(\mathbf{r}_{\text{dust},k} - \mathbf{r}_{\text{target}}), \\ \boldsymbol{\Sigma}_{k,\text{rel}}^{2\text{D}} &= \mathbf{H}(\boldsymbol{\Sigma}_{\text{pos},k} + \boldsymbol{\Sigma}_{\text{tgt,pos}}^{\text{int,ECI}}) \mathbf{H}^\top \quad (\text{Pc-inversion lane}), \\ \boldsymbol{\Sigma}_{k,\text{rel}}^{2\text{D}} &= \mathbf{H} \boldsymbol{\Sigma}_{\text{pos},k} \mathbf{H}^\top \quad (\text{shared-target lane}).\end{aligned}\tag{3.6}$$

The projected covariances are sanitized via eigenvalue clamping to prevent numerical degeneracy in subsequent inversions. The compiled production treatment of target uncertainty is deliberately isotropic: a single assumed, not observed, 100 m one-sigma position uncertainty is applied at the intercept as $\sigma^2 \mathbf{I}$, with no RIC axis-ratio anisotropy on either lane. On the Pc-inversion lane it is projected separately and added to each grain’s projected covariance in the encounter plane; on the shared-target lane it is excluded from the per-grain covariance because one target-position realization is shared by every released packet and is integrated once outside the packet-failure exponent, so that packet failures are not treated as conditionally independent of the target draw. The baseline assumes zero dust–target cross-covariance.

Third, the overlap between dust density and target uncertainty is evaluated by one of two sealed conversions, selected by lane. Production, on both lanes, uses the shared-target contact solver described below. The dust finite-disk capture-inversion method (the runtime label is *Pc-inversion*) followed by a finite-packet release floor is the earlier conversion, retained only behind a solver-qualification feature with no production caller; it is described here because it defines the capture probability the contact solver inherits. Both are dust finite-disk capture probabilities, not post-intervention conjunction collision probabilities. Under the baseline zero cross-covariance assumption, Pc-inversion computes

$$P_{\text{capture},k} = \int_{\|\mathbf{r}^\perp\| \leq R_{\text{HB},k}} \mathcal{N}(\mathbf{r}^\perp; \boldsymbol{\mu}_{k,\text{rel}}^{2\text{D}}, \boldsymbol{\Sigma}_{k,\text{rel}}^{2\text{D}}) d\mathbf{r}^\perp, \tag{3.7}$$

where $R_{\text{HB},k}$ is the finite hard-body radius. Define the dimensionless area ratio

$$\eta_k = \frac{A_k}{2\pi\sqrt{\det \Sigma_{k,\text{rel}}^{2\text{D}}}}, \quad A_k = \pi R_{\text{HB},k}^2,$$

and the second-order relative curvature correction

$$c_k = \frac{R_{\text{HB},k}^2}{8} \left(\left\| \left(\Sigma_{k,\text{rel}}^{2\text{D}} \right)^{-1} \boldsymbol{\mu}_{k,\text{rel}}^{2\text{D}} \right\|^2 - \text{tr} \left[\left(\Sigma_{k,\text{rel}}^{2\text{D}} \right)^{-1} \right] \right).$$

The small-area branch is permitted only when *both* $\eta_k \leq 10^{-2}$ and the absolute second-order curvature satisfies $|c_k| \leq 10^{-2}$. It evaluates the corrected expression in log space as

$$\log P_{\text{capture},k} = \log A_k + \log \mathcal{N}(\mathbf{0}; \boldsymbol{\mu}_{k,\text{rel}}^{2\text{D}}, \Sigma_{k,\text{rel}}^{2\text{D}}) + \ln(1 + c_k),$$

using the numerical equivalent of `ln1p`. A component that fails either gate uses the finite-disk quadrature in Equation (3.7). Component captures are combined using normalized mixture weights, with the selected branch recorded in telemetry, and the resulting capture statistic is inverted to obtain released mass.

Fourth, D2 converts geometric capture into finite physical-grain mass probability. The baseline uses spherical tungsten grains of diameter 40 μm and material density 19 250 kg/m^3 ; projected area, grain mass, and the configured A/m tuple are cross-validated dimensionally. Let m_g be grain mass and let c be the number of perfectly correlated grains in one independent packet. Then $m_{\text{pkt}} = cm_g$, and the required captured-packet count is

$$K = \left\lceil \frac{m_{\text{det}}}{m_{\text{pkt}}} \right\rceil. \quad (3.8)$$

For n released packets and geometric packet-capture probability P_{capture} , the captured count is $X \sim \text{Binomial}(n, P_{\text{capture}})$. The baseline has $c = 1$. A conservative multiplicative-Chernoff

lower-tail bound chooses n so that $\Pr(X < K) \leq 1 - P_{\text{target}}$. Defining $L = -\ln(1 - P_{\text{target}})$, this gives

$$\mu_{\text{req}} = \left(\frac{\sqrt{2L} + \sqrt{2L + 4K}}{2} \right)^2, \quad n = \left\lceil \frac{\mu_{\text{req}}}{P_{\text{capture}}} \right\rceil, \quad m_{\text{release}} = nm_{\text{pkt}}. \quad (3.9)$$

The bound above is the retained qualification-only conversion. Production does not use it on either lane: there, the released packet count is the smallest exactly representable binary64 count n whose shared-draw no-hit probability $\mathbb{E}_{\text{tgt}}[(1 - p_{\text{cond}})^n]$ falls below $(1 - P_{\text{target}})$ less a reserved numerical margin of two quadrature-tolerance terms plus a 2×10^{-8} tail term, with the final count taken as the maximum of that probability count and the deterministic-mass floor count. The baseline sets $P_{\text{target}} = 0.99$. A packet count that saturates the exact binary64 integer domain is reported as finite-domain unrepresentability, never as zero physical capture. The small-capture-probability limit of the retained bound is Poisson thinning, while the runtime bound remains finite-binomial and conservative. Seeded direct Monte Carlo checks the exceedance frequency but is not runtime authority. Historical arithmetic $m_{\text{det}}[-\ln(1 - P_{\text{target}})]/P_{\text{capture}}$ treated one capture as a deterministic-mass packet; it remains only a labeled legacy regression and cannot support physical probability claims. Packet size exposes the assumed within-packet correlation; alternate sizes require sensitivity evidence. Momentum factor κ changes m_{det} upstream and follows accepted D3: $\kappa = 1$ is the model-bounded reportable baseline and higher sensitivity values are parametric enhancement.

The configured 1000 kg dust bound is a numerical and model-validity sanity ceiling, not canister capacity or a claim that such a release is practical. Earlier campaign artifacts suggested a hundreds-of-kilograms-or-lower regime, but that magnitude remains historical context until regenerated under the current compiled science authority. Alternatives whose released mass reaches the ceiling are removed from the event front and recorded as typed failures rather than converted into a hardware-feasibility statement; the event fails only when

the ceiling removes its last alternative. At the design level, a hard-failing candidate is not dropped: it is emitted at the sealed sentinel pair (2.5 km/s, 1000 kg) when the design serviced at least one event, and otherwise at a tail-cost pair floored at those same caps; either point lies on or beyond the hypervolume reference corner and therefore contributes no dominated volume.

The distinction between these two quantities is essential. Intercepted mass is the event-level physical sufficiency test. Released mass is the event-level optimization quantity. Mission-level dust burden is not either quantity directly; it is a cohort-level unsafe-event summary derived from the released-mass objective during mission scoring.

The estimator, transfer solver, propagation kernels, coordinate transforms, matrix scheduler, optimizers, and receipts used on the production evaluation path are implemented in native Rust as one process; no Python participates in orchestration or evaluation, and Python survives only as plotting and read-only historical-oracle reference. Removed backend selectors and fallback runtime knobs fail fast rather than silently switching to a non-Rust path. Where stochastic event subset sampling is used at the optimization layer, fixed seeding preserves repeatability of sampled evidence and penalty decisions.

3.5 Simulated Space Environment

The environment model matters because conjunction geometry and dust-cloud uncertainty evolve over short horizons. Some effects, such as [RAAN](#) precession, are captured well analytically. Others, such as drag-driven changes in semimajor axis and eccentricity, require numerical propagation to represent uncertainty growth with enough fidelity.

3.5.1 Perturbations & Propagation

This work does not use a generic “promotion” rule in which every candidate begins analytically and is later escalated wholesale to a fully numerical propagator. Instead, fidelity is assigned by body role. The MF lane supplies catalogue pair screening and transfer seeds (each admitted conjunction anchor is then refined under strict HF propagation before sealing), and it is the search fidelity of every cell of the reported matrix, `mf` and `hybrid` alike. The lanes diverge only at canonical replay: in the `hybrid` cells, canonical transfer replay, fixed-canister $L \rightarrow R$ coast, and fixed-dust $R \rightarrow I$ evolution use native HF propagation; the dust physical window is capped at two hours by the timeline partition. Missing HF coefficients, ephemerides, provenance, or propagation status fail closed.

Orbital states are therefore propagated with task-scoped fidelity rather than with a one-size-fits-all lane switch. The hybrid HF lane uses a fixed 5×5 spherical-harmonic gravity field at every altitude, with force subcycling and interpolated ephemerides. The requested harmonic order is preserved rather than degraded with altitude, so a single field truncation applies uniformly across every propagated arc and no altitude-dependent model change can enter a comparison as a confound.

It retains drag through the [JB2008](#) thermospheric density model, plus [SRP](#), Sun, and Moon

perturbations after release. Two properties of that model must be stated precisely, because both are sealed and neither is a free configuration choice. First, the compiled lane evaluates the [JB2008](#) density through a fitted kernel rather than by walking the reference quadrature: the fixed quadrature plans are replaced by a degree-14 monomial fit in the exospheric temperature, and the resulting profile is gated to a maximum relative density deviation of 10^{-4} from the reference kernel over the validation grid. Second, the model consumes the measured F10, S10, M10, Y10, and Dst drivers parsed from hash-pinned Space Environment Technologies `SOLFSMY` and `DTCFILE` files, but the campaign epoch lies after those files' observed cutoff, so the reported runs are driven by a sealed last-value-persistence extension of the final validated rows. That extension is a stated atmosphere *scenario*, not a space-weather forecast and not an observation, and every density-dependent result is conditioned on it. The compiled run authority rejects any campaign whose resolved density-model identity or driver authority differs from the pinned pair, so neither can drift silently between runs. Additional terms remain available only for targeted model-form stress tests when a specific sensitivity question warrants the extra complexity.

The active set above is chosen for short-horizon JCA relevance: drag and [SRP](#) dominate dust-cloud evolution over the response horizons, while Sun and Moon terms are the retained third-body perturbations for the released cloud. Optional terms remain available when a study is explicitly designed as a sensitivity or model-form stress test. The SRP pressure is referenced to $P_{\text{1AU}} = 4.56 \times 10^{-6} \text{ N/m}^2$ at $1 \text{ AU} = 149\,597\,870.7 \text{ km}$ and applies the inverse-square satellite–Sun distance law separately from eclipse attenuation. Illumination is a binary cylindrical shadow: [SRP](#) is applied in full in sunlight and set identically to zero in shadow, with no penumbral transition. Shadow entry and exit are located as exact roots and each root ends an integration segment, because the force is discontinuous across it. This is neither a smooth transition model nor the conical occultation model used by Orekit.

Eclipse-transition result claims therefore remain outside the baseline error budget until the preregistered Task 3.4 model-form sensitivity closes.

Operationally, this propagation stage answers three questions per evaluation cycle: how hazardous-event geometry and transfer seeds are screened under a tractable analytical/ J_2 baseline, whether every canonical transfer candidate replays under HF at the fixed target epoch, and how dust uncertainty evolves from the exact release boundary over the capped post-release interval. The numerical roles are served by native Rust propagation. Encke variation keeps the numerically integrated quantity small around a two-body reference orbit, while force subcycling and ephemeris interpolation reduce repeated environmental work.

Given the epoch state $\mathbf{r}(t_0)$, $\dot{\mathbf{r}}(t_0)$, the Lightyear propagator integrates the Encke variation from a two-body reference rather than the full osculating motion [2, 48, 67]. Let the reference be $\boldsymbol{\rho}(t)$, maintained analytically in nonsingular equinoctial elements [74] through a fixed-tolerance Kepler solve at each evaluation epoch. Define

$$\delta\mathbf{r}(t) = \mathbf{r}(t) - \boldsymbol{\rho}(t), \quad (3.10a)$$

$$\delta\dot{\mathbf{r}}(t) = \dot{\mathbf{r}}(t) - \dot{\boldsymbol{\rho}}(t), \quad (3.10b)$$

$$\delta\ddot{\mathbf{r}}(t) = \ddot{\mathbf{r}}(t) - \ddot{\boldsymbol{\rho}}(t). \quad (3.10c)$$

With μ the gravitational parameter, $r = \|\mathbf{r}\|$, $\rho = \|\boldsymbol{\rho}\|$, and $\mathbf{a}_{\text{per}}$ the sum of non-Keplerian accelerations (e.g., J_2 , drag, SRP, third bodies), the perturbed and reference dynamics are

$$\ddot{\mathbf{r}} = \mathbf{a}_{\text{per}} - \mu \frac{\mathbf{r}}{r^3}, \quad (3.11)$$

$$\ddot{\boldsymbol{\rho}} = -\mu \frac{\boldsymbol{\rho}}{\rho^3}. \quad (3.12)$$

Substituting Equations (3.11) and (3.12) into Equation (3.10c) yields the Encke variation

equation

$$\delta\ddot{\mathbf{r}} = \mathbf{a}_{\text{per}} + \mu \left(\frac{\boldsymbol{\rho}}{\rho^3} - \frac{\mathbf{r}}{r^3} \right) = \mathbf{a}_{\text{per}} - \mu \left(\frac{\boldsymbol{\rho} + \delta\mathbf{r}}{\|\boldsymbol{\rho} + \delta\mathbf{r}\|^3} - \frac{\boldsymbol{\rho}}{\rho^3} \right), \quad (3.13)$$

with osculation initial conditions

$$\delta\mathbf{r}(t_0) = \mathbf{0}, \quad \delta\dot{\mathbf{r}}(t_0) = \mathbf{0}, \quad \boldsymbol{\rho}(t_0) = \mathbf{r}(t_0).$$

Because $\delta\mathbf{r}$ remains small, Equation (3.13) supports larger step sizes and improved numerical precision compared with direct Cowell integration of $\mathbf{r}$. In practice, the two-body reference $\boldsymbol{\rho}$ is maintained analytically in nonsingular equinoctial elements and the variation $\delta\mathbf{r}$ is integrated numerically by Lightyear. However, evaluating the gravity correction term $\boldsymbol{\rho}/\rho^3 - \mathbf{r}/r^3$ by direct subtraction loses approximately four significant digits when $\delta\mathbf{r}$ is small, because both terms are of comparable magnitude. The formulation therefore replaces this subtraction with Battin's $f(q)$ formulation [2]. Define

$$q = \frac{\delta\mathbf{r} \cdot (\delta\mathbf{r} + 2\boldsymbol{\rho})}{\|\boldsymbol{\rho}\|^2}, \quad f(q) = \frac{q(3 + 3q + q^2)}{1 + (1 + q)^{3/2}}. \quad (3.14)$$

The corrected Encke gravity term becomes

$$\mathbf{a}_{\text{grav}}^{\text{Encke}} = \frac{\mu}{\|\mathbf{r}\|^3} (f(q) \boldsymbol{\rho} - \delta\mathbf{r}), \quad (3.15)$$

which avoids large-quantity subtraction entirely. This formulation preserves full working precision even when the perturbation is six or more orders of magnitude smaller than the reference position (see Battin [2], §9.3, Eqs. 9.69–9.73).

Rectification (re-osculatation of the reference orbit) is triggered whenever $\|\delta\mathbf{r}\|$ exceeds 10 km. Both integration paths additionally cap each segment at 5400 s, approximately one LEO

orbital period; on the events-enabled path that cap acts alongside eclipse-root transactions, each of which also ends a segment. Together these bounds prevent long-arc propagations from accumulating unbounded perturbation growth.

For the main manuscript argument, the important distinction is not the full catalogue of force terms available in the software stack, but which are active for each body role. The MF path handles catalogue construction and transfer seeding. Production transfer candidate generation/verification, canonical transfer replay, fixed-canister coast, and fixed-dust arc use HF propagation with explicit source epochs and body-specific force contexts. The dust phase is capped at two hours and uses drag, [SRP](#), Sun/Moon third-body perturbations, fixed 5×5 gravity, force-model subcycling, and interpolated ephemerides.

3.5.2 Fidelity Allocation Rationale

The mixed-fidelity choice reflects where extra model detail actually pays off in this problem. Bulk optimizer search needs a fast event-evaluation rate because every additional candidate architecture may require dozens of event evaluations. At that stage, the main question is comparative: which designs look more promising under the same evidence? A mean-field/ J_2 -class analytical path works well there because it preserves the dominant secular geometry without paying for full numerical propagation at every call.

Two acceptance-sensitive quantities receive additional fidelity. Every canonical transfer candidate requires exact HF replay at the fixed target epoch, while the released dust cloud requires numerical propagation because short-horizon perturbations can change the encounter-plane footprint and therefore the released-mass estimate.

This is why fidelity is not handled as one blanket model choice in this work. Full numerical propagation of the complete catalogue would be prohibitively expensive, while purely ana-

lytical final evaluation would miss the physics that matters most for accepted transfers and dust mass sizing. The baseline therefore keeps catalogue construction tractable, then spends numerical budget on strict production transfer authority and released-cloud geometry.

Quantitative force-sensitivity plots are excluded from the methodology because the available asset has a narrower altitude scope and includes force terms not traceable to the active baseline. The manuscript therefore defines the operational force policy directly: MF/J_2 for catalogue screening and transfer seeding; native HF replay for every canonical transfer candidate; native HF fixed-canister and fixed-dust arcs; and optional force-menu extensions only for explicitly sourced sensitivity work.

Table 3.4: Perturbation policy by modeled object and analysis role.

Lane	Purpose	Dynamics / integrator	Active perturbations	Study baseline
Analytical (mf)	Catalogue screening, transfer-front seeding, and all search-lane evaluation	<code>equinoc_prop_j2</code> mean-element propagation	Secular J_2	Search lane for every cell; canonical-replay lane for <code>mf</code> cells
Transfer replay (HF)	Exact stored-candidate validation at fixed target epoch	Native HF replay from original base epoch	Gravity-only transfer-body context; target-corroboration bounds sealed but not yet evaluated	Final acceptance requires typed HF status and fixed-I authority
Canister coast (HF)	$L \rightarrow R$ trajectory reference	Lightyear numerical propagation	Fixed canister $A/m, C_D, C_R$ with active force model	No mass, capacity, mesh, or hardware coupling
Dust / mass validation (HF)	Dust evolution and unsafe mass validation/repair	Lightyear numerical propagation, dust phase capped at two hours	Fixed 5×5 gravity at every altitude; subcycled forces and interpolated ephemerides	HF dust lane for mass repair and released-cloud evolution
Dust propagation (HF)	Post-release dust-cloud evolution and encounter-plane density estimation	Lightyear multi-method Rust propagator (Encke variation with Battin $f(q)$ correction)	drag (JB2008 fitted kernel, hash-pinned drivers, sealed persistence scenario), srp , sun , moon	Fixed 5×5 gravity with the configured two-hour cap
Optional sensitivity configurations	Regime-specific stress testing	Same Lightyear machinery with extended force menu	Lorentz and Coulomb terms, additional third bodies, relativity	Retained for sensitivity studies rather than default evaluation

With the propagation framework established, we now turn to the constellation families that define the deployer architecture.

3.6 Constellations

This study compares three deployer-architecture encodings: Walker, Dual-Shell Walker-Delta, and Flower constellations. These families span a useful range of structured geometry, from a single symmetric shell to a second stacked shell to a resonant repeating-groundtrack lattice. Walker and Flower families provide interpretable baselines with strong geometric regularity, and the Dual-Shell Walker-Delta family tests whether a second structured shell provides meaningful benefit. An unstructured plane-constrained encoding was scoped as a fourth family but is not implemented in the reported evaluation stack; the structured-versus-unstructured comparison is accordingly identified as future work in [Section 5.4](#) rather than answered here.

3.6.1 Walker Constellation

Walker constellations are symmetric families of near-circular orbits that share a common altitude and inclination. This symmetry makes them attractive as architecture baselines because shell geometry is easy to interpret, phasing is regular, and long-term perturbation behavior is comparatively uniform across the shell. In classical $T/P/F$ notation, T total spacecraft are partitioned into P orbital planes and phased by an inter-plane offset F [\[72, 73\]](#).

Two canonical specializations are the Walker Delta and Star. In a Walker Delta, the planes' right ascensions of the ascending node span the full ring, $\Delta\Omega = 360^\circ/P$, producing shells favored for inclined, non-polar designs and uniform mid-latitude coverage; Galileo's reference layout is a $56^\circ:24/3/1$ Delta [\[20\]](#). The Delta pattern is closely related to the Ballard "rosette"

family [1]. In a Walker Star, each plane is near-polar and the RAANs occupy a half-ring, $\Delta\Omega = 180^\circ/P$, yielding strong high-latitude and global performance; the Iridium network is the archetype, approximately $86.4^\circ:66/6/2$ [65]. In both cases, satellites within planes are uniformly phased and adjacent planes observe the F -law above; software implementations typically enforce the 360° vs. 180° RAAN span for Delta and Star, respectively [45, 46]. Practically, designers tune T , P , F , and i to meet elevation-angle, dilution-of-precision, revisit, and cross-seam constraints while respecting launch-vehicle, deployment, and inter-satellite-link considerations [67, 73].

In earlier baseline comparisons, the Walker family was sometimes encoded by the reduced design vector $x = (h, i)$, with plane count and satellites per plane fixed for a lower-dimensional fair-family comparison. The reported dissertation profile instead enables bounded discrete topology variables and optimizes the Walker inter-plane phasing index, so Walker is encoded as $x = (h, i, P, S, f_{\text{rel}})$. Here h is shell altitude, i is shell inclination, P is the number of planes, S is the satellites-per-plane count, and f_{rel} is decoded as the Walker relative-spacing offset for the realized plane count. Under these bounds, $P \in [2, 8]$, $S \in [2, 5]$, and $i \in [0.5^\circ, 165^\circ]$. The resulting shell is then populated with the corresponding regular RAAN spacing and optimized in-plane phasing. If those integer bounds and phasing degrees of freedom are disabled, the design space collapses back to the older fixed-topology $x = (h, i)$ baseline.

3.6.2 Dual-Shell Walker-Delta Constellation

The Dual-Shell Walker-Delta family extends the Walker concept by staging two independent structured shells at different altitudes and inclinations. Each shell is a valid Walker constellation in its own right; the combined architecture provides altitude diversity, which expands the set of reachable scenario geometries and reduces the risk that a single-shell

staged architecture misses unsafe cases concentrated in one orbital band. In mission terms, the family tests whether a second structured shell changes access enough to matter once the natural conjunction bank is anisotropic rather than uniformly distributed.

In the reported dissertation profile, the Dual-Shell Walker-Delta family is encoded by the design vector $x = (h_1, h_2, i_1, i_2, P_1, S_1, P_2, S_2, f_1, f_2, \Delta\Omega_2, \Delta M_2)$, where each shell has its own altitude, inclination, plane count, satellites-per-plane count, and Walker phasing index. The final two variables offset the second shell's RAAN and mean anomaly relative to the first shell. Altitudes remain bounded to $[350 \text{ km}, 2000 \text{ km}]$, inclinations to $[0.5^\circ, 165^\circ]$, and each shell's plane count is bounded to $P_j \in [2, 4]$ with $S_j \in [2, 5]$. Earlier baseline comparisons fixed those shell cardinalities for fairness, but the reported production study uses bounded topology, phasing, and second-shell offset variables so the second shell can be judged both by where it is staged and by how much structured capacity it carries.

In practice, the Dual-Shell Walker-Delta family tests whether structured altitude diversity, at matched per-shell geometry, changes service access to the natural conjunction bank relative to a single structured shell.

3.6.3 Flower Constellation

Flower constellations generalize circular shells by enforcing compatibility with a chosen rotating frame so that the satellites populate a repeated noncircular path rather than independent circular planes [49, 50, 51, 59]. They are useful here because they test whether repeating-path structure can produce advantageous deployer staging geometries that a simple circular shell cannot represent. In the present study, their role is to ask whether that additional geometric flexibility improves response access while still preserving the interpretability that makes structured families attractive.

In earlier baseline comparisons, the Flower-family cell used a smaller fixed-cardinality encoding. In the reported dissertation profile, the implemented family is a bounded 2-D Lattice Flower repeat-groundtrack specialization under first-order J_2 secular dynamics. Its optimizer-facing design vector is

$$x = (h_{\text{req}}, i, e, \omega, N_o, N_{so}, N_c),$$

where h_{req} is the requested altitude before repeat-groundtrack snapping, i is inclination, e is eccentricity, ω is argument of perigee, N_o is the number of orbital planes, N_{so} is the satellites per plane, and N_c is the 2-D lattice combination number. For each integer topology, the implementation selects the nearest semi-major-axis root satisfying the J_2 -secular repeat ratio, so plotted and reported physical altitudes use the realized altitude rather than the requested value.

Table 3.5: Flower-family parameter interpretation used in the study design space.

Parameter	Interpretation in this study
h_{req}	Requested altitude before snapping to the nearest compatible J_2 -secular repeat-groundtrack root
i, e, ω	Common inclination, eccentricity, and argument of perigee for the 2-D lattice family
N_o, N_{so}	Integer plane count and satellites-per-plane topology variables
N_c	2-D lattice combination number, constrained by $0 \leq N_c < N_o$
N_p, N_d	Derived repeat-groundtrack integers satisfying the selected J_2 -secular resonance

Across families, topology and phasing domains remain family-native. Walker and Dual-Shell phasing use normalized fractions before decoding to realized integer slot indices; decoded f values remain in descriptors. Flower uses its bounded integer lattice and repeat-groundtrack domain. One hard safety screen applies to every family: every decoded design must carry a closed-form certificate that its all-time minimum pairwise separation under the secular- J_2

mean-element propagator clears the compiled 1 km floor, by a radial-gap argument for disjoint shells, an exact circular closed form for identical circular shells, or a bounded branch-and-bound lower bound for eccentric shells; designs that cannot be certified are rejected, and no sampling horizon or sample count enters the decision. A 200 km perigee guard remains in the Flower decoder but is dominated by the shared envelope. Every family also shares the same realized 350 km to 2000 km radial envelope and the same 40-satellite cap, which for Flower designs is enforced on the realized post-snap perigee and apogee.

The constellation families above define where deployers can be positioned; the debris distribution determines where they need to reach.

Table 3.6: Design-space summary for the deployer-constellation families used in this study.

Family	Searched variables	Fixed structural choices	Primary role in the study
Walker	$h, i, P, S, f_{\text{rel}}$	Optional reduction to fixed-topology h, i baseline	Reference structured shell
Dual-Shell Walker-Delta	Geometry, topology, phasing, and second-shell offsets for two shells	Optional reduction to fixed-cardinality baseline	Tests whether a second shell changes access geometry materially
Flower	Seven-variable LFC-RGT vector	2-D LFC-RGT/ J_2 specialization with altitude snapping	Tests resonant structured geometries

3.7 Debris Statistics

The baseline evidence used throughout this work is built directly from the public Space-Track catalogue via its REST [Application Programming Interface \(API\)](#) [66]. The source bytes are hash-pinned [OMM](#) and [SATCAT](#) captures, stored with their exact queries, response headers, and a retrieval receipt; queries exclude active payloads and retain resident objects whose perigees place them in [LEO](#). Those catalogue states are then screened with the same mean-element/ J_2 -class machinery used elsewhere in the evaluation pipeline. This point matters: the dissertation does not optimize against a synthetic debris environment first and then validate against catalogue data later. It starts from catalogue objects, screens the finite catalogue-conditioned natural conjunction bank under explicit filters, and reuses that fixed

bank across optimizer and architecture comparisons.

For that reason, the event-source story needs to be stated directly before the optimizer results are discussed. The canonical evidence path in this dissertation is the MF/ J_2 conjunction catalogue described next. The older surrogate studies are retained, but only as secondary sensitivity tools.

3.7.1 MF/ J_2 catalogue screening and natural conjunction-bank generation

The dissertation baseline generates its event banks offline from that pinned source population at one common anchor epoch, and a mandatory generate-twice check requires the emitted banks to be byte-identical before they are accepted. The source population comes from the captured Space-Track [LEO](#) debris objects rather than from a synthetic draw. Objects are filtered into the shell of interest first. The catalogue-generation problem is then posed locally: the search looks for pairs whose actual orbital geometries come close under the secular- J_2 lane, using adaptive semimajor-axis neighborhoods so dense regions remain physically local while sparse regions still acquire enough candidates to search.

Once a plausible pair is found, the conjunction is not stored as a single geometric snapshot. The pair's natural closest approach within the forward horizon is refined under strict HF propagation to a conjunction anchor, and the event is written as a dual-anchor object. Each event therefore carries both the common-epoch source anchors and the conjunction-state GCRS anchors for the primary and secondary objects, with one osculating Cartesian anchor per object as physical truth and a derived mean-equinoctial seed as screening state only, together with the retrieval receipt and generator and convention digests that make the bank regenerable. That choice makes later replay, transfer reasoning, and consistency checks much

cleaner than if the catalogue stored only a conjunction-point surrogate.

Unsafe-only filtering is applied after those anchors are constructed. Let t_c denote the stored conjunction epoch, with primary and secondary positions $\mathbf{r}_p(t_c)$ and $\mathbf{r}_s(t_c)$ taken directly from the conjunction anchors. The conjunction-truth miss distance is

$$d_{\text{conj}} = \|\mathbf{r}_p(t_c) - \mathbf{r}_s(t_c)\|. \quad (3.16)$$

Only events satisfying

$$d_{\text{conj}} < d_{\text{min}} \quad (3.17)$$

are admitted to Ω_{unsafe} , where d_{min} is the study’s minimum unsafe miss-distance threshold. Safe-by-default encounters are dropped at this stage rather than left in the bank to inflate later service statistics.

After unsafe-only filtering, the surviving candidate pool does not use inclination-bin rebalancing. The production banks are one seeded, population-proportional stratified sample of the complete accepted population over five encounter features: refined [TCA](#) offset, miss distance, mean geocentric radius of the two objects, magnitude of the geocentric latitude of the encounter midpoint, and relative speed. Each feature contributes a rank-half split, giving 32 strata; sampling changes membership only, rows inside each emitted bank remain in a deterministic miss-distance, [TCA](#), NORAD-pair, and identity order, and the reserve bank is the exact complement. Admission requires a conjunction-truth miss below 1 km. The result is a reusable MF/ J_2 catalogue-conditioned natural conjunction bank, explicit about its unsafe-only contract and stable enough to support matched optimizer and architecture comparisons across repeated runs. It is a controlled finite evaluation set, not an observed conjunction-rate record.

This catalogue-conditioned workflow is also why the main text uses MF/ J_2 natural-pair

screening rather than low-fidelity surrogate catalogues. The evidence used for optimizer comparison and constellation interpretation comes from the fixed natural conjunction bank generated by the mean-element screening path. Higher-fidelity dust propagation enters later, after a scenario has been admitted under the unsafe-only policy and a transfer candidate exists. That keeps the scenario source stable while still allowing higher-fidelity physics where released-mass sizing depends on it.

Debris is not uniformly distributed in [LEO](#). Instead, it clusters in inclination bands associated with common mission classes, long-lived upper stages, and major fragmentation histories [31, 33]. A particularly prominent concentration occurs near [Sun-Synchronous Orbits \(SSOs\)](#), with inclinations typically $i \approx 97^\circ$ to 99° and altitudes in the 600 km to 900 km range. The SSO condition arises from balancing J_2 -driven nodal regression with the apparent solar motion,

$$\dot{\Omega} = -\frac{3}{2} J_2 n \left(\frac{R_E}{a} \right)^2 \frac{\cos i}{(1 - e^2)^2}, \quad (3.18)$$

so that $\dot{\Omega} \approx -0.9856^\circ/\text{day}$ for typical Earth-observation shells [67]. Because atmospheric drag weakens rapidly with altitude, objects injected into these bands can persist for decades, reinforcing the anisotropy seen in the catalogue [36]. That anisotropy is central to the architecture question: deployer staging is valuable only if it aligns with the hazardous regions the catalogue actually occupies.

The use of public catalogue mean elements introduces known limitations that bound the interpretation of these statistics. The public catalogue is essentially complete for characteristic sizes above roughly 10 cm in [LEO](#), but it under-represents the millimeter-to-centimeter fragments that dominate flux and collision risk [31, 36]. Catalogue mean-element states are also optimized for short-term catalog maintenance rather than precision conjunction adjudication, which is why every admitted event is refined under strict HF propagation before it is sealed. For the present purpose, however, they remain sufficient to screen the

finite catalogue-conditioned natural conjunction bank under the stated filters; the bank is not treated as an observed conjunction-rate record.

These empirical patterns have direct implications for service design. In-plane retargeting is dominated by phasing maneuvers, whereas large plane changes are expensive at [LEO](#) speeds:

$$\Delta V_{\text{plane}} = 2 v \sin\left(\frac{\Delta i}{2}\right). \quad (3.19)$$

Consequently, a small number of deployers staged near the dominant inclination bands can repeatedly access high-density hazard regions with much lower maneuver burden than architectures that rely on frequent large plane changes. This is the physical intuition behind the structured-shell constellations compared later in the chapter.

3.7.2 Legacy Surrogate and KDE Studies

The surrogate branch is secondary to that MF/ J_2 baseline. Earlier exploratory work examined modified equinoctial coordinate views of the catalogue and tested surrogate generators, including [Gaussian Mixture Model \(GMM\)](#)- and [Kernel Density Estimation \(KDE\)](#)-based variants, as a way to create additional synthetic objects for stress testing. In those studies the [LEO](#) debris population was first mapped into equinoctial coordinates, then into a modeling space that keeps the angular coordinate continuous through $\sin L$ and $\cos L$. Several density models were fit in that space, including KDE, [GMM](#), a Gaussian-copula path, and related local-resampling baselines. Their outputs were compared against held-out real catalogue samples with multivariate and marginal metrics such as maximum mean discrepancy, energy distance, classifier two-sample tests, coverage/precision, Fréchet Gaussian distance, and marginal KS/Wasserstein diagnostics. Those experiments are still useful because they showed how anisotropic and non-Gaussian the debris population can look once mapped into a

nonsingular representation, and because they made clear that apparently simple [LEO](#) density structure is not well captured by one naive parametric family.

They do not, however, define the baseline evidence used in the present manuscript. In the reported workflow, surrogate catalogues are retained only for explicitly marked sensitivity studies. When they are used, the sampled objects still pass through the same pair-screening, local-minimum refinement, dual-anchor event construction, and unsafe-only admission filters used for public-catalogue objects. The surrogate changes the source population for a stress test; it does not replace the screening logic or the service semantics. That separation keeps the main optimizer and architecture conclusions tied to the reported evidence path while preserving a controlled place for model-form perturbation studies.

With the catalogue evidence model and optional surrogate tools in place, the next step is to choose optimization algorithms for the stochastic, nonsmooth objective landscape.

3.8 Stochastic Optimization

This section explains how optimization routines are selected, configured, and compared for a stochastic, event-driven objective landscape. The focus is on methods that preserve fairness under shared evidence and [CRN](#) controls while still being practical for expensive propagation-based evaluations.

3.8.1 Selection of Optimization Routines

The optimization routines are selected to match the structure of the debris-mitigation constellation problem posed throughout [Chapters 1 to 3](#). Optimizations are run per constellation family; the family itself is not treated as an inner decision variable. For each family, the

decision vector contains only the parameters defined by that family’s reported encoding, such as the five-variable Walker shell, the twelve-variable Dual-Shell Walker-Delta pair, or the seven-variable Flower-family 2-D LFC-RGT/ J_2 specialization.

Earlier family-level baseline comparisons often held satellites-per-plane and number-of-planes fixed to small values for tractability and fairness. The reported dissertation profile instead enables bounded discrete topology variables where the family supports them, while preserving shared family-specific limits and the same event-level semantics across the optimizer matrix. Intercept timing and transfer details are still set by an inner transfer solver (phasing + Lambert targeting) and are not direct decision variables. Objective evaluations require event-driven propagation ([Section 3.5.1](#)) to compute encounter opportunities, feasible lead times, total ΔV , and safety checks, which introduces pronounced nonsmoothness through duty-cycle windows and intercept-feasibility thresholds. Gradients are expensive here, and the landscape contains many local optima tied to J_2 precession and phasing resonances. For that reason, the reported comparison uses a mixed optimizer portfolio with strong global exploration and straightforward parallelism.

The reported optimizer matrix spans six variants: `rnsde`, `prnsde`, `mopso`, `nsga2`, `eps_nsga2`, and `age_moea2`. These are crossed with the Walker, Dual-Shell Walker-Delta, and Flower families under a common event-evaluation protocol. SMS-EMOA and simulated annealing were considered during portfolio design and are discussed in [Chapter 2](#), but neither is part of the reported matrix: neither SMS-EMOA nor simulated annealing has an implementation in the production optimizer crate. Their omission is a scope boundary of the reported study, not a finding about their suitability.

Conceptually, the portfolio is centered on two families. Differential-evolution variants provide the main continuous-search workhorses for this study because the design space is nonsmooth yet still strongly dependent on continuous orbital parameters. Dominance-based evolutionary

multiobjective algorithms provide the primary frontier-building baseline because they expose convergence-diversity tradeoffs clearly and remain widely interpretable. Swarm and indicator-based methods are then included as comparison baselines that probe alternative search behavior under the same mission semantics. Annealing-based search is not represented in the reported matrix.

Practically, these algorithms are well suited to the simulation workflow in [Section 3.5](#). Population methods support concurrent evaluation, which is essential when each design assessment requires variable-fidelity propagation, conjunction screening, and intercept geometry checks. A common robust-evaluation layer and Common Random Number seeding keep comparisons aligned across solvers.

Comparison protocol. To compare optimizers rather than combine them, each method is run independently with (i) the same configured search depth (identical population size and generation count within a campaign shape), with realized objective evaluations recorded as measured accounting rather than capped, because the canonical configuration loader refuses a hard evaluation budget, (ii) the same fixed scenario bank and the same compiled scenario order within an island, (iii) the same physical constraint definitions with algorithm-specific feasibility mechanics, and (iv) identical risk-aggregation settings for robust objectives/constraints. Constellation families are likewise benchmarked separately (Walker, dual-Walker, and Flower): each family is optimized on its own under identical budgets and then compared descriptively. Reported fronts and indicators use only feasible points under the stated constraints.

3.8.2 RNSDE and PRNSDE

[Robust Nondominated Differential Evolution \(RNSDE\)](#) is included in the active matrix because it can explore a nonsmooth event-driven landscape without assuming gradients or

a smooth feasibility boundary. The core search engine is a multi-objective NSDE variant with dithered crossover rate $CR \in (0.2, 0.9)$ and mutation scale $F \in (0.4, 1.2)$ and optional archive interaction. For cross-optimizer fairness, [RNSDE](#) shares the same population-diversity infrastructure as every other population method in the matrix: a periodic random reinitialization that, every 25 generations, replaces $\lfloor 0.1 N \rfloor$ of the population with individuals drawn uniformly within the design bounds (which adds evaluations beyond the nominal population-times-generations count, because the reported profile configures no hard evaluation cap for a guard to consult), together with front-gap-filling offspring. This shared mechanism is experiment-harness infrastructure rather than a distinguishing feature of any single algorithm; in particular, local refinement is reserved for PRNSDE, described next, so the [RNSDE](#)/PRNSDE contrast stays clean. Robustness is enforced through [CRN](#) evidence control; Beta/[HPD](#) state governs adaptive search, while final reporting uses exact canonical replay ([Section 3.9](#)).

The optimizer matrix also includes PRNSDE, a locally refined RNSDE variant that adds exploitation to the same stochastic differential-evolution backbone. Its deterministic local-refinement stage runs at initialization, on the non-random prefix of the initial population: each of at most two refined seeds draws 10 bounded Gaussian trial points at a decision-space step scale of 0.03, and a trial is kept only on a strictly positive objective gain. During the main evolutionary loop, PRNSDE enables the inherited sparse-front exploitation hook, which perturbs elite front members with bounded Gaussian local-search trials rather than refining every candidate. It is treated as a distinct comparison method because its initialization refinement plus sparse-front exploitation path changes the search mechanics under the same event evidence.

3.8.3 Nondominated Sorting Genetic Algorithm II (NSGA-II)

NSGA-II remains a core dominance-based Pareto-front generator for the study [14]. It provides the most familiar baseline for how a standard Pareto-search method behaves under the same conservative mission semantics. In the production matrix it uses simulated-binary crossover with $\eta_c = 15$ and crossover probability 0.9, together with polynomial mutation using fixed per-variable probability 0.2 and $\eta_m = 30$, with integer-safe rounding where required. Constraint handling follows Deb-style feasibility ranking, so feasible designs dominate infeasible ones before diversity tie-breaks. As with the DE family, CRN-aligned sampling is used to keep objective evidence comparable across methods.

Robust **NSGA-II**

The robust variant layers CRN sampling onto standard **NSGA-II** mechanics and uses the beta-dist controller described in [Section 3.9](#): posterior-mean, PDF-convergence, and **HPD** diagnostics control adaptive prefixes, while canonical publication separately replays the exact 64-event bank and constructs event-count coverage-box fronts. Diversity is monitored via crowding distance; an external epsilon archive provides the separate stability mechanism described next.

ε -Dominated **NSGA-II**

In noise-prone runs, an ε -dominance wrapper replaces pure nondominated sorting. Objective space is partitioned into ε -boxes with $\varepsilon = 10^{-3}$ in the $\log_{10}$ -transformed objective space used here, and only solutions that improve beyond incumbent box representatives enter an external epsilon archive. This limits rank churn and controls archive size while retaining the custom **NSGA-II** variation and algorithm-specific constraint handling.

3.8.4 Multiobjective Particle Swarm Optimization (MOPSO)

The Multiobjective Particle Swarm Optimization complements evolutionary methods with rapid global prospecting [10]. Here it is mainly a comparison point for whether a velocity-based search strategy reaches useful trade regions faster than mutation-and-selection families on the same expensive evaluation stack. Swarm sizes and archive capacities are chosen to match the evaluation budgets used by the evolutionary baselines. Velocity updates use an adaptive inertia weight that decays linearly from $w_{\max} = 0.9$ to $w_{\min} = 0.3$ across the run, with time-varying acceleration coefficients $c_1 : 1.8 \rightarrow 1.0$ and $c_2 : 1.8 \rightarrow 2.0$. A 25% per-generation Gaussian mutation and small velocity jitter (5% of range) maintain diversity and prevent premature archive saturation. Reflective boundary handling enforces variable bounds; if a reflection crosses the opposite bound, the position is clamped there and the corresponding velocity is reset to zero. Integer-coordinated parameters are rounded after updates. Shared CRN seeds keep stochastic evidence aligned with RNSDE/NSGA-II runs, so observed differences remain attributable to search dynamics rather than sampling drift.

3.8.5 Geometry-Aware Evolutionary Multiobjective Algorithms (EMOAs)

The comparison matrix also includes Adaptive Geometry Estimation Multiobjective Evolutionary Algorithm 2 (AGE-MOE2), which changes survivor selection rather than the physical evaluation stack, using its geometry-aware selection pressure [54]. It retains its algorithm-specific selection and constraint handling while sharing event physics, objective definitions, and initialization policy with the other methods.

3.8.6 Methods Considered but Not Reported

Two methods discussed in [Chapter 2](#) were considered during portfolio design and are deliberately absent from the reported matrix. [S-Metric Selection Evolutionary Multiobjective Algorithm \(SMS-EMOA\)](#) is an indicator-based alternative whose survivor selection uses hypervolume contribution [\[4\]](#); it has no implementation in the production optimizer stack, so no configuration of it was ever exercised. Simulated annealing [\[35\]](#) likewise has no implementation in the production stack; residual configuration stanzas for both survive in legacy run files but are not modeled by the loader and reach no optimizer. Neither absence is a finding about the methods themselves; both are scope boundaries of this study. One consequence should be stated plainly rather than left implicit: with annealing excluded, every method in the reported matrix is population-based, so the study carries no non-population search contrast.

Table 3.7: Conceptual roles of the active optimizer families in the comparison.

Optimizer family	Search style	Purpose in the comparison
RNSDE / PRNSDE	Differential-evolution family	Differential-evolution-family methods for mixed, event-driven design spaces
NSGA-II / ε -NSGA-II	Dominance-based EMOA family	Primary Pareto-front baseline under shared evidence and conservative scoring
AGE-MOEA2	Geometry-aware EMOA family	Tests alternative survivor-selection pressure on the same objective formulation
MOPSO	Swarm family	Contrasts velocity-based exploration with evolutionary search

3.8.7 Solution Archiving

The population-based optimizers in this study share a common external archive that maintains solution quality across generations independently of the optimizer’s internal selection logic. The epsilon-NSGA-II variant additionally maintains its external epsilon archive. This prevents

archive behavior from becoming an accidental advantage for one method over another while preserving the algorithm-specific archives required by each method.

The production archive is an `ArchiverSimple` strict-nondominated archive with capacity 4096 and a 262 144-row dominated-history safety cap. Rows are deduplicated on the exact decision vector together with its raw and transformed objectives, and the cumulative front is the strict nondominated front on the transformed objectives computed without reference to constraint violation; feasibility filtering is applied downstream, when the reportable front and the replay-candidate set are formed. Crowding selection produces only the capacity-bounded optimizer-facing subset and never trims the cumulative front itself. The optimizer’s population remains archive-unaware, so archive management does not interfere with the optimizer’s internal selection pressure.

Phenotype deduplication prevents redundant archive entries from inflating apparent frontier quality. During search, potentially feasible candidates advance through deterministic adaptive prefixes and may stop when the configured Beta PDF convergence condition is reached; a conclusive hard failure is standardized as infeasible and is never archived. After optimization, canonical replay takes the archive’s feasible strict-dominance front on the penalized search objectives, adds the feasible designs that are nondominated on the unpenalized physical pair and that the penalized front had excluded, deduplicates on exact decision bits, and re-evaluates every surviving candidate against the identical ordered full 64-scenario bank; the final population is not replayed except through the archive. Two front bases are therefore sealed per cell and never mixed: the search-archive front, which is the feasible strict front of the cumulative archive on the penalized objectives and is what the cell’s Pareto row hash binds, and the canonical replay front, which is priced on the full 64-scenario bank and is what the physical coverage-box evidence and the campaign witnesses are drawn from. Every reported value names which basis it uses. This separates adaptive search-time preservation

from full-bank canonical reporting while keeping replay common across methods.

3.8.8 Optimization Performance Metrics

To compare optimizer performance across solution quality, diversity, and computational efficiency, an offline analysis stage summarizes four complementary metrics widely used in multi-objective optimization [56], computed from the canonical fronts rather than inside canonical replay itself. In plain terms, they ask how much of the trade space a method reaches, how close it gets to a reference frontier, and how evenly it covers the solutions it finds. Hypervolume is evaluated against the sealed reference point; the remaining three are evaluated in the $\log_{10}$ objective space. All are restricted to the canonical front of one constellation family, objective mode, and fidelity group under matched evaluation controls.

Hypervolume (HV) measures the volume of objective space dominated by the reported front P relative to a common reference point $\mathbf{r}$ [80]:

$$\text{HV}(P; \mathbf{r}) = \lambda \left(\bigcup_{\mathbf{y} \in P} [\mathbf{y}, \mathbf{r}] \right), \quad (3.20)$$

where $\lambda(\cdot)$ denotes Lebesgue measure in objective space. HV is the only known unary indicator strictly monotonic with respect to Pareto dominance [79], so larger values indicate better joint convergence and frontier extent.

Inverted Generational Distance (IGD+) measures how well the reported front covers a reference front P^* [28, 68]:

$$\text{IGD}^+(P, P^*) = \frac{1}{|P^*|} \sum_{\mathbf{z} \in P^*} \min_{\mathbf{y} \in P} d^+(\mathbf{z}, \mathbf{y}), \quad (3.21)$$

where d^+ is the Pareto-compliant distance defined by Ishibuchi et al. [28]. Lower values

indicate that the computed front lies closer to the reference frontier while still covering its span. In this study P^* is not an independent frontier: it is the pooled cross-optimizer nondominated front over every cell of the same family, so each cell contributes to the reference against which it is scored and IGD+ reports relative position within a family panel rather than distance to a true Pareto front.

Front extent replaces the classical spread indicator Δ of Deb et al. [14], which is not computed in this work. In the two-objective setting used here it is the sum of the reported front’s per-axis $\log_{10}$ ranges,

$$E(P) = \left(\max_{y \in P} y_1 - \min_{y \in P} y_1 \right) + \left(\max_{y \in P} y_2 - \min_{y \in P} y_2 \right), \quad (3.22)$$

so higher values indicate a wider trade span. It measures coverage extent only and, unlike Δ , says nothing about uniformity between the extremes.

Nearest-neighbour spacing measures the variability of nearest-neighbour distances along the front. It is related to, but is not, the classical spacing indicator of Schott [61], which uses Manhattan distance and an $N - 1$ divisor; the figures label it “NN spacing (log)” for that reason. Distances are Euclidean in the $\log_{10}$ objective space used throughout:

$$S = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\hat{d}_i - \bar{\hat{d}} \right)^2}, \quad (3.23)$$

where $\hat{d}_i$ is the nearest-neighbor distance for solution i . Lower values indicate a more uniform frontier without large clustering gaps.

Two campaign shapes are reported, and neither is a paired replicate design. The *Exact36* matrix is a one-seed descriptive matrix: six optimizers, three families, and two fidelity lanes on the single root seed 41127203, giving 36 cells. The *Intersect108* matrix adds two further seeds,

41127204 and 41127205, under a cooperative three-island K_3 scheme with migration and generation barriers, giving 108 cells arranged as 36 three-island cohorts. The islands within a cohort exchange information by construction and are therefore **not** independent replicates. Migration is bounded and decision-only: at each of four generation barriers all three islands halt, freeze their fronts, and each exports at most four designs (its two per-objective extremes plus a maximum-minimum-distance diversity fill) to each of the other two islands, which accept at most eight after phenotype deduplication; a migrant travels as its decision vector alone and is re-evaluated under the receiving island’s scenario view before injection, and the last barrier coincides with the final generation. Common-random-number alignment is per island: two optimizers compared at the same island index see the identical scenario order, while the three islands of a cohort are deliberately given disjoint opening blocks and different rotations, so a cross-island difference is not a controlled comparison. Intersect108 adds one stage that Exact36 does not have: after the final barrier the cohort’s surviving decisions are replayed on all three island scenario views, only designs every view admits are kept, the result is folded into one cohort archive, and canonical replay is then performed once per cohort, so Intersect108 reports 36 cohort-level canonical fronts over its 108 search cells. The two shapes are not run at the same depth: Exact36 runs 200 generations per cell and Intersect108 runs 400 per island, so budgets are matched within a shape and never across the two. The two fidelity lanes are not two views of one evaluation: every cell searches on the same [Mean-Field \(MF\)](#)/ J_2 physics, and the lanes diverge only at canonical replay, where **hybrid** cells lower the mixed-fidelity front under strict [High-Fidelity \(HF\)](#) authority; **mf** and **hybrid** cells are therefore never pooled and only same-lane cells are compared; [Section 5.4](#) records that the declared HF acceptance replay did not execute in the reported campaign. Both shapes were last flown before the 2026-08-29 bank regeneration and the Flower fairness envelope, and [Chapter 4](#) records that every campaign-level value is historical.

All families share one sealed physical-coordinate hypervolume reference point at 2.5 km/s and 1000 kg, which is the pair of hard resource limits; because the reference does not vary by family, per-cell hypervolumes are on a common scale. Because neither reported matrix supplies independent replicates, no interval estimate or directional-agreement statistic over optimizers is computed. Exact36 provides one realization per cell, and Intersect108's three islands are cooperative by construction; in both cases a seed-paired difference and its bootstrap distribution are *undefined* rather than merely unimplemented, because there is no exchangeable set of paired differences to resample. This is a property of the campaign design and is not resolved by additional compute on these configurations. Optimizer comparison is therefore reported as a single-realization descriptive panel: per-cell hypervolume against the shared reference point, realized-work accounting, and final-front quality, with no uncertainty quantification attached and no ordering claim of any kind. Establishing an inferential comparison would require an independent-replicate campaign shape, which is identified as future work in [Section 5.4](#). Missing, duplicate, or mismatched cells fail closed; no reduced cell count or imputation is allowed. No global or per-family optimizer winner is permitted, and no analysis artifact emits a winner field of any kind.

3.9 Mission-Specific Objective Functions

The objective formulation is intentionally conservative so mission-level comparisons remain interpretable under stochastic outcomes. Search uses Beta posterior, PDF-convergence, and [HPD](#) diagnostics as operational controllers over deterministic prefixes; they are not service-feasibility statistics. Canonical archive admission and resource reconstruction are separate operations that always replay the full fixed bank.

The mission is to design a dust-deployer constellation that can remediate unsafe debris-debris conjunctions with minimal resource usage. Let $x \in \mathcal{X}$ denote the constellation design (e.g., orbital altitudes and inclinations, numbers of planes and satellites per plane, or repeating-ground-track indices depending on the family; see [Section 3.6.1](#)-[Section 3.6.3](#)). For the fixed catalogue-conditioned natural conjunction bank Ω , evaluating x produces:

- $\Delta V_{k,j}(x)$: executed maneuver cost (km/s) for event k and transfer–release alternative j , equal to phasing-burn norm plus HF-replayed transfer-burn norm plus release-control norm.
- $v_{\text{rel},k}(x)$: dust-primary relative speed (km/s) at intercept, used in deterministic momentum-transfer physics checks.
- $M_{k,j}^{\text{dust}}(x)$: raw dust finite-disk-capture-inverted released mass (kg) for the same alternative.
- $I_k(x) \in \{0, 1\}$: scenario success indicator (1 if transfer and strict deterministic intercepted-mass checks both pass; 0 otherwise). Failures include timing-window violations, transfer infeasibility under configured solver limits, intercept miss-distance tolerance violations, deterministic-mass status failures, and other non-finite numerical failures.

It is useful to read the objective construction in three layers. The first layer is scenario-level physics: each unsafe scenario returns a nondominated set of fully evaluated maneuver–mass

alternatives and a binary success or failure status. The second layer is mission bookkeeping: those fronts are aggregated into search-controller diagnostics and exact finite-bank coverage boxes. The third layer is optimizer exposure: only after the mission meaning is fixed are search regularizers applied. That ordering keeps physical objectives separate from numerical search convenience.

The same overlap geometry remains the operative interpretation here: the released-mass requirement is read from the retained contour across the likely target region on the encounter plane, not from a single nominal intercept point. That interpretation is carried into optimization without introducing a second standalone overlap figure.

Default evaluations draw from the catalogue-conditioned natural conjunction bank spanning the mission screening horizon. During search, each design follows one exact run-level CRN order at prefix targets

$$b_n = \min\{N, X + Yn\}, \quad n = 0, 1, 2, \dots,$$

where X is the initial batch size and Y is the follow-up batch size. The production values $X = 8$ and $Y = 4$ over the sealed $N = 500$ search bank yield $8, 12, 16, \dots, 500$ in 124 stages, but later prefixes are skipped after reported PDF convergence or a conclusive hard failure, and under the sealed posterior stop rule the ladder saturates by 24 events in practice, so the island's compiled balanced orthogonal-array opening block of eight events carries most of the reported search evidence. The canonical 64-event holdout bank is a separate fixed set used for replay and is never laddered. A design that reaches the last configured prefix without converging simply stops there; its row is recorded with the typed stop reason **BetaEventBankExhausted** and is reduced, archived, and reported on the same footing as a converged row. A feasible event retains every nondominated transfer-release alternative;

a provisional CVaR scalar may be recorded for search diagnostics, but it is not canonical objective authority.

Search control is handled with a beta posterior over Bernoulli scenario outcomes. If n_s and n_f are successes/failures over the currently exposed prefix and θ is success probability, then with prior $\text{Beta}(\alpha_0, \beta_0)$ (set to the uniform $\text{Beta}(1, 1)$ in the baseline configuration to avoid injecting informative prior knowledge),

$$\theta(x) \mid \mathcal{D} \sim \text{Beta}(\alpha_0 + n_s(x), \beta_0 + n_f(x)). \quad (3.24)$$

Two bookkeeping mechanisms sit alongside the ladder. Decisions repeated exactly within or across generations reuse a cached event record rather than re-running physics, so realized physics trials are reported separately from objective evaluations; and any event reported feasible whose cost falls outside the compiled feasibility contract is re-counted as a failure and forced to a worst-case constraint violation, so a malformed record can never improve a design.

The posterior mean is a search-controller statistic only:

$$s_{\text{post}}(x) = \mathbb{E}[\theta \mid \mathcal{D}] = \frac{\alpha_0 + n_s(x)}{\alpha_0 + \beta_0 + n_s(x) + n_f(x)}. \quad (3.25)$$

Highest-posterior-density intervals at confidence c (default 0.95) may control prefix advancement and identify conclusive hard failure, while the posterior PDF convergence criterion may stop search once the configured concentration threshold is reached. Neither defines final service feasibility, coverage, or archive admission. Canonical replay independently forces all $N = 64$ events for every replay candidate; final service bookkeeping uses the exact full-pass fraction

$$\phi_{64}(x) = \frac{n_{\text{successes},64}(x)}{64}.$$

Let $\mathcal{A}_e(x)$ be every feasible, fully evaluated pair of replay-verified transfer trajectory and configured release fraction for unsafe event e . Its event front is

$$\mathcal{F}_e(x) = \text{ND} \left\{ \left(\Delta V_{e,j}(x), M_{e,j}^{\text{dust}}(x) \right) : j \in \mathcal{A}_e(x) \right\}, \quad (3.26)$$

where ND denotes nondominated minimization. No transfer top-1 choice or deterministic proxy mass enters this front. For a candidate maneuver cap d , define the lowest released mass achievable for event e as

$$m_e(d; x) = \min \{m : (d', m) \in \mathcal{F}_e(x), d' \leq d\}, \quad (3.27)$$

with $m_e = +\infty$ when no alternative satisfies the cap. For $N = |\Omega|$, configured coverage q , and $k_q = \lceil qN \rceil$, let $M_{(k_q)}(d; x)$ be the k_q th order statistic of $\{m_e(d; x) : e \in \Omega\}$. The exact canonical physical front is

$$\mathcal{C}_q(x) = \text{ND} \left\{ \left(d, M_{(k_q)}(d; x) \right) : d \in \mathcal{D}(x), M_{(k_q)}(d; x) < \infty \right\}, \quad (3.28)$$

where $\mathcal{D}(x)$ is the set of executed- ΔV values represented in the event fronts. Thus each $(f_1^{\text{raw}}, f_2^{\text{raw}}) \in \mathcal{C}_q(x)$ certifies that at least k_q unsafe events can be serviced inside the same resource box. With $N = 64$ and $q = 0.9$, $k_q = 58$. A 57-of-64 design is not removed from the objective set. It is classified into the mid band and contributes a single reported $(\Delta V, M^{\text{dust}})$ pair rather than a coverage-box front, together with a positive normalized constraint violation; its positive violation excludes it from both the canonical-replay candidate set and the reported front, so it is retained as archived evidence rather than ranked behind eligible designs, and it can never masquerade as a 90%-coverage solution. During adaptive search the same reduction is applied to the exposed prefix rather than to the full bank, so a search-archive coverage box certifies $\lceil q b_n \rceil$ of b_n exposed events; only canonical replay, which forces all 64, certifies

against $k_q = 58$, and reported values therefore always name their point-set basis. The same band rule is applied twice on different evidence: during search it runs on the posterior mean over the ladder prefix and decides only archive eligibility, and during canonical replay it is recomputed on the exact 64-event pass, which is the only banding that carries reporting authority. A design below the 0.5 hard-fail success rate contributes the sentinel pair instead, or a tail-cost pair floored at the same caps when it serviced no event. Only designs at or above k_q successes yield coverage boxes, and only those can certify the canonical front. Provisional scalar summaries such as CVaR-90 remain diagnostics only. For each canonical physical pair, the optimizer-facing transformed objectives are

$$\begin{aligned} F_1(x) &= \log_{10}\left(\max(f_1^{\text{raw}}(x), 10^{-4} \text{ km/s})\right) + \Pi_{\text{size}}(x), \\ F_2(x) &= \log_{10}\left(\max(f_2^{\text{raw}}(x), 10^{-3} \text{ kg})\right) + \Pi_{\text{size}}(x). \end{aligned} \tag{3.29}$$

The log floors are fixed numerical guards, not physical lower bounds, and Π_{size} is the canonical search regularizer in [Equation \(3.31\)](#). Any topology-continuation term belongs only to provisional scalar search and is not canonical row authority. Exact coverage support, raw physical objectives, transformed search objectives, and diagnostic posterior/CVaR fields remain separate in the artifact.

The decision space $\mathcal{X}$ depends on the constellation family used in a particular experiment (parameters not listed are fixed in the baseline runs):

- Walker: in the reported profile, $x = (h, i, P, S, f_{\text{rel}})$ with bounded integer P (planes), S (satellites per plane), and a Walker relative-spacing phasing index; when those degrees of freedom are disabled the family collapses back to the older fixed-topology $x = (h, i)$ baseline.
- Dual-Shell Walker-Delta: in the reported profile, each shell carries its own bounded

altitude, inclination, integer topology, and phasing variables; the second shell also carries RAAN and mean-anomaly offsets. Earlier baseline studies fixed those shell cardinalities.

- Flower: in the reported profile, $x = (h_{\text{req}}, i, e, \omega, N_o, N_{so}, N_c)$. The integer topology variables N_o, N_{so}, N_c define the 2-D lattice layout, while the requested altitude is snapped to the nearest compatible J_2 -secular repeat-groundtrack root before evaluation.

All three families use the same objective evaluation workflow and yield (F_1, F_2) for the configured optimizer comparison, spanning differential-evolution, dominance-based, indicator-based, and swarm methods. Operationally, the logic is simple: scenario physics first, mission bookkeeping second, optimizer exposure third. The same scenario-to-mission flow reappears later in Algorithm 2.

3.9.1 Mission-Scoring Workflow

The study’s conservative scoring semantics are important enough to state procedurally. Adaptive search may stop at reported PDF convergence, at a posterior hard-failure verdict, or at the end of the ladder. A second stop mode, in which the success-band call itself ends the ladder in either direction, is implemented but is not the mode the reported campaign runs. Canonical replay is separate: every replay candidate is evaluated against the exact full bank before coverage reconstruction or canonical archive admission.

Algorithm 2 Mission-scoring workflow for one candidate constellation under the production profile.

- 1: **Input:** constellation design x , fixed bank Ω_{unsafe} of size N , batch sizes X, Y , prior $\text{Beta}(\alpha_0, \beta_0)$, confidence level c , and coverage fraction q
 - 2: Decode x into deployer orbits
 - 3: Take the sealed island-specific scenario order derived from the bank's master order and compiled rotation and head schedule
 - 4: Initialize success count $n_s \leftarrow 0$, failure count $n_f \leftarrow 0$, and hard-failure flag $h \leftarrow 0$
 - 5: **for** $n = 0, 1, 2, \dots$ with prefix length $b_n = \min\{N, X + Yn\}$ **do**
 - 6: Evaluate and cache each scenario's exact nondominated transfer-release front $\mathcal{F}_k$ and success indicator I_k
 - 7: Update n_s, n_f , form $\text{Beta}(\alpha_0 + n_s, \beta_0 + n_f)$, and compute the HPD interval for search control
 - 8: **if** the the posterior mean and the upper end of the credible interval both fall below the hard-fail success threshold **then**
 - 9: Set $h \leftarrow 1$; standardize candidate as infeasible; do not archive
 - 10: **break**
 - 11: **if** the configured posterior PDF convergence condition is met **then**
 - 12: Record search convergence and **break**
 - 13: **if** $b_n = N$ without convergence **then**
 - 14: Record **BetaEventBankExhausted** and stop on the same footing as a converged row
 - 15: **Canonical replay:** form the replay-candidate set as the union of the constraint-feasible archive front in penalized search coordinates and the constraint-feasible front nondominated on the raw physical pair, deduplicated on exact decision values
 - 16: Re-evaluate every replay candidate against the same ordered full $N = 64$ bank, independent of search stopping prefix
 - 17: Set exact service count and $\phi_{64} \leftarrow n_{\text{successes}, 64}/64$; set $k_q \leftarrow \lceil 64q \rceil$
 - 18: Sweep executed- ΔV caps over all event fronts and construct $\mathcal{C}_q(x)$ using the k_q th event-mass order statistic
 - 19: Classify each row by success band; emit coverage boxes only for qualified rows, a single reported pair for mid-band rows, and the sealed sentinel pair for hard-fail rows, retaining each replayed row's per-event evidence
 - 20: Apply log-space/search regularizers separately; admit only exact support-bearing rows to the canonical Pareto front
 - 21: **Return:** final objective pair, ϕ_{64} , posterior/HPD search diagnostics, and cached scenario-level bookkeeping
-

This workflow keeps search control separate from canonical service semantics. Posterior/Beta PDF and HPD values control adaptive prefix progression, convergence, or conclusive hard-failure termination, but canonical coverage uses exact full-bank event counts and event-front evidence. The canonical front is the nondominated union of exact $(\Delta V, M^{\text{dust}})$ coverage boxes, with every supporting event, transfer identity, and release fraction retained in the artifact.

3.10 Constraint Functions

Constraint handling is organized in three layers: design-variable bounds and integrality, transfer/intercept feasibility, and mission-level service requirements. Many checks are enforced as hard feasibility conditions that convert candidate events into failures in the beta-dist accounting, while mission-level success and mass-limit requirements accumulate into a separate normalized constraint-violation channel.

The distinction between those two roles matters. Event-level hard gates answer whether a hazardous opportunity is actually serviceable under the study’s physics and timing rules. Mission-level constraint violations answer a different question: among architectures that can service at least part of the hazardous set, which ones do so with acceptable reliability and burden? Keeping those roles separate prevents the optimizer from treating a physically invalid event as merely expensive rather than genuinely infeasible.

Design-variable bounds and integrality (per constellation family):

- Walker: $h \in [350 \text{ km}, 2000 \text{ km}]$ and $i \in [0.5 \text{ deg}, 165 \text{ deg}]$, with bounded integers $P \in [2, 8]$ and $S \in [2, 5]$. A normalized phasing variable decodes to the reported integer f .
- Dual-Shell Walker-Delta: $h_1, h_2 \in [350 \text{ km}, 2000 \text{ km}]$ and $i_1, i_2 \in [0.5 \text{ deg}, 165 \text{ deg}]$, with $P_1, P_2 \in [2, 4]$ and $S_1, S_2 \in [2, 5]$. Each normalized shell-phasing variable decodes to an

integer f ; the second shell also carries RAAN and mean-anomaly offsets.

- Flower: $h_{\text{req}} \in [350 \text{ km}, 2000 \text{ km}]$, $e \in [0, 0.3]$, $i \in [0.5 \text{ deg}, 165 \text{ deg}]$, $\omega \in [0 \text{ deg}, 360 \text{ deg}]$, $N_o \in [2, 8]$, $N_{so} \in [2, 5]$, and $N_c \in [0, 7]$ with decoded designs requiring $N_c < N_o$, a compatible repeat-groundtrack root, and snap error no larger than 50 km; incompatible or over-cap designs are rejected.

Transfer/intercept feasibility:

- Time window: feasible events must admit positive phasing, waiting, and transfer timing while preserving the minimum required lead time before conjunction.
- Maneuver and search limits: transfer feasibility is enforced by phase and transfer ΔV caps, Lambert multi-revolution limits, and bounded candidate verification depth without staged budget escalation.
- Hard maneuver gate: the accepted event transfer must satisfy $\Delta V \leq 2.5 \text{ km/s}$; this is a hard gate, not a soft objective preference.
- Intercept geometry: candidate solutions must satisfy the baseline intercept tolerance of 25 m at the dust interaction point.
- Orbit-safety checks: minimum Lambert-time, perigee, apogee, deployer–target minimum-separation, and revolution-cap guards must pass before the event can proceed to dust sizing.
- Deterministic-mass validity: the strict event-feasibility gate must return finite, status-valid intercepted-mass values; otherwise the event is infeasible.

Mission-level service requirement and penalty:

- Coverage target: canonical eligibility requires the exact full-pass count $n_{\text{successes},64} \geq \lceil 0.9 \times 64 \rceil = 58$. Search HPD/Beta state is operational controller data only; reported PDF convergence or conclusive hard failure may stop an adaptive prefix, while pool exhaustion without convergence is typed and comparison-ineligible.
- Event filtering: safe-by-default conjunctions are excluded by policy, and only unsafe events with finite $(\Delta V, v_{\text{rel}}, M^{\text{dust}})$ and valid deterministic-mass status contribute to objective aggregates.
- Resource guardrails: the accepted event transfer must pass the hard $\Delta V \leq 2.5 \text{ km/s}$ gate. Released mass must remain below the 1000 kg numerical/model sanity ceiling. Both resources remain Pareto coordinates; the mass ceiling is not hardware capacity.

Canonical coverage eligibility is the exact integer condition

$$g_q(x) = k_q - n_{\text{successes},64}(x) \leq 0, \quad k_q = \lceil qN \rceil = 58. \quad (3.30)$$

Each counted event must contain at least one exact feasible resource pair after the maneuver, orbital-safety, deterministic-mass, dust finite-disk capture, HF, and mass-sanity gates. Search may retain a normalized constraint-violation scalar for internal feasibility ordering, but canonical admission is re-derived from the exact count and support evidence rather than trusting that scalar. HPD/Beta values are likewise operational diagnostics, not canonical statistics.

Constellation Size Regularizer

With variable-topology optimization enabled, the optimizer can freely choose the number of planes P and satellites per plane S , up to the configured upper bounds. Left unconstrained,

the search naturally gravitates toward the maximum allowed constellation size because more satellites provide better geometric coverage at a purely geometric level. To encourage leaner designs when the performance benefit of additional satellites is marginal, a constellation size regularizer is added to both objectives:

$$J_i^{\text{search}}(x) = J_i^{\text{physical}}(x) n_{\text{sats}}(x)^\alpha, \quad \alpha = 0.10, \quad i \in \{1, 2\}, \quad (3.31)$$

where n_{sats} is the realized satellite count decoded from the family-specific design vector. Physical objectives and search objectives are retained as separate artifact fields. The $n_{\text{sats}}^{0.1}$ factor is a search regularizer, not a claim that satellite count changes the underlying physics; dominance during search uses J_i^{search} , while physical reporting retains J_i^{physical} .

3.10.1 Unsafe-Only Semantics and Bookkeeping

A key bookkeeping rule in this work is that safe-by-default events do not contribute to the architecture score. The mission is intended to service hazardous conjunctions, not to claim success on conjunctions that never required intervention in the first place. Including safe-by-default events in the success denominator would therefore inflate apparent reliability by rewarding the architecture for events on which it never needed to act.

The same logic constrains resource interpretation. The canonical method does not average over the full catalogue indiscriminately or give credit for safe-by-default cases. It constructs exact coverage boxes from unsafe-event fronts; infeasible events remain explicit as $+\infty$ support at every resource cap. CVaR-90 may be retained as a provisional scalar diagnostic, but it cannot replace the event-front evidence.

This separation yields a bookkeeping structure with a clear physical interpretation. Unsafe events define the service obligation. Event-level feasibility determines whether a hazardous

opportunity is serviceable. Exact coverage counts determine canonical eligibility, and the nondominated coverage-box front preserves the maneuver–mass trade space needed to service the required fraction.

3.11 Dust Propagation

Dust uncertainty is represented by the compiled single-Gaussian production state under the same two-lane dynamics setup used elsewhere. Its seven-point Julier simplex is propagated and rebuilt into the mean and covariance used in encounter-plane mass sizing and downstream reporting. The operational sequence is summarized later in Algorithm 3, while Appendix A.1 gives the mathematical details. Historical split-capable GMM K=3, Merwe-13, and Monte Carlo comparison arms remain useful only as counterfactual validation context; they do not describe the current compiled path.

Under the active baseline, catalogue construction remains on the analytical/ J_2 path; production transfer candidate generation/verification and canonical replay require strict HF authority. Dust propagation starts at the stamped post-control release state, is capped at two hours by the physical timeline, and uses fixed 5×5 spherical-harmonic gravity, drag, SRP, Sun, and Moon perturbations with interpolated ephemerides. Optional force terms remain sensitivity-study configurations.

Detailed derivations (sigma-point construction, moment reconstruction, and the historical split-axis construction) are moved to Section A.1. The production six-state transform uses the seven off-centre Julier simplex points with equal positive weights and no central point.

The Julier transform supplies a deterministic local description of how nonlinear dynamics deform the compiled Gaussian over the analysis horizon. Its propagated points are collapsed back into an updated mean and covariance; no dense Monte Carlo cloud or adaptive- α selection occurs in production. The bounded 75-sample selector belongs to a historical comparison arm.

The compiled production tuple is K1 plus Julier-7. Its split controls retain `maxvar`, scaled time-of-flight α , and rank 1 as a dormant GMM K=3 policy: with one component, no

split is applied. Historical GMM K=3, Merwe-13, adaptive- α , and Monte Carlo arms are counterfactual validation designs rather than current result authority.

In the validation protocol, Wasserstein distance is used as the primary discriminator because it measures how much probability mass must be moved to turn the approximation into the reference distribution [18, 69]. That is a natural fit for this problem: the method must preserve transported cloud shape well enough to support later encounter-plane decisions. No split-comparison outcome is asserted in this methodology chapter; numerical comparisons belong in results artifacts.

If the dormant split route is exercised in a counterfactual GMM K=3 configuration, `maxvar` selects the dominant covariance direction and `split_rank = 1` limits splitting to that mode. The sealed scaled time-of-flight policy sets $\alpha = 0.35\alpha_{\max}$ through 7200s and $\alpha = 0.60\alpha_{\max}$ above that boundary. These controls do not split the compiled K1 state.

Historical short-horizon and multihorizon designs used Wasserstein distance as a primary distributional diagnostic. They do not establish current production outcomes. Current result claims must come from the compiled K1, Julier-7 path and its current evidence records.

The force-model definition is stated directly in the lane policy above: MF/ J_2 handles catalogue screening, transfer seeds, and the full evaluation of the `mf` cells; in the `hybrid` cells, production transfer generation/verification, canonical replay, fixed-canister coast, and fixed-dust propagation use strict HF authority. Untraceable force-model and split-outcome figures are intentionally excluded from the methodology evidence chain.

Table 3.8: Derivation and validation fields for the configured UT/splitting methodology.

Evidence source	Scope	Recorded diagnostic	Method implication
Historical short-horizon criterion design	Four representative LEO scenarios at the $p90$ horizon	Proposed Wasserstein summaries, seeded repeats, and criterion diagnostics.	Counterfactual protocol only; no current outcome follows.
Historical multihorizon design	Representative scenarios over 10 min to 180 min with seeded repeats	Proposed Wasserstein, MaDEM, MCR, CvM, ELK, NISE, runtime, and coverage.	Counterfactual protocol only; rerun before making a result claim.
Compiled production tuple	Current method authority	K1 + Julier-7; dormant <code>maxvar</code> + scaled-TOF α + rank-1 split controls.	Do not infer GMM $K=3$ or split-comparison outcomes from the current configuration.

3.12 Optimization Setup

This section summarizes how the full optimization workflow connects to the simulation framework and how objectives are reconstructed for fair cross-method comparison. The workflow begins with a catalogue-conditioned natural conjunction bank, evaluates each candidate constellation through the transfer and dust-mass chain, uses Beta/HPD only for adaptive control, and publishes exact coverage-box Pareto objectives through canonical replay.

Each design is evaluated against a catalogue-conditioned natural conjunction bank spanning the mission screening horizon. Bank generation follows the catalogue-driven MF/ J_2 screening workflow and is stored for repeatability. Lambert and J_2 operations supply transfer seeds; on the **hybrid** lane, retained transfer candidates are verified under strict HF authority; canonical replay then revalidates all input pairs and carries the fixed canister and dust arcs from their stamped epochs.

For each scenario, the transfer solver executes the staged phasing-plus-transfer workflow described in [Section 3.3](#). The catalogue intercept epoch remains fixed. Search may use proxy ranking, but canonical evaluation sets the pair cap to zero, replays all native input pairs, crosses every surviving transfer with every configured release fraction, and retains the nondominated exact resource front.

At the planned intercept, the dust estimator projects both dust and target states into the encounter plane and computes released-mass requirements. Deterministic intercepted-mass validation is then enforced under the strict event-feasibility policy; failures are treated as infeasible with no fallback escalation path.

Across events, the beta-distribution posterior mean, PDF convergence, and [HPD](#) interval serve only as adaptive-search diagnostics. Canonical replay separately completes the exact 64-scenario pass for every replay candidate, where event-front evidence supplies the $k = 58$

coverage condition and the nondominated maneuver–mass resource trade-offs.

The method compares robust multi-objective routines under matched evaluation controls. Population scales, generation counts, and archive capacities are configured so that algorithm behavior is read under common budgets and evidence. CRN seeding and the fixed scenario bank enforce fairness; canonical replay removes provisional scalar aggregation as an authority boundary. One further verification stage sits below canonical replay: across the whole campaign, the two extreme canonical points, lowest maneuver cost and lowest released mass, and the exactly 58 replay events that support each of them are re-derived independently under the intercept position and velocity witness tolerances, and the result is sealed as a content-addressed witness. That witness certifies the campaign’s two extrema, not every reported row; the remaining rows carry replay provenance rather than an independent re-derivation.

The workflow separates optimizer comparison from later architecture interpretation while preserving one evidentiary regime: the MF catalogue lane supplies the natural conjunction bank, the beta-distribution objective bundle supplies the mission semantics, and the scenario bank remains fixed while optimizer and family inputs change. Any later architecture interpretation uses the same pipeline rather than introducing a second evaluation path.

This methodology chapter defines shared evidence, physics, scoring, replay, and optimizer semantics. Campaign sizes, realized fronts, and outcome claims remain outside the method definition.

3.12.1 Comparison Fairness Controls

The optimizer comparison is only interpretable because the run controls are intentionally strict. Every optimizer family sees the same natural conjunction-bank protocol, the same transfer and dust-mass semantics, and the same mission-level constraint logic. Without

shared evidence, nominally matched evaluations would still be confounded by differences in which scenarios were exposed to each method.

Three fairness controls are especially important. First, [CRN](#) sampling ensures that when unsafe prefixes are drawn, algorithms are exposed to aligned evidence streams rather than independent stochastic worlds. The scenario order is sealed per island from the bank’s master order, a compiled rotation, and a compiled orthogonal-array head; generation, optimizer, and design identity do not change it, so every method in an island sees the same evidence stream. Second, matched evaluation budgets prevent an apparent algorithm advantage from being explained mainly by larger opportunity to search. Third, reliability and resource semantics are fixed before comparison, so algorithms differ only in search behavior and not in how mission value is defined.

A fourth control governs initialization for all six optimizers, including AGE-MOEA2. Every method draws a uniform-random initial population within the active family bounds under its configured seed, so no method starts from a curated or pre-evaluated advantage. Seed-matrix blending from a prior run was designed for an earlier pipeline and is not active in the reported campaign; its configuration keys are not read by the current run loader and no seed file participates.

The production profile fixes the matrix more tightly than a generic benchmark study would. The conjunction source stays on the MF catalogue lane, adaptive prefixes use the top-level beta-dist controller, and final service/resource authority comes from canonical replay over exact coverage boxes under the current compiled science authority. One shared parallel-execution policy prevents optimizers from defining different computational experiments.

These controls are not incidental experimental details. They are what make the optimizer results interpretable. This work is not interested in whether one algorithm can appear

superior under a favorable sampling realization; it is interested in whether algorithm families behave differently when the architecture problem itself is held fixed. That is why the run design prioritizes evidence control and reporting consistency alongside the usual concerns of hypervolume, spacing, and computational cost.

Table 3.9: Common optimizer configuration principles used across studies. Exact run settings are reported with each generated study artifact.

Method	Setting	Value/Notes
RNSDE	Population budget	Moderate population and archive with $\mathcal{O}(10^2)$ generations; matched by total objective calls to peer methods
RNSDE	Variation	Differential mutation/crossover with diversity-preserving survival and periodic refresh of low-diversity regions
PRNSDE	Local refinement	RNSDE-derived search with configuration-controlled restart and local-refinement behavior for additional exploitation
NSGA-II	Core mechanics	Simulated-binary crossover, polynomial mutation, Deb-style feasibility ranking
ε -NSGA-II	Stability control	NSGA-II variant with epsilon boxes and restart controls to reduce rank churn under noisy evaluations
AGE-MOEA2	Selection pressure	Geometry-aware EMOA used as an alternative frontier-shaping baseline
MOPSO	Swarm budget	Swarm and archive sizes matched to common evaluation budgets; bounded updates with diversity-aware leader selection
Shared	Policy	CRN-aligned Beta/HPD search control, exact 64-event $q = 0.9$ coverage fronts, fixed MF catalogue scope, and one replay lane per cell: MF/ J_2 search physics in every cell, with canonical replay on MF/ J_2 in the <code>mf</code> cells and strict-HF lowering configured in the <code>hybrid</code> cells

Algorithm 3: Production Julier-7 Dust Transform

- 1: **Input:** release mean $\boldsymbol{\mu}_0 \in \mathbb{R}^6$, covariance $\boldsymbol{P}_0 \in \mathbb{R}^{6 \times 6}$, horizon τ , and dynamics map $\boldsymbol{F}_{t \rightarrow t+\tau}$
- 2: **Output:** propagated mean $\boldsymbol{\mu}^+$ and covariance $\boldsymbol{P}^+$
- 3: Bind the compiled production component count $K = 1$
- 4: Retain `maxvar`, scaled-TOF α , and rank 1 as dormant split controls; form no child

components at $K = 1$

- 5: Compute a covariance square root $\mathbf{S}$ such that $\mathbf{S}\mathbf{S}^\top = \mathbf{P}_0$
- 6: Generate the authority-pinned Julier minimal-skew simplex offsets $\{\boldsymbol{\xi}_c\}_{c=1}^7$
- 7: Omit the zero-weight centre point and assign $w_c = 1/7$ to every retained point
- 8: **for** $c = 1, \dots, 7$ **do**
- 9: $\boldsymbol{\chi}_c \leftarrow \boldsymbol{\mu}_0 + \mathbf{S}\boldsymbol{\xi}_c$
- 10: $\boldsymbol{\chi}_c^+ \leftarrow \mathbf{F}_{t \rightarrow t+\tau}(\boldsymbol{\chi}_c)$
- 11: Rebuild the propagated moments:

$$\boldsymbol{\mu}^+ = \sum_{c=1}^7 w_c \boldsymbol{\chi}_c^+, \quad (3.32)$$

$$\mathbf{P}^+ = \sum_{c=1}^7 w_c (\boldsymbol{\chi}_c^+ - \boldsymbol{\mu}^+) (\boldsymbol{\chi}_c^+ - \boldsymbol{\mu}^+)^\top. \quad (3.33)$$

- 12: **Return** $(\boldsymbol{\mu}^+, \mathbf{P}^+)$

▷ GMM K=3/Gauss-Hermite splitting, Merwe-13, adaptive- α , and Monte Carlo selection are historical counterfactual arms, not production steps.

At this point the method definition is fixed: the catalogue-conditioned natural conjunction bank is shared, adaptive search follows exact CRN targets $b_n = \min\{N, X+Yn\}$, and canonical replay independently forces the full bank. Canonical service feasibility is $n_{\text{successes},64}/64$, and canonical front admission follows completed replay and canonical-archive rules. Transfer, dust-feasibility, objective, and optimizer semantics are explicit; downstream chapters interpret outputs without redefining this workflow.

Chapter 4

Optimizer Comparison and Architecture Evidence

This chapter defines the empirical readout for the dissertation’s two-part results argument. Part A compares optimizer families descriptively under shared hazardous-event evidence, fixed mission semantics, and actual-work reporting controls. Part B interprets deployer architectures from the complete provenance-bearing evidence pool without selecting an optimizer. The purpose of this chapter is therefore not merely to display fronts, but to keep optimizer evidence and architecture evidence separated so dissertation-level claims remain attached to the reported campaign evidence.

4.1 Two-Part Results Structure

The results structure follows the logic established in Chapters 1–3. Part A is a method-comparison question: how do optimizer families describe the maneuver–mass trade space when the event bank, transfer gates, dust-mass semantics, reliability scoring, and resource aggregation are held fixed? Part B is an architecture question: across the complete supported design evidence, what do deployer constellations reveal about staging geometry, topology, and service burden?

That separation protects the interpretation. A search method can navigate the configured

landscape effectively without proving that one constellation family is universally sufficient. Likewise, a constellation pattern is meaningful only when its supporting rows retain complete method, seed, family, and physical provenance. Architecture interpretation therefore compares evidence exposed by all methods descriptively; it does not depend on a global or per-family optimizer choice.

4.2 Part A: Optimizer-Family Comparison

Part A reports optimizer behavior through a fixed evidence contract. Every optimizer-family cell is interpreted against the same catalogue-derived unsafe-event protocol, the same beta-distribution mission semantics, the same transfer and dust-feasibility gates, the same [CRN](#) policy where event sampling is stochastic, and the same preference for realized objective-evaluation accounting over raw wall-clock timing.

The maneuver axis of the reported trade space is an *intercept* ΔV . As defined in [Chapter 3](#), it is the sum of the executed phasing burn, the replayed transfer burn, and the release-control burn. It excludes the arrival, or rendezvous, ΔV that would be required to match the target’s velocity; the runtime records that quantity as a diagnostic and never sums it into any reported total. The exclusion follows from the intercept concept rather than from a reporting choice, but its magnitude is large enough that it must be stated rather than left implicit. Across the three reference events of the sealed Stage-1 physics fixture (`crates/nd_pipeline/tests/fixtures/physics_3event`, frozen at commit `51dec426`, 2026-07-20), the recorded arrival delta- V magnitude is 13.33 km/s, 15.12 km/s, and 14.21 km/s, roughly $350\times$ the executed maneuver total for the same events. That fixture predates the current compiled-science authority (atmosphere model 8, Vern7 integrator, Julier-7 sigma set) by eight shipped reseals, one of which is recorded as moving every strict

high-fidelity endpoint, so the ratio is reported as an order-of-magnitude statement about the excluded term and not as a current campaign value. Every ΔV reported in this chapter is nonetheless an intercept ΔV —phasing, transfer, and release control only—and is not comparable to any published figure that includes an arrival or capture burn.

The exact $6 \times 3 \times 2$ cell grid—six optimizer methods, three constellation families, and two fidelity lanes—belongs at the end of this evidence chain. Two campaign shapes are reported over that grid. The *Exact36* matrix evaluates it on the single root seed 41127203, giving 36 cells and one realization per cell. The *Intersect108* matrix adds the seeds 41127204 and 41127205 under a cooperative three-island scheme with migration, giving 108 cells arranged as 36 three-island cohorts; the islands within a cohort exchange information by construction and are therefore not independent replicates. Because neither shape supplies independent replicates, every pairwise value remains strictly descriptive and no interval or agreement statistic is computed over the optimizer axis at all. No result selects, promotes, or carries one optimizer into Part B; older ordering diagnostics remain historical and non-authoritative.

Both shapes were last flown as campaign r55t (the internal label of the last matrix flight), closed out on 2026-08-28, on the third-generation (v3) event banks then in force and under the pre-correction Flower radial envelope. The representative banks (the 500-event search bank B500, the 64-event canonical holdout bank H64, and the reserve complement) were regenerated on 2026-08-29, and the shared-altitude Flower fairness envelope landed after the campaign; no matrix has been flown on either, so every campaign-level value in this chapter is historical, is scoped to that campaign, and supports no current family ordering. No figure in this chapter is drawn from a current campaign. The rendering suite in `tools/dissertation_plots` is operable and its outputs carry per-directory manifests, but no matrix rendered after the bank regeneration exists, so the reported evidence is presented in tabular and textual form only. Historical renders retained in the presentation material are labelled there as retired

configurations and back no claim in this chapter.

4.2.1 Screening-Lane Ranking Stability

The mixed-fidelity screening lane earns its role only if the candidate ordering it produces survives a change of fidelity, and that was measured rather than assumed. Every front candidate on the fixture’s three reference events was submitted for offline re-convergence under the mandated high-fidelity acceptance model, outside the reported campaign, and its ordering compared against the mixed-fidelity one. The measurement was made on 2026-07-25 by a diagnostic harness retired from the tree on 2026-08-12 (commit `dc4536`); it is recoverable only at its frozen source commit `6c91ed9f` and has not been re-run against the representative event banks regenerated on 2026-08-29. The target is held on the mixed-fidelity path in both arms, so the difference isolates transfer-arc fidelity and says nothing about catalogue-target error.

The ordering largely survives and the values do not. Rank correlation is high, with Spearman ρ between 0.93 and 0.97 and Kendall τ between 0.80 and 0.89. But the mixed-fidelity optimum is the high-fidelity optimum on only two of the three events; on the binding event—index 1 of the fixture’s three—it falls to rank 28 of that event’s 187 priced candidates, so a shortlist of at least $k = 28$ —about 15 % of those priced candidates, and 11.5 % of its 243-member front—is needed to retain the high-fidelity optimum across all three events. Per-candidate intercept ΔV moves by a median of 2.5 % to 9.2 % and a maximum of 264 % to 651 % across the three events. No per-event reported ΔV is therefore defensible at better than roughly 10 %, even where the ranking holds.

Four limits attach to that result and are stated because the numbers flatter the screening lane if quoted bare. First, the re-convergence is conditional: only the transfer burn is free, while

the phasing burn and all three timings stay pinned at their mixed-fidelity values, so each high-fidelity cost is an upper bound and the reported ranking disturbance is a lower bound rather than an estimate. Second, of the 704 front candidates across the three events, 105 failed to re-converge, and the failures concentrate in the expensive tail (best failed mixed-fidelity rank 24, 25 and 10 respectively), so they cannot overturn the first event's $k = 1$; the third event nonetheless carries one unpriced top-ten candidate, so its $k = 1$ is conditional and the binding event's $k = 28$ is a floor rather than an estimate. Third, the acceptance model is gravity-only 5×5 , matching the declared acceptance contract but omitting drag over these 4 h to 31 h horizons. Fourth, the measurement covers three events on one constellation design and does not generalize to the population sweep the optimizer actually runs. Like every other campaign-level value in this chapter, this ranking-stability result is historical and supports no current claim; the $k = 28$ shortlist is a property of the retired fixture, not live guidance. This measurement does not rescue the cross-optimizer comparison, and must not be read as doing so. It is a statement about candidate ordering within an event, which is the only level at which an ordering statistic is currently computable; the optimizer axis fails for the prior reason given above, namely that neither reported campaign shape supplies independent replicates.

4.3 Evidence Controls and Descriptive Completeness Gate

The bridge from Part A to Part B is an explicit descriptive-completeness gate. Architecture rows may be interpreted only when results remain traceable to common hazardous-event exposure, shared nominal controls, actual-work accounting, complete campaign-shape coverage,

and the same transfer and dust-feasibility semantics used elsewhere in the manuscript. This gate protects the architecture read from a favorable sampling realization, an inconsistent solver path, or a presentation-only front without choosing among optimizers.

Three checks are decisive. First, the compared optimizers see aligned stochastic evidence through the configured event-sampling and [CRN](#) policy. Second, the reported fronts reflect the same mission semantics, including strict deterministic intercepted-mass feasibility and conservative beta-distribution scoring. Third, every interpreted design remains attached to complete campaign-shape coverage and exact physical front evidence, with the campaign’s two extreme canonical points additionally re-derived under an independent witness, so siting discussion is traceable to the reported campaign rather than to an isolated artifact. Because both reported shapes provide one realization per cell, that traceability is a provenance guarantee and not a replication guarantee, and the siting discussion is worded accordingly.

4.4 Part B: Constellation-Siting Interpretation

Part B is the architecture-interpretation contract, not a report of architecture results. No such interpretation is supported by the evidence in this chapter: the last flown matrix predates both the bank regeneration and the Flower fairness envelope ([Section 4.2](#)), so no design-level, family-level, or siting statement is made here. When a post-correction matrix exists, Part B will interpret deployer architectures through the complete provenance-bearing design pool admitted by the descriptive-completeness gate, asking where supported designs stage their satellites, how topology variables are used, and whether structured families capture most of the useful trade space under that evidence. All optimizer methods will remain represented; architecture comparisons will not imply an optimizer winner. The interpretation will be organized at three levels:

- design-level patterns, such as altitude, inclination, plane count, phasing, and family membership among supported front members;
- family-level patterns, such as whether Walker, Flower, or Dual-Shell Walker-Delta designs occupy particular front regions;
- campaign-scope limits, including catalogue scope, event-sampling policy, solver depth, sensitivity settings, and objective semantics.

The phrase “pooled non-dominated set” is reserved for an explicitly defined set being plotted or summarized. The manuscript avoids retired universal-front wording and avoids implying a universal optimum over architectures outside the studied families and campaign controls.

4.4.1 Topology and Family Use

The topology discussion will report how the evidence uses bounded topology variables. Walker, Dual-Shell Walker-Delta, and Flower designs will be interpreted through their Chapter 3 variables and bounds, under the shared radial envelope and separation certificate now enforced on all three. All three evaluated families are analytically structured; an unstructured plane-constrained encoding was scoped for this study but was not evaluated, so no evidence in this chapter bounds what an unstructured architecture could achieve. Claims about median satellite count, bound saturation, or family sufficiency will be tied to post-correction campaign tables and are not imported from earlier diagnostic runs; in particular, no pre-correction Flower-versus-Walker ordering is carried forward, because the earlier Flower encoding did not share the realized radial envelope.

4.4.2 Reference and Sensitivity Comparisons

Reference-grid, worked-case, covariance-sensitivity, size-penalty, and archive-retention material will have distinct roles once such evidence exists; none is reported in this chapter:

- a worked case illustrates objective construction and is not treated as representative unless the relevant provenance supports that claim;
- a reference grid contextualizes optimizer performance under matched event, solver, and scoring semantics;
- covariance and size-penalty sweeps provide sensitivity evidence with explicit run-shape limits;
- archive-retention or cross-optimizer transfer tests are mechanics controls, not optimizer-comparison evidence unless the descriptive protocol defines them that way.

4.5 Chapter Evidence Discipline

The result interpretation is strongest when every numerical claim traces to an archived artifact, every figure caption names the plotted set’s scope, and every Part A or Part B conclusion stays at the strength warranted by the evidence. The chapter therefore treats stale process language, old campaign labels, and internal diagnostic names as evidence-quality problems rather than stylistic details.

Table 4.1: Evidence fields required for the Part A optimizer comparison.

Evidence field	Interpretation role
Campaign provenance	Identifies the result root, configuration family, population and generation shape, seed count, and archive/provenance sidecars used for reporting.
Realized-work accounting	Records realized objective evaluations, physics event trials, and configured generation depth per cell. Work is measured, not capped: the canonical loader refuses a hard evaluation budget, so no cell is made comparison-ineligible on budget grounds.
Canonical replay work	Canonical replay is accounted separately and never pads or reconciles optimizer-budget consumption.
Descriptive work milestones	Reports realized objective evaluations at common front-quality milestones as descriptive trajectories, never as a method-selection threshold.
Final-front quality	Reports archive quality after matched evidence and mission semantics as descriptive evidence with no selection threshold.
Campaign-shape completeness	Requires every declared optimizer-family-fidelity cell to be present for the reported matrix shape: 36 cells for the Exact36 matrix, and 108 cells arranged as 36 three-island cohorts for the Intersect108 matrix. Missing, duplicate, or mismatched cells fail closed.
Family-shared hypervolume	Reports single-realization hypervolume values in $\log_{10}$ -transformed maneuver-mass coordinates against one global reference corner fixed at the compiled feasibility caps (2.5 km/s, 1000 kg) with no margin. The reference is shared across all families and cells so box-fractions are mutually comparable; it is not derived from the data and is not a per-family reference. Every reported value additionally names its point-set basis—canonical $N = 64$ holdout front, or reconstructed search-archive front—because the two differ.
Pairwise description	Reports per-cell differences as single-realization descriptives. No interval estimate and no directional-agreement statistic is computed, because neither reported matrix supplies independent replicates.
Winner policy	No global or per-family optimizer winner is permitted, and no analysis artifact emits a winner field.

Chapter 5

Discussion and Interpretation Boundaries

This chapter defines how the empirical evidence is interpreted across optimizer and architecture reads. The stable discussion point is methodological: optimizer families and deployer architectures are compared under shared hazardous-event evidence, fixed mission semantics, and explicit uncertainty-aware scoring. Outcome-specific statements about descriptive optimizer behavior, architecture sufficiency, topology preference, and sensitivity margins remain attached to the evidence surfaces that support them.

5.1 How the Evidence Is Read

The discussion preserves the separation between method evidence and architecture evidence. Part A reports single-realization hypervolume against one global reference corner, realized-work accounting, and final-front quality descriptively, with no optimizer winner recorded anywhere in the artifacts and with no uncertainty quantification attached to any optimizer comparison. That restriction is not a stylistic conservatism. Neither reported campaign shape supplies independent replicates: the Exact36 matrix carries one seed per cell, and the three islands of an Intersect108 cohort are cooperative by construction and exchange information during the search. A seed-paired difference and the bootstrap distribution over such differences

are therefore *undefined* for the campaign shape actually flown, rather than defined and left unimplemented. No amount of additional compute on these configurations produces an interval estimate or a directional-agreement statistic, because there is no exchangeable set of paired differences to estimate one from; obtaining such a statistic would require an independent-replicate campaign shape, which is identified as future work in [Section 5.4](#). Part B supports an architecture claim only after the complete provenance-bearing design pool and relevant design sets have been scoped; it does not inherit a selected optimizer.

This structure prevents two common overreads. First, a strong optimizer result does not by itself imply that one architecture family is best everywhere. Second, a recurring siting pattern is not universal unless the evidence tests the catalogue, scoring, and sensitivity changes needed to support that strength of language. The discussion therefore distinguishes dissertation-level findings from campaign-scoped findings and from sensitivity or diagnostic evidence.

5.2 Astrodynamics Evidence in the Optimization Context

The optimizer-facing evidence matters because it remains tied to the astrodynamics developed in Chapter 3. A candidate architecture receives credit only when it passes the staged transfer solve, preserves usable lead time, satisfies the deterministic intercepted-mass gate, and produces a released-mass estimate that is defensible on the encounter plane. Those physical checks make the Pareto fronts interpretable as mission evidence rather than abstract optimizer outputs.

Each empirical pattern is therefore read through the physics that produced it. If a deployer

family stages near particular altitude-inclination bands, the interpretation asks whether that pattern follows from hazardous-event density, transfer-cost structure, topology constraints, or uncertainty-driven dust burden. If a sensitivity sweep changes released mass without materially changing maneuver cost, the discussion ties that result to the encounter-plane conversion rather than to generic optimizer behavior.

5.3 Contribution Closure Map

The dissertation’s research questions close through an evidence map rather than through unsupported outcome language, summarized in [Table 5.1](#).

The map keeps the discussion aligned with the manuscript’s structure. It shows that the dissertation has a coherent evaluation framework while tying outcome-specific closure to the appropriate evidence surface.

5.4 Limits on Present Claims

Several boundaries remain visible in the interpretation. Comparative statements are tied to the hazardous-event bank, shared nominal controls, actual-work accounting rules, family encodings, and scoring rules used in the reported study. Model-form conclusions are tied to the defended mixed-fidelity and dust-propagation baseline, and to a stated atmosphere scenario rather than to observed space weather. The reported campaign epoch lies beyond the observed solar and geomagnetic record, so the [JB2008](#) drivers are held at their last validated observed values past that cutoff under a declared persistence policy. Every density-dependent result—orbital decay, drag-driven dispersion of the released cloud, and any lifetime statement—is therefore conditioned on that scenario and is not a space-weather forecast.

Table 5.1: Evidence used to close each dissertation research question.

Question	Stable methodological basis	Outcome evidence
Item 1	Catalogue-first unsafe-event workflow and reusable hazardous-event evidence bank	Event-bank provenance and sampling scope
Item 2	Staged transfer, strict event feasibility, and released-mass objective construction	Maneuver-mass fronts and reference comparisons
Item 3	Shared-evidence optimizer-family matrix and actual-work reporting controls	Single-realization descriptive differences under per-island common-random-number alignment; no interval or directional-agreement statistic is defined for the flown campaign shapes
Item 4	Uniform-random initialization and the PRNSDE local-refinement contrast	Seeded and refined method behavior under the reported matrix
Item 5	Layered dynamics policy and Rust-backed evaluation path from Chapter 3	Run-shape, solver-depth, and fidelity-scope reporting
Item 6	Beta/HPD adaptive search diagnostics, and the exact $q = 0.9$ event-coverage front priced on the canonical $N = 64$ replay bank (58 of 64 events)	Sensitivity and uncertainty-scaling evidence; search-lane rows are single reported pairs
Item 7	Three-family architecture comparison and bounded family encodings	Part B family and topology interpretation

Architecture conclusions are tied to the studied Walker, Flower, and Dual-Shell Walker-Delta families, all three of which are analytically structured. An unstructured plane-constrained encoding was scoped as a fourth family but is not implemented in the reported evaluation stack and was not evaluated, so no evidence here bounds an unstructured architecture in either direction and the structured-sufficiency question is answered only within the structured set. Operational conclusions remain outside flight-readiness claims and respect the scope boundaries stated in Chapter 1. One methodological extension is identified here rather than

left implicit. Neither reported campaign shape supplies independent replicates, so no interval estimate and no directional-agreement statistic over the optimizer axis is defined for the evidence actually flown. Obtaining one requires a distinct campaign shape—independent seeds per cell with no migration between islands—which is a separate flight rather than a re-analysis of the reported package.

Intercept precision is not established. The sub-metre intercept closure the solver reports is self-consistency of one reduced model rather than physical closure: the declared high-fidelity acceptance replay does not execute in the reported campaign, so the acceptance check re-propagates with the same model that produced the candidate. When a legacy gravity-only 5×5 diagnostic replay is run offline against the fixture’s three reference events—matching the declared acceptance force contract but omitting drag, and explicitly not production Hybrid acceptance physics—the intercept miss is 82 km to 1713 km against a declared 25 m tolerance. The cause is the mean-versus-osculating element boundary at catalogue ingestion—the transfer propagator is secular J_2 on mean elements while catalogue targets enter as osculating states—and not a defect in the transfer solver. The reported per-event ΔV values remain valid as comparative search output, which is the role they play in this work, but neither the 25 m tolerance nor the sub-metre closure is a physical statement, and no per-event trajectory may be presented as flight-feasible.

The released-cloud model carries no grain-property spread. A prior and larger limitation attaches to the sizing threshold itself: the released mass is solved against a 1 km miss goal that traces to no requirement document, and the deterministic mass is approximately linear in that goal, so an operational screening threshold of 5 km to 10 km would move the reported mass floor by the same factor. The runtime additionally separates the immutable raw solver mass from a commanded mass equal to the larger of the raw mass and a compiled minimum-practical floor, so a reported mass at that floor is a commanded value rather than a solved

one. Absolute released-mass magnitudes are therefore conditioned on an unprovenanced threshold and must not be read as a requirement. Area-to-mass ratio, drag coefficient, and reflectivity coefficient are single scalars applied uniformly to the entire released cloud on the mixed-fidelity and hybrid paths alike, and no per-grain size or property distribution is modelled. Because differential drag and [SRP](#) scale with area-to-mass, that spread is a physical source of covariance growth over the free flight, so its omission is inherited by every lane rather than resolved by raising fidelity. Covariance realism is therefore not established at any fidelity. This limitation does not have a knowable direction: because collision probability is non-monotone in covariance size whenever the encounter-plane miss distance is nonzero, an under-realistic covariance does not bias the released mass one way or the other, and the reported values cannot be described as conservative or optimistic without characterizing where the true miss-distance and covariance pair sits relative to the modelled one. The inversion machinery itself is not implicated: the probability kernel is a short-encounter encounter-plane disk integral evaluated where its assumptions hold, and the mass solve inverts against packet count at fixed covariance, which is a monotone map. The released-mass figures should accordingly be read as relative diagnostics across scenarios rather than as calibrated absolute mass requirements.

Those limits strengthen the dissertation by defining how far each conclusion travels. The discussion is strongest where evidence, physics, and mission semantics agree, and most cautious where a result depends on one campaign shape or one sensitivity setting.

Chapter 6

Conclusions and Forward Work

This dissertation establishes dust-enabled just-in-time collision avoidance as a coupled architecture-under-uncertainty problem. Hazardous-event evidence, transfer physics, dust-cloud uncertainty, and optimizer comparison are evaluated within one traceable workflow rather than as separate analyses.

6.1 Core Contributions

The dissertation contributes four linked methodological pieces. First, it builds a catalogue-first unsafe-event evidence pipeline so candidate constellations are judged against reusable hazardous conjunction opportunities rather than disconnected synthetic examples. Second, it formulates dust-enabled [JCA](#) as a deployer-architecture problem, connecting fixed-epoch transfer replay, release control, deterministic intercepted-mass checks, raw-cloud capture sizing, and mission scoring in one traceable chain. Third, it develops a layered uncertainty-propagation and scoring framework, enabled by a custom multi-method Rust propagator using Battin’s Encke formulation, that separates adaptive Beta/[HPD](#) search diagnostics from exact canonical authority: the canonical $N = 64$ replay bank prices publication coverage at the exact $q = 0.9$ quantile, that is 58 of 64 unsafe events, and retains multiple maneuver–mass solutions on that lane, while search-lane rows are reported as single maneuver–mass pairs by construction because the sealed coverage-front band is [Qualified](#), which no search row

reaches under the sealed posterior stop rule. Fourth, it defines a shared-evidence descriptive framework across three architecture families and six optimizer methods while prohibiting global and per-family optimizer winners. Current attainment and operational risk-reduction claims remain held until evidence sealed under the current compiled science authority and receipt chain is independently reconstructed.

These contributions are independent of any one presentation of pairwise optimizer diagnostics or topology counts. The empirical conclusions attach to the reported evidence package, while the methodological contribution is the framework that makes those conclusions readable.

6.2 Empirical Closure Points

The empirical close of the dissertation rests on three questions:

1. what descriptive differences among optimizer methods are supported by realized-work and final-front evidence from a single-realization campaign, why the reported campaign shapes cannot support an interval or directional-agreement statement, and what independent-replicate shape would be needed to obtain one;
2. which deployer architecture patterns are supported by the Part B constellation read, and how strongly those patterns depend on catalogue scope, family encoding, and solver/scoring settings;
3. which sensitivity findings are robust enough to state as dissertation conclusions rather than as diagnostic or follow-on evidence.

This framing keeps pairwise optimizer diagnostics, exact front counts, architecture sufficiency percentages, and sensitivity margins tied to the evidence that produces them.

6.3 Limitations and Boundary Conditions

The manuscript carries explicit limits. Comparative statements remain tied to the hazardous-event bank, mission-scoring semantics, family encodings, solver settings, and campaign shape used for the reported evidence package. Fidelity and uncertainty-transport choices remain important near acceptance boundaries, so claims stay attached to the reported baseline rather than drifting into model-form generalizations. Constellation-level conclusions remain scoped to the families studied here; they do not exhaust every possible remediation architecture. All three evaluated families are analytically structured, so whether an unstructured, plane-constrained architecture could outperform them remains open: that encoding was scoped for this study but is not implemented in the reported evaluation stack, and no evidence here bounds it in either direction.

Those limits tell the reader how far the conclusions travel. The dissertation defends its framework and problem formulation without turning campaign-specific margins into universal statements.

6.4 Forward Work

The natural next layer is broader empirical stress testing. Extensions include richer policy coupling, additional catalogue epochs, broader traffic-management implications, deeper force-model sensitivity sweeps, and further validation of the dust-propagation representation. These extensions build on the same evidence chain rather than replacing it.

Several pieces of that chain are implemented and sealed but not reported empirically here, and they bound what a follow-on campaign must re-measure rather than re-build: the closed-form minimum-separation certificate that replaced the sampled constellation screen,

the regenerated representative event banks and their byte-identical generation receipt, the shared-altitude fairness envelope now enforced on the Flower encoding, and the post-Hybrid verification and validation lane. A campaign flown on that stack supersedes every empirical value quoted in [Chapter 4](#).

6.5 Closing Perspective

Taken as a methodological whole, the dissertation gives dust-enabled just-in-time collision avoidance a concrete and conservative design form. It evaluates deployer architectures through hazardous-event evidence, transfer feasibility, dust-cloud uncertainty, and shared optimizer controls rather than through isolated coverage or average-case surrogates. That discipline is the through-line from problem formulation to empirical interpretation.

Chapter 7

Summary

This short final chapter gathers the dissertation into one synthesis.

7.1 Study Synthesis

The dissertation treats dust-based just-in-time collision avoidance as an architecture-under-uncertainty problem grounded in reusable hazardous-event evidence, staged transfer feasibility, encounter-plane dust sizing, and conservative mission-level scoring. Within that formulation, the manuscript defines how candidate constellations and optimizer families are to be compared under shared physical and statistical semantics, and demonstrates that comparison on historical campaigns. Quantitative attainment, family ordering, and optimizer ranking all remain held: the representative event banks and the Flower fairness envelope were both corrected after the reported campaigns, and no post-correction comparison matrix has yet been run. Optimizer evidence is common-random-number aligned and descriptive, with no global or per-family winner. The central contribution is the disciplined coupling of event evidence, astrodynamics, uncertainty propagation, and multiobjective search.

7.2 Final Takeaway and Next Layer

Dust-enabled JCA is most persuasive when it is posed as a selective, uncertainty-aware response architecture and judged through strict event-level physics together with mission-level bookkeeping. The next research layer extends that foundation through larger catalogues, longer campaigns, richer policy coupling, and targeted sensitivity expansion.

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Appendices

Appendix A

Supplementary Dust-Covariance Details

A.1 Production Transform and Historical Split Details

This appendix records the operational covariance transport summarized in [Section 3.11](#) and labels the older split studies explicitly. The main text focuses on workflow and justification; this section gives the compact mathematical form used in the analysis.

A.1.1 Production K1 and Historical Mixture Recombination

Production carries one Gaussian, so the following generic mixture identity is evaluated at $K = 1$ and $w_1 = 1$. The $K > 1$ form is retained only to define historical split studies:

$$p(\mathbf{x}) = \sum_{j=1}^K w_j \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_j, \mathbf{P}_j), \quad w_j \geq 0, \quad \sum_{j=1}^K w_j = 1. \quad (\text{A.1})$$

The mixture mean and covariance used for reporting and downstream scoring are

$$\bar{\boldsymbol{\mu}} = \sum_{j=1}^K w_j \boldsymbol{\mu}_j, \quad (\text{A.2})$$

$$\bar{\mathbf{P}} = \sum_{j=1}^K w_j \left[\mathbf{P}_j + (\boldsymbol{\mu}_j - \bar{\boldsymbol{\mu}})(\boldsymbol{\mu}_j - \bar{\boldsymbol{\mu}})^\top \right]. \quad (\text{A.3})$$

This decomposition preserves both within-component spread and between-component separation.

For the historical rank-1 three-component comparison arm, the component weights are not equal: $\mathbf{w} = (1/6, 2/3, 1/6)$, with standardized axial nodes $(-\sqrt{3}, 0, +\sqrt{3})$. Thus $\sum_j w_j \mathbf{z}_j = \mathbf{0}$ and $\sum_j w_j \mathbf{z}_j \mathbf{z}_j^\top = \mathbf{I}$, so the split recovers the parent first two moments when the child covariance is $\boldsymbol{\Sigma} = (1 - \alpha^2) \mathbf{P}_\parallel + \mathbf{P}_\perp$. A counterfactual rank-2 split uses the tensor product of these nodes and weights, giving nine components with $W_{ab} = w_a w_b$. These weighted identities replace the equal-component approximation sometimes used in generic mixture descriptions. Neither arm is active production authority: production is sealed at (K=1), so no child offsets are formed. More explicitly, with $\mathbf{z}_{ab} = (z_a, z_b)^\top$,

$$\sum_{a,b} W_{ab} \mathbf{z}_{ab} = \mathbf{0}, \quad \sum_{a,b} W_{ab} \mathbf{z}_{ab} \mathbf{z}_{ab}^\top = \mathbf{I}_2, \quad (\text{A.4})$$

$$\sum_{a,b} W_{ab} (\boldsymbol{\mu}_{ab} - \boldsymbol{\mu}_0) \mathbf{z}_{ab}^\top = \alpha \mathbf{L}_2, \quad \sum_{a,b} W_{ab} (\boldsymbol{\mu}_{ab} - \boldsymbol{\mu}_0)(\boldsymbol{\mu}_{ab} - \boldsymbol{\mu}_0)^\top = \alpha^2 \mathbf{P}_\parallel. \quad (\text{A.5})$$

The third identity is the weighted M-UT cross-covariance between standardized split coordinates and physical mean offsets; it holds for rank 1 as well after replacing (a, b) by i , and fixes the tensor-product weighting used by the moment check and by recombination.

A.1.2 Per-Component Unscented Propagation

Each component is propagated independently with an unscented transform (UT). Production uses Julier's minimal-skew simplex at $W_0 = 0$. For the six-state transform, the zero-weight centre point is dropped and the retained $n + 1 = 7$ whitened offsets $\boldsymbol{\xi}_i$ carry equal mean and covariance weights $W_i = 1/(n + 1)$. With $\mathbf{S}_j \mathbf{S}_j^\top = \mathbf{P}_j$,

$$\boldsymbol{\chi}_{j,i} = \boldsymbol{\mu}_j + \mathbf{S}_j \boldsymbol{\xi}_i, \quad i = 1, \dots, n + 1. \quad (\text{A.6})$$

With scenario dynamics map $\Phi_{t_0 \rightarrow t}$, propagated sigma points are $\hat{\boldsymbol{\chi}}_{j,i}(t) = \Phi_{t_0 \rightarrow t}(\boldsymbol{\chi}_{j,i})$, and component moments become

$$\boldsymbol{\mu}_j^+ = \sum_i W_i^{(m)} \hat{\boldsymbol{\chi}}_{j,i}, \quad (\text{A.7})$$

$$\mathbf{P}_j^+ = \sum_i W_i^{(c)} (\hat{\boldsymbol{\chi}}_{j,i} - \boldsymbol{\mu}_j^+) (\hat{\boldsymbol{\chi}}_{j,i} - \boldsymbol{\mu}_j^+)^\top + \mathbf{Q}. \quad (\text{A.8})$$

Here $\mathbf{Q}$ denotes optional process-noise augmentation when enabled; baseline short-horizon runs typically use deterministic propagation with fixed environmental drivers. The scaled-UT triple and its 13-point Van der Merwe construction are historical comparison authority, not production inputs. They are distinct from the DART momentum-enhancement κ in [Equation \(3.5\)](#).

For encounter-plane sizing, let $\mathbf{H} \in \mathbb{R}^{2 \times 3}$ contain the orthonormal in-plane basis rows. The target position covariance is a fixed, intercept-local RIC covariance, not a covariance propagated by the dust model. If $\mathbf{R}_{\text{RIC} \rightarrow \text{ECI}}^{\text{int}}$ maps that intercept-local frame into ECI, define

$\Sigma_t^{\text{int,ECI}} = \mathbf{R}_{\text{RIC} \rightarrow \text{ECI}}^{\text{int}} \Sigma_t^{\text{int,RIC}} \mathbf{R}_{\text{RIC} \rightarrow \text{ECI}}^{\text{int}\top}$. The relative projected mean and covariance are

$$\boldsymbol{\mu}_{\text{rel}}^{2D} = \mathbf{H}(\boldsymbol{\mu}_d - \boldsymbol{\mu}_t), \quad (\text{A.9})$$

$$\Sigma_{\text{rel}}^{2D} = \mathbf{H} \left(\Sigma_d + \Sigma_t^{\text{int,ECI}} \right) \mathbf{H}^\top \quad (\text{Pc-inversion lane}). \quad (\text{A.10})$$

The two covariances are projected separately and summed in the encounter plane. The shared-target lane omits Σ_t here and integrates it once outside the packet exponent. The baseline assumes zero dust–target cross-covariance. A future correlated model would add the corresponding cross terms before projection; it is not silently folded into the current target covariance.

In Chapter 3 terms, this is the local nonlinear transport step used in the dust lane. Production carries one Gaussian component and propagates its seven Julier simplex points deterministically through the scenario dynamics. Recombination restores the first two moments needed for downstream encounter-plane calculations. Historical adaptive- α and bounded-Monte-Carlo split-selection arms are counterfactual studies; they do not run in production and do not define current results.

A.1.3 Split-Criterion Family and Baseline Choice

The compiled authority retains a dormant split recipe even though production ($K=1$) does not form child components. It is sealed in the compiled science constant, not in run configuration: the criterion token `maxvar` and policy token `scaled` live in `mf_transfer.native_policy`, and `split_rank = 1` with `gmm_components = 1` in `mf_lowering`. No run-configuration key can override them, and the implemented kernel is max-variance selection with the scaled time-of-flight schedule unconditionally; the one-variant selector enums were removed once they could no longer select anything. In a counterfactual ($K>1$) arm, the `maxvar` rule selects the

dominant covariance direction, that is, the eigenvector associated with the largest eigenvalue of the component covariance. With `split_rank = 1`, only that single dominant mode is split. The sealed scaled-TOF policy sets $\alpha = 0.35\alpha_{\max}$ through 7200 seconds and $\alpha = 0.60\alpha_{\max}$ above that boundary, setting how much principal-direction variance remains inside the child covariance

$$\Sigma = (1 - \alpha^2) \mathbf{P}_{\parallel} + \mathbf{P}_{\perp}$$

and how much appears through the separation of the child means. At production (K=1), those offsets are identically absent, so the criterion, rank, and scaled-TOF schedule are sealed but dormant. The historical (K=3) arm used the three-point standard-normal Gauss–Hermite layout with weights $1/6, 2/3, 1/6$.

Current production is therefore (K=1) + Julier-7. The historical (K=3), Van der Merwe-13, adaptive- α , Monte Carlo, and alternate-criterion arms are retained only as labeled counterfactual context, not as current runtime paths or current-result authority.

A.1.4 Selection-Evidence Workflow

Historical split-policy studies used the following workflow; this is not the current production selector:

- a short-horizon $p90$ criterion comparison over representative [LEO](#) scenarios with Wasserstein distance as the primary discriminator and MaDEM/MCR/CvM/ELK/NISE as diagnostics;
- multi-seed ranking and tie-breaking on pooled distributional diagnostics, with criterion and regime labels retained for audit;
- a direct split-versus-no-split multihorizon HF comparison over horizons from 10 to 180 min,

recording accuracy, runtime, and scenario coverage;

- a historical fixed-default decision rule that records aggregate diagnostics, horizon coverage, variability, and the counterfactual cost of retaining a split-capable representation rather than relying on one pooled average.

These procedures describe counterfactual study design only. Numerical outcomes are outside this methods appendix and are not inferred here.